Basic Mathematics

Mathematics

Lecture Notes

A friendly path through algebra, geometry, and calculus.

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Chapter 1

Introduction

These notes are an invitation to look at mathematics in a different way. For many students, mathematics has long appeared as a collection of formulas, rules, and mnemonic patterns: one recognizes the type of problem, chooses a remembered procedure, performs the required operations, and moves on. Such skills have their place, but they are not the heart of mathematics. Mathematics begins when we stop for a moment and ask what we are really looking at, what kind of object has appeared, what question is being asked, what is known, what is unknown, and why a proposed step is justified.

The purpose of this course is therefore not only to practise calculations. We will learn how to read mathematical expressions, how to formulate precise questions, how to distinguish an object from its notation, how to notice structure, and how to build simple but meaningful chains of reasoning. A solved problem is not valuable only because it produces an answer. It is valuable because it can reveal a method of thought: how to begin, what to observe, which assumptions matter, what can be transformed, what remains unchanged, and how a conclusion follows from earlier facts.

The course will focus on three fundamental parts of mathematics: linear algebra, analytic geometry, and calculus. Linear algebra introduces vectors, matrices, systems of equations, and transformations. Analytic geometry shows how algebraic equations describe points, lines, planes, directions, distances, and angles. Calculus gives a language for functions, limits, continuity, derivatives, integrals, change, and accumulation. These topics are not isolated chapters placed next to one another. Together they form a basic language for describing structure, space, and change.

We will try to study this language from the inside. A definition should not be treated as a sentence to memorize, but as a way of introducing a new kind of object. A theorem should not be treated as a decorative statement, but as the result of asking a clear question under precise assumptions. An example should not be treated as a template to copy blindly, but as a small experiment in reasoning. Even a simple calculation can show something important if we ask why it works and what it tells us.

For this reason, some arguments will deliberately return in different places. The same habit of asking what is fixed and what is variable appears when we study functions, vectors, equations, and transformations. The same need for precision appears when we define a line, compute a derivative, solve a system of equations, or interpret an integral. The same mathematical question can often be seen algebraically, geometrically, and conceptually. This repetition is intentional: it helps the reader recognize patterns of reasoning rather than only patterns of exercises.

The number of solved tasks is not the main measure of progress. What matters is how the tasks are solved. Did we understand the question? Did we identify the object correctly? Did we choose notation consciously? Did we know why each step was allowed? Did we pause to interpret the result? Did the calculation teach us something about the idea behind it? These questions are part of the work. They are not an addition to mathematics; they are mathematics being done carefully.

Mathematics is useful in physics, engineering, computer science, data analysis, modelling, and many other fields, but its value is not only practical. It is also a way of seeing. It teaches precision, discipline, imagination, and the ability to move from simple observations to general conclusions. These notes are meant to show a piece of that beauty, strength, and richness. The goal is not merely to survive mathematical techniques, but to begin to appreciate mathematics as an art of asking good questions and giving well-reasoned answers.

Chapter 2

Matrices

2.1 What matrices are

A matrix is a rectangular arrangement of mathematical objects, usually numbers. This description is correct, but it is not yet very informative. If we stop at this description, a matrix looks like a table, and matrix algebra looks like a collection of rules for moving numbers around. In this chapter we will try to see more than that. A matrix is a way of organizing information. It can describe data, relations between quantities, coefficients of equations, or a rule that transforms one vector into another.

The first skill is therefore not calculation, but reading. When we see a matrix, we should ask what its shape is, what its entries represent, and how its rows and columns are meant to be interpreted. The same array of numbers may be only a table in one problem, but in another problem it may describe a transformation, a system of equations, or a geometric operation.

Definition

A matrix with m rows and n columns is called an $m \times n$ matrix. It has the form

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

The entry a_{ij} is located in row i and column j.

This notation is simple, but it contains an important discipline. The first index tells us where to look vertically, and the second index tells us where to look horizontally. A student who confuses these indices will often make mistakes later when multiplying matrices or describing systems of equations.

Example

Consider the matrix

$$M = \begin{pmatrix} 4 & -2 & 0 \\ 7 & 1 & 5 \end{pmatrix}.$$

This is a 2×3 matrix. The entry m_{11} is 4, the entry m_{12} is -2, and the entry m_{23} is 5. We are not doing any operation yet. We are only learning to read the object.

The size of a matrix is part of its identity. A row matrix, a column matrix, and a square matrix may contain similar numbers, but they play different roles. A column matrix can represent a vector. A square matrix can sometimes represent a transformation from a space to itself. A rectangular matrix may describe a relation between two collections of quantities of different sizes.

Definition

A matrix is called square if it has the same number of rows and columns. An $n \times n$ matrix is a square matrix of order n.

We will often begin with square matrices because many basic ideas are easiest to see there. However, rectangular matrices are just as important. The point is not to memorize a list of shapes, but to develop the habit of asking: what kind of object am I holding, and what operations make sense for this object?

2.2 Basic matrix operations

Once we know how to read a matrix, we can begin to operate on it. The simplest operations are addition, subtraction, and multiplication by a number. These operations are deliberately simple because they preserve the shape of the matrix. They do not change a table into a different kind of object; they change its entries while keeping the same positions.

Definition

Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices of the same size $m \times n$. Their sum is the matrix

$$A + B = (a_{ij} + b_{ij}),$$

and their difference is the matrix

$$A - B = (a_{ij} - b_{ij}).$$

If λ is a number, then

$$\lambda A = (\lambda a_{ij}).$$

The phrase "of the same size" is not a technical detail. It is the reason the operation is meaningful. When two matrices have the same size, each entry of one matrix has a partner in the other matrix. Addition and subtraction then happen position by position.

Example

Let

$$A = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 5 & 4 \\ -2 & 1 \end{pmatrix}.$$

Then

$$A + B = \begin{pmatrix} 2+5 & -1+4 \\ 0+(-2) & 3+1 \end{pmatrix} = \begin{pmatrix} 7 & 3 \\ -2 & 4 \end{pmatrix}.$$

Also,

$$A - B = \begin{pmatrix} 2-5 & -1-4 \\ 0-(-2) & 3-1 \end{pmatrix} = \begin{pmatrix} -3 & -5 \\ 2 & 2 \end{pmatrix}.$$

Finally,

$$-3A = \begin{pmatrix} -6 & 3 \\ 0 & -9 \end{pmatrix}.$$

The calculation is not difficult. The important thing is the discipline of positions. We do not add rows to columns, and we do not rearrange entries because it looks convenient. We respect the shape of the object.

These operations satisfy the familiar rules one expects from ordinary arithmetic. For matrices of the same size,

$$A + B = B + A, \qquad (A + B) + C = A + (B + C).$$

This should not be surprising: addition is performed entry by entry, and ordinary addition of numbers is commutative and associative.

Remark

Whenever a matrix operation is defined entry by entry, many properties of ordinary arithmetic pass directly to matrices. But this principle must be used carefully. Matrix multiplication will not be entry-by-entry multiplication, and therefore it will behave differently.

2.3 Matrix multiplication

Matrix multiplication is the first operation in this chapter that is not simply performed position by position. This is why it deserves careful attention. Many mistakes with matrices come from treating multiplication as if it were only another version of addition. It is not. Addition compares matching positions. Multiplication combines rows of the first matrix with columns of the second matrix.

Definition

Let $A = (a_{ij})$ be an $m \times n$ matrix and let $B = (b_{ij})$ be an $n \times p$ matrix. The product AB is the $m \times p$ matrix $C = (c_{ij})$, where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

In words, the entry c_{ij} is obtained by multiplying row i of A by column j of B.

The condition on sizes is built into the definition. If A has size $m \times n$, then each row of A has n entries. To multiply this row by a column of B, each column of B must also have n entries. Therefore B must have n rows. We summarize this by writing

$$(m \times n)(n \times p) = m \times p.$$

The inner dimensions must match; the outer dimensions describe the size of the result.

Example

Let

$$A = \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix}.$$

The first entry of AB comes from the first row of A and the first column of B:

$$1 \cdot 2 + 3 \cdot 5 = 17.$$

The first row and second column give

$$1 \cdot 0 + 3 \cdot (-1) = -3.$$

Continuing in the same way,

$$AB = \begin{pmatrix} 17 & -3 \\ 16 & -4 \end{pmatrix}.$$

The example is small, but the point is general. Each entry of the product answers one local question: how does a chosen row of the first matrix interact with a chosen column of the second matrix?

Why the order matters

For ordinary numbers, $ab = ba$. For matrices this is not generally true. The reason is conceptual, not only computational. The product AB asks rows of A to interact with columns of B. The product BA asks a different question.

Using the matrices from the previous example,

$$BA = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 7 & 11 \end{pmatrix}.$$

Thus $AB \neq BA$. The matrices have not changed, but the order has changed, and the question asked by the multiplication has changed.

Remark

The statement $AB \neq BA$ does not mean that two matrices can never commute. Some matrices do commute. The point is that commutativity is not automatic. If it matters, it must be checked or justified from the structure of the matrices.

Matrix multiplication is associative:

$$(AB)C = A(BC),$$

whenever the products are defined. This means that if several compatible matrices are multiplied in a fixed order, parentheses do not change the result. But the order of the matrices themselves still matters. Associativity tells us how to group operations. It does not allow us to swap them.

2.4 Special matrices and visible structure

Some matrices are important because their structure is easy to see. They help us learn a useful lesson: a matrix is not only a collection of entries. The position of the entries matters, and certain patterns can make an operation simple or meaningful.

The zero matrix and the identity matrix

The zero matrix is the matrix whose entries are all zero. It behaves like zero for matrix addition. If A is an $m \times n$ matrix and 0 is the zero matrix of the same size, then

$$A + 0 = A.$$

The identity matrix plays a similar role for multiplication, but only for square matrices.

Definition

The $n \times n$ identity matrix is

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

For compatible matrices,

$$I_n A = A, \qquad A I_n = A.$$

The identity matrix is important because it represents doing nothing. This phrase is informal, but helpful. Multiplying by the identity preserves the object, just as multiplying a number by 1 preserves the number.

Diagonal matrices

A diagonal matrix is a square matrix whose non-zero entries, if any, lie only on the main diagonal.

Definition

A square matrix $D = (d_{ij})$ is diagonal if

$$d_{ij} = 0 \quad \text{whenever } i \neq j.$$

We often write

$$D = \operatorname{diag}(d_1, d_2, \ldots, d_n).$$

Diagonal matrices are useful because they act independently on different positions. There is no mixing between coordinates. This is why many calculations with diagonal matrices are transparent.

Example

Let

$$D = \operatorname{diag}(3, -1, 2).$$

Then

$$D^2 = \operatorname{diag}(9, 1, 4), \qquad D^3 = \operatorname{diag}(27, -1, 8).$$

The power is computed by taking powers of the diagonal entries. This works because all off-diagonal entries are zero, so the coordinates do not mix.

If two diagonal matrices have the same size, then they commute. Indeed,

$$\operatorname{diag}(a_1, \ldots, a_n) \operatorname{diag}(b_1, \ldots, b_n) = \operatorname{diag}(a_1 b_1, \ldots, a_n b_n),$$

and the same result is obtained in the opposite order because ordinary numbers commute. This is a good example of structure explaining an algebraic fact.

Patterns worth noticing

When a matrix appears in a problem, it is useful to ask whether it has visible structure. Is it diagonal? Does it contain many zeros? Are two rows equal? Are rows or columns multiples of one another? Does it look symmetric? Such observations can simplify computations, but more importantly they help us understand the object.

Remark

A calculation often becomes easier after we notice structure. But noticing structure is not a trick; it is part of mathematical understanding. We learn not only to compute with objects, but also to see them.

2.5 Row operations and problem solving

Matrices are often used because they allow us to transform a problem without rewriting every sentence of the problem. One of the most important examples is the use of row operations. Row operations will become essential when we solve systems of linear equations, but it is useful to meet them already here.

A row operation changes a matrix in a controlled way. We do not change entries randomly. We perform an operation that has a clear description and a clear purpose.

Definition

The basic elementary row operations are:

- swapping two rows,
- multiplying a row by a non-zero number,
- adding a multiple of one row to another row.

The value of row operations is not that they make matrices look different. Their value is that they help reveal structure. For example, they can help us simplify a matrix, detect whether equations are dependent, or prepare a system for solving.

Example

Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 0 & 1 & 5 \end{pmatrix}.$$

Suppose we replace the second row by the second row minus twice the first row. Then

$$(2, 4, 3) - 2(1, 2, -1) = (0, 0, 5).$$

The new matrix is

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 0 & 5 \\ 0 & 1 & 5 \end{pmatrix}.$$

This operation did not merely produce new numbers. It revealed that part of the second row was already explained by the first row, except for the last entry.

This example should be read slowly. The operation has three layers. First, there is arithmetic. Second, there is a transformation of the row. Third, there is interpretation: the new row tells us something about dependence between rows.

Column vectors as matrices

A column vector may be viewed as a matrix with one column. This viewpoint will be used constantly in linear algebra. It allows us to write vector equations and matrix equations in the same language.

For instance,

$$u = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$$

is a 3×1 matrix. Its transpose is the row matrix

$$u^\top = \begin{pmatrix} 2 & -1 & 4 \end{pmatrix}.$$

The distinction between a row and a column is not cosmetic. The products $u^\top u$ and $uu^\top$ have different sizes and different meanings:

$$u^\top u$$

is a 1×1 matrix, while

$$uu^\top$$

is a 3×3 matrix.

This is another example of a recurring principle: before calculating, read the shapes.

How to approach matrix exercises

When solving matrix exercises, the aim is not to rush toward the answer. The answer is only the last line of a process. A good solution should make the process visible.

Summary

When working with matrices, ask:

- What is the size of each matrix?
- Which operations are defined?
- Am I adding entries position by position, or multiplying rows by columns?
- Does the order of multiplication matter here?
- Is there a visible structure that simplifies the problem?
- What does the result tell me about the original objects?

A student who asks these questions is not merely applying a recipe. They are beginning to use matrices as mathematical objects. That is the real purpose of this chapter.

Chapter 3

Determinants

3.1 Why determinants matter

In the previous chapter we learned how to read matrices and how to perform basic operations on them. We can add matrices, multiply them by numbers, multiply compatible matrices, and recognize simple structures such as the identity matrix or a diagonal matrix. But after learning these operations, a natural question appears.

Are all square matrices equally good?

This question is intentionally vague at first. Mathematics often begins in this way: we notice that some objects behave differently from others, and only later do we find the right words and the right definitions. Consider the following matrices:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

The first matrix is the identity matrix. It preserves everything under multiplication. The second matrix stretches the first coordinate by a factor of 2 and the second coordinate by a factor of 3. The third matrix is different. Its second row is exactly twice its first row:

$$(2, 4) = 2(1, 2).$$

So the two rows do not contain two independent pieces of information. One row is only a repetition of the other, scaled by a number.

This observation suggests a deeper question. Can a matrix lose information? Can a matrix collapse different inputs into outputs that are no longer distinguishable? Can we detect this collapse by assigning one carefully chosen number to the matrix?

The determinant is such a number.

This statement should not be read as a definition yet. At this stage it is only a promise. We are looking for a number attached to a square matrix, but not just any number. We want a number that reacts to the structure of the matrix. It should notice when rows or columns become dependent. It should distinguish matrices that preserve two-dimensional or three-dimensional structure from matrices that collapse it. It should also help us decide, later, whether a matrix can be inverted.

Remark

A determinant is not a summary of all entries of a matrix in the sense of an average or a sum. It is a number designed to detect structural behaviour. In particular, the value 0 will become extremely important: it will mean that the matrix has lost a certain kind of independence.

There is also a geometric way to think about determinants. A 2×2 matrix can transform the plane. Some matrices stretch areas, some reverse orientation, and some flatten the plane onto a line. The determinant measures this behaviour. In dimension 2, its absolute value describes an area scale factor. In dimension 3, its absolute value describes a volume scale factor. A determinant equal to zero means that area or volume has been flattened completely.

For the purposes of this course, however, we will not begin from geometry alone. We will move between several viewpoints:

- determinants as numbers computed from square matrices,
- determinants as tests for structural collapse,
- determinants as tools for recognizing invertibility,
- determinants as quantities affected in predictable ways by row operations.

The chapter will therefore follow a path of discovery. First we study small matrices, where the formulas can be seen directly. Then we introduce a systematic method for larger matrices: minors, cofactors, and Laplace expansion. After that we collect the properties that make determinants useful in actual calculation. Finally we connect determinants with singular matrices and prepare the transition to the next chapter, where invertibility becomes the central topic.

3.2 Determinants of small matrices

The determinant is defined only for square matrices. This is already meaningful: we are not attaching this number to every rectangular table. We are studying matrices that can be interpreted as transforming a space into itself, or as square coefficient matrices in systems of equations.

We begin with small cases. This is not only because small cases are easier. Small cases allow us to see the idea before it becomes hidden behind notation.

3.2.1 The determinant of a 1×1 matrix

The simplest square matrix has only one entry:

$$A = \begin{pmatrix} a \end{pmatrix}.$$

There is no structure to compare, no rows to exchange, and no area or volume to measure. The determinant is simply the entry itself:

$$\det A = a.$$

This case looks trivial, but it is useful because larger definitions will reduce determinants to smaller determinants. Eventually, a long computation ends at 1×1 determinants.

3.2.2 The determinant of a 2×2 matrix

Now consider a general 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

The determinant is defined by

$$\det A = ad - bc.$$

The formula should be read carefully. We multiply along the main diagonal and subtract the product along the other diagonal:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

At first this may look like an arbitrary rule. It is not arbitrary. It is a rule that reacts very strongly to dependence between rows or columns.

Example

For
$$A = \begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix}$$
we get
$$\det A = 3 \cdot 5 - 1 \cdot 2 = 15 - 2 = 13.$$
The determinant is non-zero. This suggests that the two rows are not repetitions of each other and that the matrix has not collapsed its two-dimensional structure.

Example

For
$$B = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}$$
we get
$$\det B = 1 \cdot 6 - 2 \cdot 3 = 6 - 6 = 0.$$
The second row is three times the first row:
$$(3, 6) = 3(1, 2).$$
Thus the determinant detects that the two rows are dependent.

This example gives us the first important interpretation: when the determinant of a 2×2 matrix is zero, the two rows or the two columns fail to provide two independent directions of information.

3.2.3 What changes the determinant?

Let us observe how the formula reacts to simple operations. For
$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$
we have
$$\det A = ad - bc.$$
If we exchange the two rows, we obtain
$$A' = \begin{pmatrix} c & d \\ a & b \end{pmatrix}.$$
Then
$$\det A' = cb - da = -(ad - bc) = -\det A.$$
So exchanging two rows changes the sign of the determinant.

If the two rows are equal,
$$A = \begin{pmatrix} a & b \\ a & b \end{pmatrix},$$
then
$$\det A = ab - ba = 0.$$
This is exactly what we should expect: two equal rows certainly do not provide two independent directions.

Remark

The determinant is sensitive to order and independence. It changes sign when rows are swapped, and it becomes zero when rows become dependent. These facts will later become general properties for determinants of all sizes.

3.2.4 The determinant of a 3×3 matrix and Sarrus' rule

For a 3×3 matrix

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix},$$

one useful formula is

$$\det A = aei + bfg + cdh - ceg - bdi - afh.$$

This formula can be remembered using Sarrus' rule. Repeat the first two columns after the matrix:

$$\begin{array}{ccc|cc} a & b & c & a & b \\ d & e & f & d & e \\ g & h & i & g & h \end{array}$$

Then add the products along the three downward diagonals and subtract the products along the three upward diagonals:

$$\det A = (aei + bfg + cdh) - (ceg + bdi + afh).$$

Example

Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 2 & 0 & 5 \end{pmatrix}.$$

Using Sarrus' rule,

$$\det A = (1 \cdot 3 \cdot 5 + 2 \cdot 4 \cdot 2 + 0 \cdot (-1) \cdot 0) - (0 \cdot 3 \cdot 2 + 2 \cdot (-1) \cdot 5 + 1 \cdot 4 \cdot 0).$$

Hence

$$\det A = (15 + 16 + 0) - (0 - 10 + 0) = 31 - (-10) = 41.$$

The determinant is non-zero.

The computation above is mechanical, but it is still worth reading. A non-zero determinant tells us that the rows and columns of the matrix do not collapse into a lower-dimensional structure. Later this will imply that the matrix is invertible.

Remark

Sarrus' rule is convenient, but it is special. It works for 3×3 matrices only. It is not a method for 4×4 or larger matrices. For larger matrices we need a more general idea.

3.2.5 A warning about calculation

When computing determinants, students often try to memorize many separate rules. A better habit is to ask three questions:

- What is the size of the matrix?

- Which method is appropriate for this size and this structure?
- Is there visible structure that can simplify the calculation?

For 2×2 matrices, the formula $ad - bc$ is usually fastest. For 3×3 matrices, Sarrus' rule may be useful. But as soon as the matrix becomes larger, or as soon as many zeros appear, the method of expansion by minors and cofactors becomes much more important.

3.3 Minors, cofactors, and Laplace expansion

Sarrus' rule gives a quick method for 3×3 matrices, but it does not generalize to larger matrices. If we want a method that works for every square matrix, we need a different idea. The key idea is reduction: we compute a large determinant by reducing it to smaller determinants.

This is a recurring mathematical pattern. When an object is too large to understand at once, we try to break it into smaller objects of the same kind. For determinants, the smaller objects are called minors.

3.3.1 Minors

Let $A = (a_{ij})$ be an $n \times n$ matrix. Choose an entry a_{ij}. Remove row i and column j. The determinant of the remaining $(n-1) \times (n-1)$ matrix is called the minor of a_{ij}, and is denoted by M_{ij}.

Example

Let
$$A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 5 & 2 & 1 \end{pmatrix}.$$
To find M_{12}, we remove row 1 and column 2. The remaining matrix is
$$\begin{pmatrix} 0 & 4 \\ 5 & 1 \end{pmatrix}.$$
Therefore
$$M_{12} = \det \begin{pmatrix} 0 & 4 \\ 5 & 1 \end{pmatrix} = 0 \cdot 1 - 4 \cdot 5 = -20.$$

A minor is not an entry of the original matrix. It is a determinant obtained after deleting a row and a column. This distinction matters: the entry a_{ij} is a number already present in the matrix; the minor M_{ij} is computed from what remains after removing the row and column of that entry.

3.3.2 Cofactors

The minor still needs a sign. The signs follow a checkerboard pattern:
$$\begin{pmatrix} + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$
This pattern is encoded by the factor $(-1)^{i+j}$.

Definition

The cofactor of the entry a_{ij} is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

Example

Using the matrix from the previous example, we found

$$M_{12} = -20.$$

Since $(-1)^{1+2} = (-1)^3 = -1$, the cofactor is

$$C_{12} = (-1)M_{12} = 20.$$

For the entry in position $(1,1)$, the sign is positive. Removing row 1 and column 1, we get

$$M_{11} = \det \begin{pmatrix} -1 & 4 \\ 2 & 1 \end{pmatrix} = (-1) \cdot 1 - 4 \cdot 2 = -9.$$

Thus

$$C_{11} = M_{11} = -9.$$

The alternating signs are not decoration. Without them, the determinant would not have the correct behaviour under row exchange, and it would not detect orientation properly.

3.3.3 Laplace expansion along a row

Laplace expansion expresses a determinant as a sum of entries multiplied by their cofactors. If we expand along row i, then

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}.$$

Equivalently,

$$\det A = \sum_{j=1}^{n} a_{ij}C_{ij}.$$

Example

Let

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 5 & 2 & 1 \end{pmatrix}.$$

Expand along the first row. We need the cofactors C_{11}, C_{12}, and C_{13}.
First,

$$C_{11} = +\det \begin{pmatrix} -1 & 4 \\ 2 & 1 \end{pmatrix} = (-1) \cdot 1 - 4 \cdot 2 = -9.$$

Second,

$$C_{12} = -\det \begin{pmatrix} 0 & 4 \\ 5 & 1 \end{pmatrix} = -(0 \cdot 1 - 4 \cdot 5) = 20.$$

Third,

$$C_{13} = +\det \begin{pmatrix} 0 & -1 \\ 5 & 2 \end{pmatrix} = 0 \cdot 2 - (-1) \cdot 5 = 5.$$

Therefore

$$\det A = 2C_{11} + 1C_{12} + 3C_{13} = 2(-9) + 1(20) + 3(5).$$

Hence

$$\det A = -18 + 20 + 15 = 17.$$

Notice how the computation works. A 3×3 determinant was reduced to three 2×2 determinants. For a 4×4 determinant, Laplace expansion reduces the problem to 3×3 determinants, and so on. The method is recursive.

3.3.4 Laplace expansion along a column

We may also expand along a column. If we expand along column j, then

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}.$$

Equivalently,

$$\det A = \sum_{i=1}^{n} a_{ij}C_{ij}.$$

The freedom to choose a row or a column is important. A wise choice can make a determinant much easier to compute.

Example

Consider

$$B = \begin{pmatrix} 4 & 0 & 1 & 2 \\ 0 & 3 & 0 & -1 \\ 2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 6 \end{pmatrix}.$$

The second column contains three zeros:

$$\begin{pmatrix} 0 \\ 3 \\ 0 \\ 0 \end{pmatrix}.$$

So we expand along the second column. Only one term survives:

$$\det B = 3C_{22}.$$

Since $(-1)^{2+2} = 1$,

$$C_{22} = \det \begin{pmatrix} 4 & 1 & 2 \\ 2 & 5 & 0 \\ 1 & 0 & 6 \end{pmatrix}.$$

Now compute the remaining 3×3 determinant. Using Sarrus' rule,

$$\det \begin{pmatrix} 4 & 1 & 2 \\ 2 & 5 & 0 \\ 1 & 0 & 6 \end{pmatrix} = (4 \cdot 5 \cdot 6 + 1 \cdot 0 \cdot 1 + 2 \cdot 2 \cdot 0) - (2 \cdot 5 \cdot 1 + 1 \cdot 2 \cdot 6 + 4 \cdot 0 \cdot 0).$$

Thus

$$C_{22} = 120 - (10 + 12) = 98.$$

Therefore

$$\det B = 3 \cdot 98 = 294.$$

This example shows why Laplace expansion is not merely a theoretical formula. It gives a strategy. Before computing, inspect the matrix. A row or column with many zeros is usually the best place to expand.

3.3.5 A method, not a ritual

Laplace expansion can be used along any row or any column. The result is always the same determinant. But the amount of work may be very different. The mathematical question is not only "Can I compute it?" but also "How can I compute it intelligently?"

Summary

When using Laplace expansion, follow this order.

- Inspect the matrix before calculating.
- Look for a row or column with many zeros.
- Write the sign pattern carefully.
- Compute the minors as smaller determinants.
- Keep the structure visible; do not compress too many steps at once.

3.4 Properties of determinants

At this point we know how to compute determinants of small matrices, and we know a general method: Laplace expansion. But if we use Laplace expansion blindly, determinant calculations can become long and unpleasant. Mathematics is not only about having a method. It is also about knowing when a method can be simplified.

The determinant has several important properties. These properties are not just facts to memorize. They are tools for reading a matrix before calculating.

3.4.1 Triangular matrices

A square matrix is upper triangular if all entries below the main diagonal are zero. It is lower triangular if all entries above the main diagonal are zero.

For example,

$$U = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{pmatrix}$$

is upper triangular.

Theorem

The determinant of a triangular matrix is the product of its diagonal entries.

$$\det U = u_{11}u_{22}\cdots u_{nn}.$$

For the matrix above,

$$\det U = 2 \cdot 3 \cdot (-2) = -12.$$

This is much faster than expanding the determinant directly.

The reason is visible from Laplace expansion. If we expand a triangular matrix along a row or column with many zeros, the determinant keeps reducing to smaller triangular matrices.

Eventually only diagonal entries remain.

3.4.2 Swapping two rows

If two rows of a matrix are exchanged, the determinant changes sign.

Theorem

If B is obtained from A by swapping two rows, then

$$\det B = -\det A.$$

The same is true for swapping two columns.

For a 2×2 matrix this can be checked directly:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc,$$

while

$$\det \begin{pmatrix} c & d \\ a & b \end{pmatrix} = cb - da = -(ad - bc).$$

The general case follows from the same alternating nature of the determinant.

This property explains why the sign pattern in cofactors matters. The determinant remembers orientation. Exchanging two rows reverses that orientation.

3.4.3 Equal rows and proportional rows

If two rows of a matrix are equal, then the determinant is zero.

Indeed, suppose two rows are equal. If we swap these two rows, the matrix does not change. Therefore the determinant should remain the same. But swapping two rows changes the sign of the determinant. Hence

$$\det A = -\det A,$$

so

$$2 \det A = 0,$$

and therefore

$$\det A = 0.$$

This is a good example of a short argument. We did not compute the determinant. We used a property to force its value.

If two rows are proportional, the determinant is also zero. For example,

$$A = \begin{pmatrix} 1 & 3 & -2 \\ 2 & 6 & -4 \\ 0 & 1 & 5 \end{pmatrix}$$

has the second row equal to twice the first row. The determinant is zero because the first two rows do not contain independent information.

Remark

A determinant equal to zero often means that some hidden dependence is present. Repeated rows and proportional rows are the easiest visible cases of this phenomenon.

3.4.4 Multiplying a row by a number

If one row of a matrix is multiplied by a number λ, then the determinant is also multiplied by λ.

Theorem

If B is obtained from A by multiplying one row by λ, then

$$\det B = \lambda \det A.$$

The same is true for multiplying one column by λ.

For example,

$$\det \begin{pmatrix} 2 & 4 \\ 3 & 5 \end{pmatrix} = 2 \cdot 5 - 4 \cdot 3 = 10 - 12 = -2.$$

If we multiply the first row by 3, we get

$$\det \begin{pmatrix} 6 & 12 \\ 3 & 5 \end{pmatrix} = 6 \cdot 5 - 12 \cdot 3 = 30 - 36 = -6.$$

The new determinant is three times the old one:

$$-6 = 3(-2).$$

This property is often used when simplifying determinants or when factoring common numbers out of rows or columns.

3.4.5 Adding a multiple of one row to another

One of the most useful properties is the following.

Theorem

Adding a multiple of one row to another row does not change the determinant.

This property is extremely important because it allows us to use row operations to simplify a determinant without changing its value. It is the determinant version of a powerful idea: transform the object into a simpler one while preserving the quantity we care about.

Example

Compute

$$\det \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 0 & 4 \end{pmatrix}.$$

We simplify the matrix using row operations that do not change the determinant. Replace

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - R_1.$$

Then

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 0 & 4 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & -2 & 3 \end{pmatrix}.$$

Now replace

$$R_3 \leftarrow R_3 + 2R_2.$$

We get

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 5 \end{pmatrix}.$$

This matrix is upper triangular, so its determinant is the product of the diagonal entries:

$$\det A = 1 \cdot 1 \cdot 5 = 5.$$

This calculation is much more efficient than expanding immediately. It also demonstrates how determinant properties support intelligent computation.

3.4.6 The determinant of a product

Another fundamental property is

$$\det(AB) = \det(A)\det(B),$$

for square matrices A and B of the same size.

This formula says that determinants multiply under matrix multiplication. If one matrix scales area or volume by one factor and another matrix scales it by another factor, then doing both transformations multiplies the scale factors.

One important consequence is immediate. If A has determinant zero, then for every compatible square matrix B,

$$\det(AB) = 0.$$

A matrix that has already collapsed structure cannot be repaired by multiplying by another matrix on one side.

3.4.7 How to use the properties

The properties of determinants are not separate tricks. They form a strategy.

Summary

Before computing a determinant, ask:

- Is the matrix triangular?
- Are two rows or columns equal or proportional?
- Can I create zeros using row operations that preserve the determinant?
- If I swap rows, did I change the sign?
- If I multiply a row by a number, did I multiply the determinant by that number?
- Is Laplace expansion now easier after simplification?

A determinant should not be expanded mechanically unless there is no better path. Often the best computation begins with looking at the matrix and asking what structure is already visible.

3.5 Singularity and parameters

The determinant is especially important because of what happens when it is equal to zero. A zero determinant is not just a numerical accident. It tells us that the matrix has lost a certain kind of independence.

Definition

A square matrix A is called singular if

$$\det A = 0.$$

It is called non-singular if

$$\det A \neq 0.$$

The word "singular" means exceptional or special. In this context, it means that the matrix does not behave like a fully independent square matrix. Something has collapsed. Rows or columns may be dependent, the associated transformation may flatten space, or a system of equations may fail to have a unique solution.

3.5.1 Reading singularity from rows

Consider

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & -1 \end{pmatrix}.$$

The second row is twice the first row:

$$(2, 4, 6) = 2(1, 2, 3).$$

Therefore the rows are dependent, and the determinant is zero:

$$\det A = 0.$$

We do not need to expand the determinant to know this. The structure is visible.

This is an important habit: before calculating, inspect. Sometimes the answer is already present in the shape of the object.

3.5.2 A determinant as a test for invertibility

A central theorem, which will become essential in the next chapter, is the following.

Theorem

A square matrix A is invertible if and only if

$$\det A \neq 0.$$

Equivalently, A is non-invertible if and only if

$$\det A = 0.$$

This theorem is one of the reasons determinants matter. It says that a single number can answer a structural question: can the matrix be reversed? If the determinant is non-zero, the matrix has not collapsed its structure, and an inverse matrix exists. If the determinant is zero, the matrix has lost information, and no inverse matrix can recover it.

We will study inverse matrices in the next chapter. Here we should understand the direction of thought:

$$\det A \neq 0 \quad \Longrightarrow \quad \text{the matrix can be inverted,}$$

while

$$\det A = 0 \quad \Longrightarrow \quad \text{the matrix cannot be inverted.}$$

3.5.3 Matrices depending on a parameter

Determinants are often used when a matrix contains a parameter. Then the determinant becomes an expression depending on that parameter. Solving

$$\det A = 0$$

tells us for which parameter values the matrix becomes singular.

Example

Let

$$A(t) = \begin{pmatrix} t & 1 \\ 4 & t \end{pmatrix}.$$

Then

$$\det A(t) = t \cdot t - 1 \cdot 4 = t^2 - 4.$$

The matrix is singular when

$$t^2 - 4 = 0.$$

Therefore

$$t = 2 \quad \text{or} \quad t = -2.$$

For all other values of t, the matrix is non-singular.

This example is simple, but it shows the main idea. A parameter does not merely make the computation longer. It changes the question. We are no longer asking for one determinant value. We are asking when the structure of the matrix changes.

Example

Let

$$B(a) = \begin{pmatrix} 1 & a & 0 \\ 0 & 1 & a \\ a & 0 & 1 \end{pmatrix}.$$

Using Sarrus' rule,

$$\det B(a) = (1 \cdot 1 \cdot 1 + a \cdot a \cdot a + 0 \cdot 0 \cdot 0) - (0 \cdot 1 \cdot a + a \cdot 0 \cdot 1 + 1 \cdot a \cdot 0).$$

Thus

$$\det B(a) = 1 + a^3.$$

The matrix is singular when

$$1 + a^3 = 0,$$

so

$$a = -1.$$

Thus $B(a)$ is singular only for $a = -1$, and non-singular for all other real values of a.

The determinant turns a structural question about a matrix into an algebraic equation. This is one of the most useful aspects of the concept.

3.5.4 What should be concluded?

When a problem asks for parameter values for which a matrix is singular, the answer is not just a list of numbers. The reasoning has a structure:

1. compute $\det A$ as a function of the parameter,
2. solve the equation $\det A = 0$,
3. interpret the roots as values where the matrix loses invertibility.

Summary

The determinant separates square matrices into two broad classes.

- If $\det A \neq 0$, the matrix is non-singular and invertible.
- If $\det A = 0$, the matrix is singular and not invertible.
- If A depends on a parameter, the equation $\det A = 0$ identifies the exceptional parameter values.

This prepares the next chapter. Once we know how to detect whether a matrix is invertible, we can ask a new question: if the inverse exists, how do we find it?

3.6 Determinant patterns

A student who has just learned determinants often tries to compute every determinant by direct expansion. This is understandable, but it is not the best habit. Determinants reward attention to structure. Before expanding, we should ask what kind of matrix we are looking at.

In this final section we collect several patterns that help with determinant calculations. The purpose is not to create a list of tricks. The purpose is to train the eye.

3.6.1 Zeros are invitations

A row or column with many zeros usually tells us where to expand.

Example

Compute

$$\det \begin{pmatrix} 2 & 0 & 1 & 3 \\ 0 & 0 & 5 & 1 \\ 4 & 0 & -1 & 2 \\ 1 & 7 & 0 & 6 \end{pmatrix}.$$

The second column is

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 7 \end{pmatrix}.$$

So expansion along the second column gives only one non-zero term:

$$\det A = 7C_{42}.$$

The sign is

$$(-1)^{4+2} = 1,$$

so

$$\det A = 7 \det \begin{pmatrix} 2 & 1 & 3 \\ 0 & 5 & 1 \\ 4 & -1 & 2 \end{pmatrix}.$$

Now the problem has been reduced to a 3×3 determinant.

The point is not that this computation is finished immediately. The point is that the matrix itself suggested a good first move.

3.6.2 Triangular form is a goal

Because triangular matrices have easy determinants, it is often useful to transform a determinant into triangular form using row operations that preserve the determinant.

Example

Let
$$A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 3 & 7 \\ 3 & 5 & 12 \end{pmatrix}.$$
Use row operations that do not change the determinant:
$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - 3R_1.$$
Then
$$A \longrightarrow \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 2 & 6 \end{pmatrix}.$$
Now use
$$R_3 \leftarrow R_3 - 2R_2.$$
We obtain
$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}.$$
This triangular matrix has determinant
$$1 \cdot 1 \cdot 0 = 0.$$
Therefore $\det A = 0$. The final zero row reveals that the original rows were dependent.

This example is valuable because the calculation and the interpretation agree. We did not merely obtain zero. We saw why zero appeared.

3.6.3 Common factors

If every entry in one row has a common factor, that factor can be taken out of the determinant. The same is true for columns.

For example,
$$\det \begin{pmatrix} 2a & 2b \\ c & d \end{pmatrix} = 2 \det \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$
This is often useful when determinants contain parameters or expressions.

Example

Compute
$$\det \begin{pmatrix} 3x & 6x \\ 2 & 5 \end{pmatrix}.$$

The first row has a common factor $3x$:

$$\det \begin{pmatrix} 3x & 6x \\ 2 & 5 \end{pmatrix} = 3x \det \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}.$$

Now

$$\det \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix} = 1 \cdot 5 - 2 \cdot 2 = 1.$$

Hence

$$\det \begin{pmatrix} 3x & 6x \\ 2 & 5 \end{pmatrix} = 3x.$$

Factoring is not only a simplification of arithmetic. It may reveal parameter values for which the determinant becomes zero.

3.6.4 Vandermonde-type structure

Some determinants have recognizable patterns. One famous example is the Vandermonde determinant:

$$V = \begin{pmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{pmatrix}.$$

Its determinant is

$$\det V = (y - x)(z - x)(z - y).$$

This formula is not something we need to memorize at this stage. What matters is the meaning: the determinant becomes zero when two of the parameters are equal. For example, if $x = y$, then the first two columns are equal:

$$\begin{pmatrix} 1 \\ x \\ x^2 \end{pmatrix} = \begin{pmatrix} 1 \\ y \\ y^2 \end{pmatrix}.$$

Therefore the determinant must be zero. The factor $(y - x)$ records this fact algebraically.

This is a powerful idea. A determinant with parameters often contains factors that describe when rows or columns become dependent.

3.6.5 Looking before expanding

The following checklist is often more valuable than a formula.

Summary

Before computing a determinant, ask:

- Is the matrix triangular or almost triangular?
- Is there a row or column with many zeros?
- Are two rows or columns equal or proportional?
- Can I create zeros using row operations that preserve the determinant?
- Is there a common factor in a row or column?
- If parameters are present, for which values could rows or columns become dependent?

- Which method gives the shortest honest calculation?

The last question is important. We do not want cleverness that hides the reasoning. We want calculations that are both efficient and understandable.

What to remember

The determinant is a number attached to a square matrix, but it is not merely a computational ornament. It detects structural information. It becomes zero when independence is lost. It changes predictably under row and column operations. It can be computed by formulas for small matrices, by Laplace expansion in general, and often more efficiently by using determinant properties.

Most importantly, determinants prepare the next question. If $\det A \neq 0$, then the matrix A is invertible. We can now ask how such an inverse matrix is found and how it can be used. That question leads directly to the next chapter.

Chapter 4

Matrix Inversion

4.1 The meaning of an inverse matrix

In the previous chapter we learned that the determinant can distinguish between singular and non-singular square matrices. The condition

$$\det A \neq 0$$

means that the matrix has not collapsed its structure. This raises a natural question: if a matrix has not collapsed information, can we reverse its action?

This question leads to the idea of an inverse matrix.

For ordinary numbers, the inverse of a non-zero number a is the number a^{-1} such that

$$aa^{-1} = 1.$$

For example, the inverse of 5 is $\frac{1}{5}$, because

$$5 \cdot \frac{1}{5} = 1.$$

The number 1 is special because multiplication by 1 changes nothing. In matrix algebra, the identity matrix plays the same role.

Definition

Let A be a square matrix of order n. A matrix B is called an inverse of A if

$$AB = I_n \quad \text{and} \quad BA = I_n.$$

If such a matrix exists, it is denoted by A^{-1}, and we write

$$AA^{-1} = I_n, \qquad A^{-1}A = I_n.$$

The two equations are both important. Matrix multiplication is not generally commutative, so the statements $AB = I$ and $BA = I$ are not automatically the same piece of information. The inverse must undo the matrix on both sides.

Remark

The notation A^{-1} does not mean $\frac{1}{A}$. Matrices are not numbers, and matrix division is not defined in the same way as numerical division. The symbol A^{-1} means: the matrix which reverses the action of A, if such a matrix exists.

4.1.1 Undoing an operation

The idea of an inverse is the idea of undoing. If a matrix A transforms a vector x into a vector y, then

$$Ax = y.$$

If A^{-1} exists, we can apply it to both sides:

$$A^{-1}Ax = A^{-1}y.$$

Since $A^{-1}A = I$, this becomes

$$x = A^{-1}y.$$

Thus the inverse matrix recovers the original vector from the transformed vector.

This explains why singular matrices cannot have inverses. If a matrix collapses different inputs into the same output, then there is no way to recover the original input uniquely. Information has been lost.

Example

Consider the diagonal matrix

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

Multiplication by D doubles the first coordinate and triples the second coordinate:

$$D \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 3y \end{pmatrix}.$$

To reverse this action, we should divide the first coordinate by 2 and the second coordinate by 3. Therefore

$$D^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}.$$

Indeed,

$$DD^{-1} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2.$$

This example is simple because the matrix is diagonal. Each coordinate is changed independently. More general matrices mix coordinates, so their inverses are harder to find. But the meaning stays the same: an inverse matrix reverses the action of the original matrix.

4.1.2 When does an inverse exist?

The determinant gives the fundamental test.

Theorem

A square matrix A is invertible if and only if

$$\det A \neq 0.$$

If $\det A = 0$, then A is singular and has no inverse.

This theorem connects the present chapter with the previous one. Determinants do not merely produce numbers. They answer the structural question of whether a matrix can be reversed.

Summary

The inverse matrix should be understood through three questions:

- What operation does the original matrix perform?

- Has this operation lost information?
- If no information was lost, what matrix reverses the operation?

The rest of this chapter develops methods for finding inverse matrices and for using them correctly.

4.2 Inverses of 2×2 matrices

The first non-trivial case is a 2×2 matrix. It is small enough that the inverse can be written explicitly, but rich enough to show the main ideas: the determinant appears, the order of entries matters, and the inverse exists only under a clear condition.

Let
$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$
We want to find a matrix
$$A^{-1} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$$
such that
$$AA^{-1} = I_2.$$
Multiplying, we get
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{pmatrix}.$$
For this to be the identity matrix, we need
$$\begin{cases} ap + br = 1, \\ cp + dr = 0, \end{cases} \qquad \begin{cases} aq + bs = 0, \\ cq + ds = 1. \end{cases}$$
Thus finding the inverse is already a problem about solving systems of linear equations. Instead of solving these systems every time, we use the known formula.

Theorem

Let
$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$
If
$$\det A = ad - bc \neq 0,$$
then
$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$
If $ad - bc = 0$, then A has no inverse.

The formula has two visible parts. First, the determinant appears in the denominator. This is not accidental: if the determinant is zero, the inverse cannot exist. Second, the matrix
$$\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$
is obtained by switching the diagonal entries a, d and changing signs of the off-diagonal entries b, c.

Example

Find the inverse of

$$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}.$$

First compute the determinant:

$$\det A = 2 \cdot 3 - 1 \cdot 5 = 6 - 5 = 1.$$

Since the determinant is non-zero, the inverse exists. Therefore

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}.$$

We should verify the result:

$$AA^{-1} = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 6-5 & -2+2 \\ 15-15 & -5+6 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Verification is not a decorative step. It helps us remember the meaning of the inverse. We are not only producing a matrix; we are producing a matrix that must multiply with the original one to give the identity.

Example

Consider

$$B = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}.$$

The determinant is

$$\det B = 1 \cdot 6 - 2 \cdot 3 = 6 - 6 = 0.$$

The formula cannot be used because it would require division by zero. More importantly, the inverse does not exist. The second row is three times the first row, so the matrix has lost independent information.

This example shows why the determinant test is not just a technical condition. A zero determinant indicates structural collapse. There is no inverse because there is no unique way to undo the action of the matrix.

4.2.1 A formula should still be read

It is tempting to memorize the 2×2 inverse formula mechanically. But it should be read as a statement with meaning:

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The determinant controls whether the inverse exists and how strongly the inverse scales the result. The second matrix rearranges the entries in the only way that makes multiplication return the identity.

Summary

For a 2×2 inverse:

- compute $\det A = ad - bc$,
- if $\det A = 0$, stop: the matrix is not invertible,
- if $\det A \neq 0$, use the inverse formula,
- verify the result by multiplying with the original matrix.

4.3 The Gauss–Jordan method

The formula for a 2×2 inverse is useful, but it does not give a practical method for larger matrices. For a 3×3 matrix, and especially for larger matrices, we need a systematic procedure. The most important procedure is the Gauss–Jordan method.

The idea is simple and powerful. If we want to find A^{-1}, we want to transform A into the identity matrix. Every row operation we apply to A must also be applied to the identity matrix beside it. When the left side becomes I, the right side becomes A^{-1}.

We write this as an augmented matrix:

$$[A \mid I].$$

The goal is

$$[A \mid I] \longrightarrow [I \mid A^{-1}].$$

Remark

The method does not work by magic. Row operations correspond to multiplying by elementary matrices. If a sequence of row operations transforms A into I, then the same sequence transforms I into the matrix that reverses A, namely A^{-1}.

4.3.1 A complete 2×2 example

Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}.$$

We form the augmented matrix

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{array}\right].$$

Our goal is to turn the left block into the identity matrix.

First eliminate the entry below the first pivot:

$$R_2 \leftarrow R_2 - 3R_1.$$

Then

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{array}\right].$$

Now eliminate the entry above the second pivot:

$$R_1 \leftarrow R_1 - 2R_2.$$

This gives

$$\left[\begin{array}{cc|cc} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{array}\right].$$

The left side is now I_2, so

$$A^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$

We verify:

$$\begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix} \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 7-6 & -2+2 \\ 21-21 & -6+7 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

4.3.2 A complete 3×3 example

Now consider

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}.$$

We start with

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right].$$

Use the first row to eliminate the entry below the first pivot:

$$R_3 \leftarrow R_3 - R_1.$$

Then

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{array}\right].$$

Use the second row to eliminate the entry below the second pivot:

$$R_3 \leftarrow R_3 + R_2.$$

This gives

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 & 1 & 1 \end{array}\right].$$

Normalize the third row:

$$R_3 \leftarrow \frac{1}{2} R_3.$$

Thus

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right].$$

Now eliminate the entry above the third pivot in row 2:

$$R_2 \leftarrow R_2 - R_3.$$

We get

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right].$$

Finally eliminate the entry above the second pivot in row 1:

$$R_1 \leftarrow R_1 - R_2.$$

Therefore

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right].$$

So

$$A^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

This example is longer than the 2×2 case, but the logic is the same. We create pivots, eliminate entries below and above them, and keep track of every operation on the identity matrix.

4.3.3 What if the method fails?

The Gauss–Jordan method may fail to produce the identity matrix on the left. This happens precisely when the matrix is singular.

Example

Let

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

Start with

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array}\right].$$

Apply

$$R_2 \leftarrow R_2 - 2R_1.$$

We obtain

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right].$$

The left side cannot become the identity matrix, because the second row of the left block is zero. Thus B has no inverse.

This is not a failure of the method. It is information. The method reveals that the matrix is singular.

Summary

The Gauss–Jordan method for finding A^{-1}:

1. Form the augmented matrix $[A \mid I]$.
2. Use elementary row operations to reduce the left block.
3. If the left block becomes I, the right block is A^{-1}.
4. If the left block cannot become I, then A is not invertible.

4.4 The adjugate method

The Gauss–Jordan method is often the most practical method for finding inverse matrices by hand. There is also another method, built from determinants and cofactors. This method is called the adjugate method.

The adjugate method is important for two reasons. First, it gives an explicit formula for the inverse of a matrix. Second, it shows a deep connection between determinants, cofactors, and invertibility.

4.4.1 The cofactor matrix and the adjugate matrix

Let $A = (a_{ij})$ be an $n \times n$ matrix. For each entry a_{ij}, we can compute its cofactor

$$C_{ij} = (-1)^{i+j} M_{ij},$$

where M_{ij} is the minor obtained by deleting row i and column j.

The matrix of cofactors is

$$C = \begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{pmatrix}.$$

The adjugate matrix is the transpose of the cofactor matrix.

Definition

The adjugate of a square matrix A, denoted by $\operatorname{adj}(A)$, is defined as

$$\operatorname{adj}(A) = C^{\top},$$

where C is the cofactor matrix of A.

The transpose is essential. The adjugate is not simply the cofactor matrix; it is the cofactor matrix with rows and columns exchanged.

4.4.2 The inverse formula

The central formula is the following.

Theorem

If A is a square matrix and $\det A \neq 0$, then

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A).$$

This formula should be read slowly. It says that the inverse exists only when $\det A \neq 0$. It also says that the inverse can be built from cofactors. Thus the same objects used to compute determinants also describe the inverse matrix.

A compact way to express the reason behind the formula is

$$A \operatorname{adj}(A) = (\det A) I.$$

If $\det A \neq 0$, we divide by $\det A$ and obtain

$$A \left(\frac{1}{\det A} \operatorname{adj}(A) \right) = I.$$

So

$$\frac{1}{\det A}\operatorname{adj}(A)$$

is the inverse matrix.

4.4.3 A 3×3 example

Let

$$A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 0 & 2 & 1 \end{pmatrix}.$$

First compute the determinant. Expanding along the first row,

$$\det A = 1 \det \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} + 2 \det \begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix}.$$

The first minor gives

$$3 \cdot 1 - 1 \cdot 2 = 1,$$

and the second minor gives

$$(-1) \cdot 2 - 3 \cdot 0 = -2.$$

Therefore

$$\det A = 1 + 2(-2) = -3.$$

Since $\det A \neq 0$, the inverse exists.

Now compute the cofactors. We list them carefully:

$$C_{11} = \det \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} = 1,$$

$$C_{12} = -\det \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix} = 1,$$

$$C_{13} = \det \begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix} = -2,$$

$$C_{21} = -\det \begin{pmatrix} 0 & 2 \\ 2 & 1 \end{pmatrix} = 4,$$

$$C_{22} = \det \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = 1,$$

$$C_{23} = -\det \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = -2,$$

$$C_{31} = \det \begin{pmatrix} 0 & 2 \\ 3 & 1 \end{pmatrix} = -6,$$

$$C_{32} = -\det \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} = -3,$$

$$C_{33} = \det \begin{pmatrix} 1 & 0 \\ -1 & 3 \end{pmatrix} = 3.$$

Thus the cofactor matrix is

$$C = \begin{pmatrix} 1 & 1 & -2 \\ 4 & 1 & -2 \\ -6 & -3 & 3 \end{pmatrix}.$$

Therefore

$$\operatorname{adj}(A) = C^\top = \begin{pmatrix} 1 & 4 & -6 \\ 1 & 1 & -3 \\ -2 & -2 & 3 \end{pmatrix}.$$

Finally,

$$A^{-1} = \frac{1}{-3}\begin{pmatrix} 1 & 4 & -6 \\ 1 & 1 & -3 \\ -2 & -2 & 3 \end{pmatrix}.$$

So

$$A^{-1} = \begin{pmatrix} -\frac{1}{3} & -\frac{4}{3} & 2 \\ -\frac{1}{3} & -\frac{1}{3} & 1 \\ \frac{2}{3} & \frac{2}{3} & -1 \end{pmatrix}.$$

This method is longer than Gauss–Jordan for this example, but it shows exactly how determinants and cofactors build the inverse.

4.4.4 When should this method be used?

For hand calculations, the adjugate method is convenient for 2×2 matrices and sometimes manageable for 3×3 matrices. For larger matrices it becomes computationally heavy because many cofactors must be computed.

Its conceptual value is very high: it explains why the determinant appears in inverse formulas and why zero determinant prevents invertibility.

Summary

The adjugate method:

1. Compute $\det A$.
2. If $\det A = 0$, the inverse does not exist.
3. Compute all cofactors C_{ij}.
4. Form the cofactor matrix C.
5. Transpose it to obtain $\operatorname{adj}(A)$.
6. Use $A^{-1} = \frac{1}{\det A}\operatorname{adj}(A)$.

4.5 Invertibility and structure

It is possible to treat invertibility as a purely computational question: compute the determinant, or try Gauss–Jordan elimination, and see what happens. This works, but it is not the only way to think. Often the structure of a matrix already tells us something about whether an inverse should exist.

The central rule remains the same:

$$A \text{ is invertible} \iff \det A \neq 0.$$

But before computing the determinant, we should learn to look at the matrix.

4.5.1 Diagonal matrices

A diagonal matrix is one of the clearest cases. Let

$$D = \operatorname{diag}(d_1, d_2, \ldots, d_n).$$

Then

$$\det D = d_1 d_2 \cdots d_n.$$

Therefore D is invertible exactly when none of the diagonal entries is zero.

If every $d_i \neq 0$, then

$$D^{-1} = \operatorname{diag}\left(\frac{1}{d_1}, \frac{1}{d_2}, \ldots, \frac{1}{d_n}\right).$$

Example

Let

$$D = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Then

$$D^{-1} = \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{5} \end{pmatrix}.$$

This inverse is easy to understand: each coordinate was scaled independently, so the inverse simply rescales each coordinate back.

If a diagonal entry is zero, one coordinate is collapsed to zero. That information cannot be recovered.

4.5.2 Triangular matrices

Triangular matrices are also easy to test. If T is triangular, then

$$\det T = \text{product of diagonal entries}.$$

Thus T is invertible if and only if every diagonal entry is non-zero.

Example

Let

$$T = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \\ 0 & 0 & 6 \end{pmatrix}.$$

Since

$$\det T = 1 \cdot (-1) \cdot 6 = -6 \neq 0,$$

the matrix is invertible.

On the other hand,

$$S = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 0 & 5 \\ 0 & 0 & 3 \end{pmatrix}$$

has determinant

$$\det S = 2 \cdot 0 \cdot 3 = 0.$$

So S is not invertible.

This is not merely a shortcut. A zero on the diagonal of a triangular matrix means that one pivot is missing. Without all pivots, the matrix cannot be reduced to the identity matrix.

4.5.3 Repeated or dependent rows

If two rows of a matrix are equal, the determinant is zero. If one row is a multiple of another, the determinant is also zero. More generally, if one row can be built from the others, the determinant is zero.

Example

Consider

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 6 & 3 \\ 0 & 1 & 4 \end{pmatrix}.$$

The second row is three times the first row. Therefore the rows are dependent, and

$$\det A = 0.$$

So A is not invertible.

The interpretation is important. If one row repeats information already contained in another row, then the matrix does not contain enough independent information to be reversed.

4.5.4 Products of matrices

The determinant rule

$$\det(AB) = \det A \det B$$

has a direct consequence for invertibility.

Theorem

If A and B are invertible square matrices of the same size, then AB is invertible, and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

The order in this formula is essential. To undo the product AB, we must undo the last action first.

Suppose a vector x is transformed by B and then by A:

$$x \longmapsto Bx \longmapsto A(Bx) = ABx.$$

To recover x, first undo A, and then undo B:

$$ABx \longmapsto Bx \longmapsto x.$$

Thus the inverse is

$$B^{-1}A^{-1}.$$

We can verify algebraically:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I.$$

Similarly,

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I.$$

4.5.5 Matrices satisfying special identities

Sometimes a matrix is invertible because it satisfies an identity that already reveals its inverse.

Example

Suppose a square matrix A satisfies

$$A^2 = I.$$

Then A is its own inverse, because

$$AA = I.$$

Therefore

$$A^{-1} = A.$$

Such a matrix is called an involution.

Example

Suppose a square matrix A satisfies

$$A^2 - 3A + 2I = 0.$$

Then

$$A^2 - 3A = -2I.$$

Factor the left side:

$$A(A - 3I) = -2I.$$

Multiplying by $-\frac{1}{2}$, we get

$$A\left(-\frac{1}{2}(A - 3I)\right) = I.$$

Therefore

$$A^{-1} = -\frac{1}{2}(A - 3I) = \frac{1}{2}(3I - A).$$

This argument finds the inverse without performing Gauss–Jordan elimination.

These examples show that invertibility is not only a computational topic. It is also a structural topic. Sometimes the matrix tells us how it can be undone.

Summary

To recognize invertibility, look for structure:

- diagonal matrices are invertible when no diagonal entry is zero,
- triangular matrices are invertible when all diagonal entries are non-zero,
- repeated or dependent rows imply non-invertibility,
- products of invertible matrices are invertible,
- special identities may reveal the inverse directly.

4.6 Matrix equations

One of the main reasons for studying inverse matrices is that they allow us to solve matrix equations. The idea is similar to solving ordinary equations, but there is one crucial difference: matrix multiplication is not commutative. Therefore the side on which we multiply by an inverse matters.

4.6.1 Equations of the form $AX = B$

Suppose

$$AX = B,$$

where A is a known square matrix, B is a known matrix, and X is unknown. If A is invertible, we multiply on the left by A^{-1}:

$$A^{-1}AX = A^{-1}B.$$

Since $A^{-1}A = I$, this gives

$$X = A^{-1}B.$$

The inverse must be placed on the left because A is on the left of X. Writing XA^{-1} would not undo AX.

Example

Solve

$$AX = B,$$

where

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}, \qquad B = \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix}.$$

From an earlier example,

$$A^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$

Therefore

$$X = A^{-1}B = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix}.$$

Compute the product:

$$X = \begin{pmatrix} 28 - 10 & 7 - 4 \\ -12 + 5 & -3 + 2 \end{pmatrix} = \begin{pmatrix} 18 & 3 \\ -7 & -1 \end{pmatrix}.$$

Thus

$$X = \begin{pmatrix} 18 & 3 \\ -7 & -1 \end{pmatrix}.$$

4.6.2 Equations of the form $XA = B$

Now suppose

$$XA = B.$$

Here A is on the right of X. To undo multiplication by A, we must multiply on the right by A^{-1}:

$$XAA^{-1} = BA^{-1}.$$

Since $AA^{-1} = I$, we get

$$X = BA^{-1}.$$

Example

Solve
$$XA = B,$$
where
$$A = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}, \qquad B = \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix}.$$
Again,
$$A^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$
But now
$$X = BA^{-1} = \begin{pmatrix} 4 & 1 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}.$$
Thus
$$X = \begin{pmatrix} 28 - 3 & -8 + 1 \\ 35 - 6 & -10 + 2 \end{pmatrix} = \begin{pmatrix} 25 & -7 \\ 29 & -8 \end{pmatrix}.$$
This is not the same answer as in the previous example. The equation is different because the unknown matrix is multiplied by A on a different side.

4.6.3 Equations with matrices on both sides

Some equations have the unknown matrix between two known matrices:
$$AXB = C.$$
If A and B are invertible, then we undo A on the left and B on the right:
$$A^{-1}(AXB)B^{-1} = A^{-1}CB^{-1}.$$
Therefore
$$X = A^{-1}CB^{-1}.$$

The order is not optional. Each inverse must be placed on the side where it can cancel the corresponding matrix.

4.6.4 Solving vector equations

The equation
$$Ax = b$$
where x and b are column vectors is a special case of $AX = B$. If A is invertible, then
$$x = A^{-1}b.$$
This gives a conceptual method for solving linear systems. In the next chapter we will study systems of linear equations in detail, and we will compare different methods: elimination, inverse matrices, and Cramer's rule.

Example

Solve
$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

Let
$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$
The determinant is
$$\det A = 2 \cdot 1 - 1 \cdot 1 = 1,$$
so
$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$
Therefore
$$\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1}b = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 5-3 \\ -5+6 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$
Thus
$$x = 2, \qquad y = 1.$$

4.6.5 Common mistakes

The most common mistake is to move matrices as if they were numbers. For example, from
$$AX = B$$
one may be tempted to write
$$X = BA^{-1}.$$
This is wrong in general. Multiplying by A^{-1} on the right does not cancel the A on the left of X. The correct step is
$$A^{-1}AX = A^{-1}B,$$
so
$$X = A^{-1}B.$$

Summary

When solving matrix equations:

- identify where the unknown matrix is placed,
- multiply by inverses on the side where cancellation is valid,
- preserve the order of multiplication,
- check that the matrices involved are invertible,
- verify the final answer by substitution when possible.

Inverse matrices are powerful because they allow equations to be solved by undoing operations. But the power comes with a responsibility: the order of multiplication must be respected at every step.

Chapter 5

Systems of Linear Equations

5.1 What systems of linear equations describe

A system of linear equations is not only a collection of equations to solve. It is a collection of conditions that must be satisfied at the same time. Each equation gives one condition. A solution is an assignment of values to the unknowns that satisfies all conditions simultaneously.

This point is important. When we solve a system, we are not solving the equations one after another as separate problems. We are looking for values that make the whole system true.

Example

Consider the system

$$\begin{cases} x + y = 5, \\ 2x - y = 1. \end{cases}$$

The pair $(2, 3)$ satisfies the first equation because

$$2 + 3 = 5.$$

It also satisfies the second equation because

$$2 \cdot 2 - 3 = 1.$$

Therefore $(2, 3)$ is a solution of the system.
The pair $(1, 4)$ satisfies the first equation, since $1 + 4 = 5$, but it does not satisfy the second equation, since $2 \cdot 1 - 4 = -2$. Therefore $(1, 4)$ is not a solution of the system.

The example should be read slowly. A solution of a system must pass every test in the system. Passing one equation is not enough.

Unknowns, coefficients, and constants

A linear equation in the unknowns $x_1, x_2, \ldots, x_n$ has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b.$$

The numbers $a_1, a_2, \ldots, a_n$ are coefficients. The number b is the constant term or right-hand side.

A system of linear equations is made of several such equations:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2, \\ \qquad\qquad\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m. \end{cases}$$

Here m is the number of equations and n is the number of unknowns.

Remark

The coefficients describe how the unknowns enter the equations. The right-hand side describes what values the linear combinations must equal. A system compares these two pieces of information.

Solution sets

The solution set of a system is the set of all solutions. Sometimes this set contains one element. Sometimes it is empty. Sometimes it contains infinitely many elements.

These are the three basic possibilities for a linear system:

- exactly one solution,
- no solution,
- infinitely many solutions.

The goal of solving a system is not only to find numbers. The goal is to determine which of these cases occurs and to describe the solution set accurately.

Matrix form

Systems of linear equations are naturally connected with matrices. The coefficients form a matrix, the unknowns form a column vector, and the right-hand side forms another column vector.

For example, the system

$$\begin{cases} 2x + y = 5, \\ x - y = 1 \end{cases}$$

can be written as

$$\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}.$$

Using the notation

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \qquad x = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad b = \begin{pmatrix} 5 \\ 1 \end{pmatrix},$$

we write the system compactly as

$$Ax = b.$$

This notation is not merely shorter. It reveals the structure of the problem. The matrix A transforms the unknown vector x into the right-hand side vector b. Solving the system means finding which input vector x, if any, is sent to b.

The augmented matrix

When solving a system, it is often convenient to write the coefficients and the right-hand side together in an augmented matrix:

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & -1 & 1 \end{array}\right].$$

The vertical line reminds us that the last column is not another column of coefficients. It is the right-hand side.

The augmented matrix is a compact record of the system. Row operations on this matrix correspond to controlled transformations of the equations.

Summary

When reading a system of linear equations, ask:

- What are the unknowns?
- What are the equations, and what condition does each one impose?
- What are the coefficients and right-hand sides?
- What would it mean for one assignment to satisfy all equations at once?
- Can the system be written as $Ax = b$?
- What does the augmented matrix record?

A system of equations should first be read as a structured mathematical object. Only then should we choose a method for solving it.

5.2 Geometric meaning of solutions

Linear systems have an algebraic form, but they also have a geometric meaning. This geometric interpretation is often the easiest way to understand why a system may have one solution, no solution, or infinitely many solutions.

The central idea is simple: each equation describes a set of points. The solution set of the system is the intersection of all these sets.

Two variables: lines in the plane

In two variables, a linear equation

$$ax + by = c$$

usually describes a line in the plane. A system of two linear equations describes two lines, and a solution is a point that lies on both lines.

Example

Consider

$$\begin{cases} x + y = 5, \\ 2x - y = 1. \end{cases}$$

The first equation describes one line. The second equation describes another line. The pair $(2, 3)$ is a solution because the point $(2, 3)$ lies on both lines.

This is the geometric meaning of solving the system: find the common point of the two lines.

One solution

If two lines in the plane intersect in one point, the system has exactly one solution.

Example

The system

$$\begin{cases} x + y = 5, \\ 2x - y = 1 \end{cases}$$

has exactly one solution. Solving gives

$$x = 2, \qquad y = 3.$$

Geometrically, the two lines meet at the point $(2, 3)$.

When a system has exactly one solution, the conditions are independent enough to determine the unknowns completely.

No solution

If two lines are parallel and distinct, they never meet. Then the system has no solution.

Example

Consider
$$\begin{cases} x + y = 1, \\ x + y = 4. \end{cases}$$
The left-hand sides are the same, but the right-hand sides are different. The same expression $x + y$ cannot equal both 1 and 4 at the same time.
Geometrically, these are two parallel distinct lines. Therefore the system has no solution.

A system with no solution is called inconsistent. The equations impose conditions that cannot all be satisfied simultaneously.

Infinitely many solutions

If two equations describe the same line, then every point on that line is a solution. The system has infinitely many solutions.

Example

Consider
$$\begin{cases} x + y = 3, \\ 2x + 2y = 6. \end{cases}$$
The second equation is just twice the first equation. It gives the same condition, not a new independent condition.
Therefore the solution set is the whole line

$$x + y = 3.$$

There are infinitely many solutions, for example $(0, 3)$, $(1, 2)$, and $(3, 0)$.

This case is not a failure of mathematics. It means that the equations do not contain enough independent information to determine one point.

Three variables: planes in space

In three variables, a linear equation

$$ax + by + cz = d$$

usually describes a plane in space. A system of linear equations describes the common points of several planes.

Several things may happen:

- three planes may meet at one point,
- they may have no common point,
- they may share a line,
- they may all be the same plane.

The algebraic classification remains the same: one solution, no solution, or infinitely many solutions. The geometry becomes richer because planes can intersect in more ways than lines in the plane.

Reading geometry from algebra

The algebraic form of a system often reveals the geometric situation. For example, proportional left-hand sides may indicate parallel or identical lines. A contradiction such as

$$0 = 5$$

means that the conditions cannot all be satisfied. A row of zeros such as

$$0 = 0$$

means that one equation became redundant.

These patterns will appear naturally during Gaussian elimination.

Summary

When interpreting a system geometrically, ask:

- What geometric object does each equation describe?
- Is the solution set an intersection of lines, planes, or other linear objects?
- Do the objects meet in one point?
- Are they parallel or inconsistent?
- Is one condition a repetition of another condition?
- Does the geometry suggest one solution, no solution, or infinitely many solutions?

The geometric viewpoint does not replace algebraic methods. It explains what those methods are detecting.

5.3 Gaussian elimination

Gaussian elimination is a method for simplifying a system of linear equations without changing its solution set. The method uses elementary row operations on the augmented matrix. These operations are not random changes of notation. They correspond to replacing equations by equivalent equations.

The aim is to transform a complicated system into a readable one.

Equivalent systems

Two systems are equivalent if they have the same solution set. Gaussian elimination works because elementary row operations produce equivalent systems.

The allowed row operations are:

- swapping two rows,
- multiplying a row by a non-zero number,
- adding a multiple of one row to another row.

These are the same operations introduced earlier for matrices. Here their meaning is sharper: they change the written form of the equations, but not the set of solutions.

Remark

The purpose of row operations is not to make the matrix look different. The purpose is to reveal the solution set by simplifying the conditions.

A first example

Consider the system

$$\begin{cases} x + y = 5, \\ 2x - y = 1. \end{cases}$$

Its augmented matrix is

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 2 & -1 & 1 \end{array}\right].$$

We use the first row to eliminate the x-coefficient in the second row:

$$R_2 \leftarrow R_2 - 2R_1.$$

This gives

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -3 & -9 \end{array}\right].$$

The second row now means

$$-3y = -9,$$

so

$$y = 3.$$

Substitute into the first equation:

$$x + 3 = 5,$$

so

$$x = 2.$$

Therefore the solution is

$$(x, y) = (2, 3).$$

This example shows the basic strategy. We simplify the system until the unknowns can be read one by one.

Pivots

A pivot is an entry used to eliminate other entries in its column. In the previous example, the first pivot was the 1 in the upper-left corner. It allowed us to remove the 2 below it.

Pivots are not only computational markers. They identify independent information in the system. Each pivot usually corresponds to a variable that can be solved from the system.

Example

Consider

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 2 & 6 \end{array}\right].$$

The pivot positions are in the first, second, and third variable columns. The last row gives

$$2z = 6,$$

so $z = 3$. The second row then gives

$$y + 4z = 5,$$

so $y + 12 = 5$, hence $y = -7$. The first row gives

$$x + 2y - z = 3,$$

so $x + 2(-7) - 3 = 3$, and therefore $x = 20$.

This process is called back substitution. We solve from the bottom upward after the system has been simplified.

Row echelon form

A matrix is in row echelon form when the leading non-zero entries move to the right as we move downward, and zero rows, if any, are placed at the bottom. Row echelon form is useful because it makes the structure of the system visible.

For example,

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 4 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

is in row echelon form. The last row represents the equation

$$0 = 0,$$

which gives no new restriction.

Detecting no solution

During elimination, a row may become

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 5 \end{array}\right].$$

This row represents

$$0 = 5.$$

That is impossible. Therefore the system has no solution.

Example

Consider

$$\begin{cases} x + y = 1, \\ 2x + 2y = 5. \end{cases}$$

The augmented matrix is

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 2 & 5 \end{array}\right].$$

Apply

$$R_2 \leftarrow R_2 - 2R_1.$$

We get

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 0 & 3 \end{array}\right].$$

The second row says

$$0 = 3,$$

which is impossible. The system has no solution.

The contradiction is not a technical detail. It is the algebraic sign that the conditions are inconsistent.

Free variables and infinitely many solutions

If a variable column has no pivot, that variable is free. A free variable can be chosen as a parameter. The remaining variables are expressed in terms of it.

Example

Consider the row echelon form

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & 4 \end{array}\right].$$

This represents

$$\begin{cases} x + 2y - z = 3, \\ z = 4. \end{cases}$$

The variable y has no pivot, so let

$$y = t.$$

Then $z = 4$, and the first equation gives

$$x + 2t - 4 = 3,$$

so

$$x = 7 - 2t.$$

Thus the solution set is

$$(x, y, z) = (7 - 2t, t, 4), \qquad t \in \mathbb{R}.$$

There are infinitely many solutions.

A parameter is not a decoration. It records the freedom that remains after all independent conditions have been used.

What elimination tells us

Gaussian elimination does more than compute answers. It classifies the system.

Summary

After row reduction, look for these patterns:

- A pivot in every variable column and no contradiction: exactly one solution.
- A row of the form $0 =$ non-zero: no solution.
- At least one free variable and no contradiction: infinitely many solutions.

Gaussian elimination is the most flexible method in this chapter. It works for square and non-square systems, and it reveals not only the solution but also the structure of the solution set.

5.4 Cramer's rule

Cramer's rule is a method for solving certain square systems of linear equations using determinants. It is not the most efficient method for large systems, but it is conceptually important because it shows a direct connection between determinants and solvability.

The method applies to systems where the number of equations equals the number of unknowns and the coefficient matrix has non-zero determinant.

The two by two case

Consider the system

$$\begin{cases} ax + by = e, \\ cx + dy = f. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

and its determinant is

$$\det A = ad - bc.$$

If $\det A \neq 0$, the system has exactly one solution.

Cramer's rule says that the solution can be computed by replacing one column of the coefficient matrix at a time by the right-hand side column.

Define

$$A_x = \begin{pmatrix} e & b \\ f & d \end{pmatrix}, \qquad A_y = \begin{pmatrix} a & e \\ c & f \end{pmatrix}.$$

Then

$$x = \frac{\det A_x}{\det A}, \qquad y = \frac{\det A_y}{\det A}.$$

Remark

The denominator $\det A$ is not a technical detail. If $\det A = 0$, the coefficient matrix is singular, and Cramer's rule does not apply.

A worked example

Example

Solve the system

$$\begin{cases} 2x + y = 5, \\ x - y = 1. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}.$$

Its determinant is

$$\det A = 2 \cdot (-1) - 1 \cdot 1 = -2 - 1 = -3.$$

Since $\det A \neq 0$, Cramer's rule applies.
For x, replace the first column by the right-hand side:

$$A_x = \begin{pmatrix} 5 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then

$$\det A_x = 5 \cdot (-1) - 1 \cdot 1 = -5 - 1 = -6.$$

So

$$x = \frac{\det A_x}{\det A} = \frac{-6}{-3} = 2.$$

For y, replace the second column:

$$A_y = \begin{pmatrix} 2 & 5 \\ 1 & 1 \end{pmatrix}.$$

Then

$$\det A_y = 2 \cdot 1 - 5 \cdot 1 = 2 - 5 = -3.$$

So

$$y = \frac{\det A_y}{\det A} = \frac{-3}{-3} = 1.$$

Thus

$$(x, y) = (2, 1).$$

This result should be checked by substitution. We have

$$2 \cdot 2 + 1 = 5, \qquad 2 - 1 = 1.$$

Both equations are satisfied.

The general square case

For a square system

$$Ax = b,$$

where A is an $n \times n$ matrix and b is a column vector, Cramer's rule applies if

$$\det A \neq 0.$$

For each unknown x_i, form the matrix A_i by replacing the i-th column of A with b. Then

$$x_i = \frac{\det A_i}{\det A}.$$

The rule is elegant, but the number of determinant computations grows quickly. For this reason, Gaussian elimination is usually more practical for larger systems.

Why the determinant condition matters

The condition

$$\det A \neq 0$$

means that the coefficient matrix is invertible. Geometrically and structurally, the equations contain enough independent information to determine one solution.

If

$$\det A = 0,$$

then the matrix is singular. This does not automatically mean that the system has no solution. It means that the system cannot have a unique solution determined by Cramer's rule. The system must be analyzed further: it may have no solution or infinitely many solutions.

Example

Consider

$$\begin{cases} x + y = 2, \\ 2x + 2y = 4. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix},$$

and

$$\det A = 1 \cdot 2 - 1 \cdot 2 = 0.$$

Cramer's rule does not apply. But the system is not inconsistent. The second equation is twice the first, so there are infinitely many solutions.

This example is important. A zero determinant means loss of uniqueness, not automatically loss of existence.

Summary

When using Cramer's rule, ask:

- Is the system square?
- What is the coefficient matrix A?
- Is $\det A \neq 0$?
- Which column must be replaced to compute each unknown?
- Does the final answer satisfy the original equations?
- If $\det A = 0$, what further classification is needed?

Cramer's rule shows how determinants can solve systems, but it must be used only when its structural condition is satisfied.

5.5 The inverse matrix method

Some systems of linear equations can be solved by using the inverse of the coefficient matrix. This method is based on the matrix equation

$$Ax = b.$$

Here A is the coefficient matrix, x is the column vector of unknowns, and b is the right-hand side vector.

The method is simple in form, but its meaning should be understood carefully. If A is invertible, then multiplying by A^{-1} undoes the action of A.

Solving $Ax = b$

Suppose

$$Ax = b$$

and A is invertible. Multiply both sides on the left by A^{-1}:

$$A^{-1}Ax = A^{-1}b.$$

Since

$$A^{-1}A = I,$$

we get

$$Ix = A^{-1}b.$$

Therefore

$$x = A^{-1}b.$$

Theorem

If A is an invertible square matrix, then the system

$$Ax = b$$

has the unique solution

$$x = A^{-1}b.$$

The word "invertible" is essential. If A^{-1} does not exist, this method cannot be used.

A worked example

Example

Solve

$$\begin{cases} 2x + y = 5, \\ x + y = 3. \end{cases}$$

Write the system as

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

Thus

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

The determinant is

$$\det A = 2 \cdot 1 - 1 \cdot 1 = 1,$$

so A is invertible. For a 2×2 matrix,

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} x \\ y \end{pmatrix} = A^{-1}b = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

Compute the product:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5-3 \\ -5+6 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Thus

$$x = 2, \qquad y = 1.$$

The inverse matrix method gives the same result as other correct methods. The difference is the language: here solving means applying the inverse transformation.

Why the method works

The equation

$$Ax = b$$

can be read as follows: the matrix A acts on the unknown vector x and produces the vector b. If A is invertible, then this action can be reversed. Applying A^{-1} to b recovers the original vector x.

This interpretation is important. The formula

$$x = A^{-1}b$$

is not magic. It is the matrix version of undoing an operation.

Connection with determinants

For a square matrix A, invertibility is equivalent to

$$\det A \neq 0.$$

Therefore the inverse matrix method applies exactly when the coefficient matrix has non-zero determinant.

If

$$\det A = 0,$$

then A^{-1} does not exist. The system may have no solution or infinitely many solutions, but it cannot be solved by multiplying by A^{-1}.

Example

Consider

$$\begin{cases} x + y = 2, \\ 2x + 2y = 4. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix},$$

and

$$\det A = 0.$$

The inverse matrix method cannot be used. However, the system does have solutions: every pair (x, y) satisfying $x + y = 2$ is a solution. There are infinitely many such pairs.

Again, the failure of invertibility means loss of uniqueness, not automatically inconsistency.

When this method is useful

The inverse matrix method is conceptually clear, but it is not always the most practical computational method. Finding A^{-1} can be more work than solving the system directly by elimination.

The method is especially useful when:

- the inverse matrix is already known,
- several systems have the same coefficient matrix A but different right-hand sides,
- we want to understand the system as a matrix equation.

For routine computation, Gaussian elimination is usually more flexible.

Summary

When using the inverse matrix method, ask:

- Can the system be written as $Ax = b$?
- Is A square?
- Is A invertible, equivalently $\det A \neq 0$?
- Have I multiplied on the correct side by A^{-1}?
- Does the vector $A^{-1}b$ satisfy the original system?

The inverse matrix method is powerful because it connects solving systems with the idea of reversing a linear transformation.

5.6 Parametric systems and classification

Some systems contain a parameter. Then the question is not only "What is the solution?" but also "For which parameter values does the structure of the system change?"

This is where determinants, row reduction, and geometric interpretation come together.

The three possible outcomes

Every linear system has one of the following outcomes:

- exactly one solution,
- no solution,
- infinitely many solutions.

For a square system

$$Ax = b,$$

one important test is the determinant of A. If

$$\det A \neq 0,$$

then A is invertible and the system has exactly one solution. If

$$\det A = 0,$$

then the system does not have a unique solution obtained by inversion. But we must still decide whether it has no solution or infinitely many solutions.

Remark

The condition $\det A = 0$ is a warning sign, not a complete answer. It says that uniqueness is lost. It does not by itself decide whether solutions exist.

A simple parameter example

Consider the system

$$\begin{cases} x + y = 1, \\ 2x + 2y = a. \end{cases}$$

The left-hand side of the second equation is twice the left-hand side of the first equation. Therefore consistency depends on the right-hand side.

If we multiply the first equation by 2, we get

$$2x + 2y = 2.$$

So for the second equation to describe the same condition, we need

$$a = 2.$$

If $a = 2$, the two equations describe the same line, and there are infinitely many solutions:

$$x + y = 1.$$

If $a \neq 2$, the equations describe parallel distinct lines, and there is no solution.

Summary

For

$$\begin{cases} x + y = 1, \\ 2x + 2y = a, \end{cases}$$

we have:

- if $a = 2$, infinitely many solutions,
- if $a \neq 2$, no solution.

This example shows that a parameter may affect only the right-hand side, but that can still change the solution set completely.

Using determinants to find exceptional values

Now consider a parameter in the coefficient matrix. Let

$$A(t) = \begin{pmatrix} t & 1 \\ 4 & t \end{pmatrix}.$$

The determinant is

$$\det A(t) = t^2 - 4.$$

The exceptional values occur when

$$t^2 - 4 = 0,$$

so

$$t = 2 \qquad \text{or} \qquad t = -2.$$

For all other values of t, the matrix is invertible. Therefore any system

$$A(t)x = b$$

has exactly one solution for $t \neq 2$ and $t \neq -2$.

The exceptional values $t = 2$ and $t = -2$ require separate analysis. At those values, the determinant test tells us that the system is not uniquely solvable, but not whether it is inconsistent or has infinitely many solutions.

A full classification example

Example

Classify the system depending on the parameter a:

$$\begin{cases} x + y = 3, \\ 2x + ay = 6. \end{cases}$$

The coefficient matrix is

$$A(a) = \begin{pmatrix} 1 & 1 \\ 2 & a \end{pmatrix}.$$

Its determinant is

$$\det A(a) = a - 2.$$

If

$$a \neq 2,$$

then $\det A(a) \neq 0$, so the system has exactly one solution.
Now consider the exceptional value $a = 2$. The system becomes

$$\begin{cases} x + y = 3, \\ 2x + 2y = 6. \end{cases}$$

The second equation is twice the first. Therefore the system has infinitely many solutions.
Thus:

- if $a \neq 2$, the system has exactly one solution,
- if $a = 2$, the system has infinitely many solutions.

The determinant found the value where the structure changes. Row interpretation then classified what happens at that value.

Example

Classify the system depending on a:

$$\begin{cases} x + y = 3, \\ 2x + 2y = a. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix},$$

so

$$\det A = 0.$$

There is no value of a that makes this coefficient matrix invertible. We must compare the equations.
Twice the first equation gives

$$2x + 2y = 6.$$

Therefore:

- if $a = 6$, the two equations describe the same line, so there are infinitely many solutions,
- if $a \neq 6$, the two equations are inconsistent, so there is no solution.

This example shows that a singular coefficient matrix may still lead to different outcomes depending on the right-hand side.

Parameters and row reduction

For larger systems, row reduction is often the safest method. It allows us to see contradictions, zero rows, and free variables directly.

A row such as

$$0 = 0$$

indicates redundancy. A row such as

$$0 = 5$$

indicates inconsistency. A missing pivot indicates a free variable. These patterns remain valid even when parameters are present.

The difficulty is that some row operations may require dividing by an expression containing a parameter. Before dividing by such an expression, we must consider when it might be zero.

Remark

When a parameter appears, do not divide by an expression such as $a - 2$ without asking whether $a - 2 = 0$ is possible. Parameter values that make a denominator zero are usually exceptional cases and must be treated separately.

A classification strategy

A good solution to a parameter system should not only list parameter values. It should explain why the classification changes.

Summary

For a square parameter system $A(t)x = b(t)$, a useful strategy is:

1. Compute $\det A(t)$.
2. Solve $\det A(t) = 0$ to find exceptional parameter values.
3. For non-exceptional values, conclude that there is exactly one solution.
4. For exceptional values, analyze the system separately using row reduction or equation comparison.
5. Classify each case as exactly one solution, no solution, or infinitely many solutions.

The determinant identifies where uniqueness can fail. Elimination and interpretation decide what happens there.

What the chapter has built

Systems of linear equations bring together the earlier algebraic tools.

Matrices organize the coefficients. Row operations simplify the system. Determinants detect whether a square coefficient matrix is invertible. Inverse matrices give the formula $x = A^{-1}b$ when the inverse exists. Cramer's rule gives a determinant-based expression for the solution when $\det A \neq 0$.

These methods are not isolated tricks. They are different ways of reading the same object: a system of linear conditions.

Summary

At the end of this chapter, remember:

- A system is a set of conditions to be satisfied simultaneously.
- The augmented matrix records the coefficients and the right-hand side.
- Gaussian elimination is the most flexible method for solving and classifying systems.
- If $\det A \neq 0$, a square system $Ax = b$ has exactly one solution.
- If $\det A = 0$, uniqueness is lost, but existence must still be checked.
- A linear system may have exactly one solution, no solution, or infinitely many solutions.

The goal is not only to solve systems. The goal is to understand what the equations say about structure, dependence, and compatibility.

Chapter 6

Vectors and Coordinates

6.1 Points and coordinates

Analytic geometry begins with a simple but powerful idea: geometric objects can be described by numbers. A point on a line can be described by one number. A point in a plane can be described by two numbers. A point in space can be described by three numbers. This numerical description allows us to use algebra to answer geometric questions.

The important word here is description. A point is not the same thing as the symbols used to describe it. The symbols depend on a coordinate system. Once a coordinate system is chosen, the symbols become a precise way to say where the point is.

Definition

In the plane, a point is usually written as an ordered pair

$$P = (x, y).$$

The first coordinate x describes horizontal position, and the second coordinate y describes vertical position. In three-dimensional space, a point is usually written as an ordered triple

$$P = (x, y, z).$$

The order of the coordinates matters. The points $(2, 5)$ and $(5, 2)$ are generally different points. The numbers are the same, but their roles are not the same. The first number answers one question, and the second number answers another question.

Example

The point

$$P = (3, -2)$$

means the following: starting from the origin, move 3 units in the positive horizontal direction and 2 units in the negative vertical direction.

The point

$$Q = (-2, 3)$$

uses the same two numbers, but in different positions. It describes a different point.

This is a small example of a general habit that will appear throughout analytic geometry. Before doing a calculation, we should read the object. A pair of numbers is not just a pair of numbers. It is an ordered description of position.

The origin and the coordinate axes

The origin is the point from which positions are measured. In the plane it is

$$O = (0, 0),$$

and in space it is

$$O = (0, 0, 0).$$

The coordinate axes give directions of measurement. In the plane, the x-axis records horizontal displacement from the origin, while the y-axis records vertical displacement. In space, the z-axis adds a third independent direction.

A coordinate system therefore does two things at once. It gives a point of reference, and it gives directions in which numbers are measured. Without this structure, a coordinate tuple has no geometric meaning.

Remark

Coordinates are not only labels. They are labels chosen relative to a system of measurement. Changing the coordinate system may change the coordinates of a point, even though the point itself has not moved.

Coordinates as variables

Sometimes coordinates are fixed numbers. For example,

$$A = (1, 4), \qquad B = (-3, 2)$$

are particular points. But often coordinates are variables. When we write

$$P = (x, y),$$

we may be describing an arbitrary point in the plane. The letters x and y are not yet known numbers. They are placeholders for possible positions.

This distinction is essential. A fixed point is one object. A point with variable coordinates can represent many possible points. Equations will later use this idea. For instance, an equation in x and y may describe all points whose coordinates satisfy a certain condition.

Example

The expression

$$P = (x, 2)$$

does not describe one fixed point unless x is known. It describes all points whose second coordinate is 2. Geometrically, these points lie on a horizontal line.
Similarly,

$$P = (1, y)$$

describes all points whose first coordinate is 1. These points lie on a vertical line.

This example already hints at the main idea of analytic geometry: an algebraic condition can describe a geometric object. The condition $y = 2$ describes a horizontal line. The condition $x = 1$ describes a vertical line. Later we will use more complicated conditions to describe more complicated objects.

Reading before calculating

The first task in coordinate geometry is not to apply a formula. It is to understand what kind of object is being described.

Summary

When reading coordinates, ask:

- Am I looking at a fixed point or a point with variable coordinates?

- Which coordinate describes which direction?
- Is the order of the coordinates important in this problem?
- Is an equation describing one point or a whole set of points?
- What geometric information is encoded in the algebraic notation?

The aim is to develop a controlled language. Coordinates allow us to speak about geometry with numbers, but the numbers must be read with care.

6.2 Vectors as displacements

A point describes a position. A vector describes a displacement. This difference is one of the most important distinctions in analytic geometry.

If we say that a point is $(3, 2)$, we are saying where something is. If we say that a vector is $(3, 2)$, we are saying how something moves: three units in one coordinate direction and two units in another coordinate direction. The same pair of numbers may appear in both situations, but the meaning is different.

Definition

Let

$$A = (a_1, a_2), \qquad B = (b_1, b_2)$$

be points in the plane. The vector from A to B is

$$\overrightarrow{AB} = B - A = (b_1 - a_1,\, b_2 - a_2).$$

In space, if

$$A = (a_1, a_2, a_3), \qquad B = (b_1, b_2, b_3),$$

then

$$\overrightarrow{AB} = (b_1 - a_1,\, b_2 - a_2,\, b_3 - a_3).$$

The formula is simple, but it should be read carefully. We subtract the starting coordinates from the ending coordinates. A displacement is not only about where we end. It is about how we get from one position to another.

Example

Let

$$A = (1, 4), \qquad B = (6, 2).$$

Then

$$\overrightarrow{AB} = B - A = (6 - 1,\, 2 - 4) = (5, -2).$$

This means that to move from A to B, we move 5 units in the positive x-direction and 2 units in the negative y-direction.

The vector $(5, -2)$ does not remember the absolute position of the starting point. It remembers the displacement. The same displacement could begin somewhere else.

Example

Let
$$C = (-3, 1), \qquad D = (2, -1).$$
Then
$$\overrightarrow{CD} = D - C = (2 - (-3),\ -1 - 1) = (5, -2).$$
Thus
$$\overrightarrow{AB} = \overrightarrow{CD},$$
even though the points A, B, C, D are not the same. The arrows begin in different places, but they describe the same displacement.

This is why vectors are often called free objects in elementary geometry. A vector can be represented by an arrow, but the arrow may be moved parallel to itself without changing the vector. What matters is direction, length, and orientation, not the particular location of the drawing.

Position vectors

There is one special way in which points and vectors are connected. If a vector begins at the origin and ends at a point P, then the coordinates of the vector and the coordinates of the point are the same.

If
$$P = (p_1, p_2),$$
then
$$\overrightarrow{OP} = (p_1, p_2),$$
where $O = (0, 0)$.

Remark

The equality of coordinates does not mean that points and vectors are the same kind of object. It means that a point can be represented by the vector from the origin to that point. This representation is useful, but the distinction should not be forgotten.

This distinction will help us later. A line can be described by starting from one point and moving in a vector direction. A plane can be described by a point and two independent directions, or by a point and one normal vector. In every case, we must know whether an expression describes a position, a displacement, or a direction.

Opposite displacements

The vector from B to A is the opposite of the vector from A to B:
$$\overrightarrow{BA} = A - B = -(B - A) = -\overrightarrow{AB}.$$
This is not a new formula to memorize. It simply says that going back reverses the displacement.

Example

For the points
$$A = (1, 4), \qquad B = (6, 2),$$
we found
$$\overrightarrow{AB} = (5, -2).$$

In the opposite direction,

$$\overrightarrow{BA} = A - B = (1-6,\, 4-2) = (-5, 2).$$

Thus

$$\overrightarrow{BA} = -\overrightarrow{AB}.$$

The sign of a vector therefore carries geometric information. It tells us orientation. The vectors $(5, -2)$ and $(-5, 2)$ lie along the same direction line, but they point in opposite directions.

Summary

When working with vectors as displacements, ask:

- Which point is the starting point?
- Which point is the ending point?
- Am I describing a position or a displacement?
- Would the same displacement be obtained if the arrow were moved elsewhere?
- What changes when the direction of the arrow is reversed?

A vector is not merely a pair or triple of numbers. It is a controlled way of recording movement from one position to another.

6.3 Vector operations

Vectors can be added, subtracted, and multiplied by numbers. These operations look algebraic, but each of them has a geometric meaning. If we treat them only as coordinate formulas, we lose much of their value.

The guiding idea is this: vectors describe displacements, and vector operations describe how displacements are combined or changed.

Adding vectors

Suppose a first displacement moves an object by u, and a second displacement moves it by v. The total displacement is $u+v$. In coordinates, addition is performed component by component.

Definition

For vectors

$$u = (u_1, u_2), \qquad v = (v_1, v_2),$$

their sum is

$$u + v = (u_1 + v_1,\, u_2 + v_2).$$

In space, for

$$u = (u_1, u_2, u_3), \qquad v = (v_1, v_2, v_3),$$

we define

$$u + v = (u_1 + v_1,\, u_2 + v_2,\, u_3 + v_3).$$

The formula says that horizontal changes add to horizontal changes, vertical changes add to vertical changes, and third-coordinate changes add to third-coordinate changes. We do not mix the roles of the coordinates.

Example

Let

$$u = (3, -1), \qquad v = (-2, 5).$$

Then

$$u + v = (3 + (-2),\ -1 + 5) = (1, 4).$$

Geometrically, moving by u and then by v produces the same final displacement as moving once by $(1, 4)$.

This interpretation explains why vector addition is commutative:

$$u + v = v + u.$$

If the same two displacements are applied, the final total displacement is the same, even though the path used to draw the two arrows may look different.

Subtracting vectors

Subtraction measures the difference between displacements. If u and v are two vectors, then

$$u - v = u + (-v).$$

Here $-v$ is the vector with the same length and opposite orientation.

Definition

For vectors

$$u = (u_1, u_2), \qquad v = (v_1, v_2),$$

their difference is

$$u - v = (u_1 - v_1,\ u_2 - v_2).$$

Example

Let

$$u = (4, 2), \qquad v = (1, 5).$$

Then

$$u - v = (4 - 1,\ 2 - 5) = (3, -3).$$

This vector records how the displacement v must be changed to become the displacement u.

Subtraction should be read with care. The vectors $u - v$ and $v - u$ are opposites:

$$v - u = -(u - v).$$

Changing the order changes the orientation.

Multiplying a vector by a number

Multiplying a vector by a number changes its scale. If the number is positive, the orientation is preserved. If the number is negative, the orientation is reversed. If the number is zero, the vector collapses to the zero vector.

Definition

If
$$u = (u_1, u_2),$$
and λ is a real number, then
$$\lambda u = (\lambda u_1,\ \lambda u_2).$$
In space,
$$\lambda(u_1, u_2, u_3) = (\lambda u_1,\ \lambda u_2,\ \lambda u_3).$$

Example

Let
$$u = (2, -3).$$
Then
$$3u = (6, -9), \qquad -2u = (-4, 6), \qquad 0u = (0, 0).$$
The vector $3u$ points in the same direction as u, but is three times as long. The vector $-2u$ lies on the same direction line, but points in the opposite direction.

Parallel vectors

Scalar multiplication gives a simple way to recognize parallel vectors. Two non-zero vectors are parallel when one is a scalar multiple of the other.

Definition

Non-zero vectors u and v are parallel if there exists a real number $\lambda \neq 0$ such that
$$v = \lambda u.$$

Example

The vectors
$$u = (2, 3), \qquad v = (6, 9)$$
are parallel because
$$v = 3u.$$
The vector
$$w = (-4, -6)$$
is also parallel to u, because
$$w = -2u.$$
The negative scalar means that w points in the opposite orientation along the same direction line.

This idea will be central in the next chapter. A line can be described by a point and a direction vector. All vectors parallel to that direction vector describe the same direction line, although they may have different lengths or orientations.

The zero vector

The zero vector is the displacement that changes nothing:
$$0 = (0, 0)$$

in the plane, and

$$0 = (0, 0, 0)$$

in space.

It is useful, but it must be handled carefully. The zero vector has no direction. For this reason, when we discuss directions or parallelism, we usually require direction vectors to be non-zero.

Remark

A zero displacement is meaningful: it says that the starting point and ending point coincide. But it should not be used as a direction vector for a line or a plane, because it does not point anywhere.

Summary

When using vector operations, ask:

- Am I combining displacements or comparing them?
- Does scalar multiplication preserve or reverse orientation?
- Is a vector being used as a direction?
- If so, is it non-zero?
- Are two vectors parallel because one is a scalar multiple of the other?

The purpose of vector operations is not only to produce new coordinates. The purpose is to control displacements, directions, and geometric relationships.

6.4 Length, distance, and midpoints

Coordinates allow us to measure geometric quantities. Once a displacement is written as a vector, its length can be computed. Once two points are written in coordinates, the distance between them can be computed. The formulas are familiar, but they should not be treated as isolated rules. They come from the Pythagorean theorem applied to coordinate differences.

Length of a vector

A vector tells us how far and in which direction to move. Its length measures only the size of the displacement, not its orientation.

Definition

For a vector

$$v = (v_1, v_2)$$

in the plane, its length is

$$\|v\| = \sqrt{v_1^2 + v_2^2}.$$

For a vector

$$v = (v_1, v_2, v_3)$$

in space, its length is

$$\|v\| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

The squares appear because coordinate directions are perpendicular. A displacement in the horizontal direction and a displacement in the vertical direction form the legs of a right triangle. The vector is the diagonal.

Example

Let
$$v = (3, 4).$$
Then
$$\|v\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$$
The vector has length 5. The coordinates $(3, 4)$ describe how the displacement is decomposed into coordinate directions; the length 5 describes the total size of the displacement.

A vector and its opposite have the same length:
$$\|-v\| = \|v\|.$$
This is natural. Reversing a displacement changes its orientation, but not how long it is.

Distance between two points

The distance between two points is the length of the vector joining them. If
$$A = (a_1, a_2), \qquad B = (b_1, b_2),$$
then
$$\overrightarrow{AB} = (b_1 - a_1,\, b_2 - a_2).$$
Therefore the distance between A and B is
$$d(A, B) = \|\overrightarrow{AB}\|.$$

Definition

For points
$$A = (a_1, a_2), \qquad B = (b_1, b_2)$$
in the plane,
$$d(A, B) = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2}.$$
For points in space,
$$A = (a_1, a_2, a_3), \qquad B = (b_1, b_2, b_3),$$
we have
$$d(A, B) = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2 + (b_3 - a_3)^2}.$$

The formula is not new information. It is the length formula applied to the displacement from A to B.

Example

Let
$$A = (1, -2), \qquad B = (5, 1).$$
Then
$$\overrightarrow{AB} = (5 - 1,\, 1 - (-2)) = (4, 3).$$

Hence

$$d(A,B) = \|\overrightarrow{AB}\| = \sqrt{4^2+3^2} = 5.$$

This calculation should be read in two layers. Algebraically, we subtract coordinates and take a square root. Geometrically, we first form the displacement from one point to the other, and then measure its length.

Midpoints

A midpoint is the point halfway between two given points. In coordinates, it is obtained by averaging corresponding coordinates.

Definition

For points

$$A = (a_1, a_2), \qquad B = (b_1, b_2),$$

the midpoint of the segment AB is

$$M = \left(\frac{a_1+b_1}{2}, \frac{a_2+b_2}{2}\right).$$

In space, for

$$A = (a_1, a_2, a_3), \qquad B = (b_1, b_2, b_3),$$

the midpoint is

$$M = \left(\frac{a_1+b_1}{2}, \frac{a_2+b_2}{2}, \frac{a_3+b_3}{2}\right).$$

The formula is easy to remember, but again we should understand it. Each coordinate is placed halfway between the corresponding coordinates of A and B.

Example

Let

$$A = (-1, 4), \qquad B = (5, 2).$$

The midpoint is

$$M = \left(\frac{-1+5}{2}, \frac{4+2}{2}\right) = (2, 3).$$

The first coordinate 2 is halfway between -1 and 5. The second coordinate 3 is halfway between 4 and 2.

There is also a vector way to read the midpoint. Starting from A, the full displacement to B is $B - A$. Half of that displacement is $\frac{1}{2}(B-A)$. Therefore

$$M = A + \frac{1}{2}(B - A).$$

After simplifying coordinates, this gives the same midpoint formula. This form is often more meaningful: the midpoint is reached by starting at A and moving halfway toward B.

What is being measured?

Length, distance, and midpoint calculations are common, but they answer different questions.

A length belongs to a vector. It asks how large a displacement is. A distance belongs to a pair of points. It asks how far apart two positions are. A midpoint is itself a point. It asks where the halfway position lies.

Summary

When measuring with coordinates, ask:

- Am I measuring a vector or comparing two points?
- Have I first formed the correct displacement?
- Does the result represent a number, such as a distance, or a point, such as a midpoint?
- Can the calculation be interpreted geometrically after it is done?

Analytic geometry becomes clearer when every formula is connected to the object it measures.

6.5 Dot product and angles

Some geometric questions are not only about length. We often want to know whether two directions are similar, opposite, or perpendicular. We may want to measure the angle between two vectors. For this purpose, one of the most important operations is the dot product.

The dot product takes two vectors and produces a number. This number is not a new vector. It is a measurement of how the two vectors interact directionally.

Definition

For vectors
$$u = (u_1, u_2), \qquad v = (v_1, v_2),$$
the dot product is
$$u \cdot v = u_1 v_1 + u_2 v_2.$$
In space, for
$$u = (u_1, u_2, u_3), \qquad v = (v_1, v_2, v_3),$$
we define
$$u \cdot v = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

The formula multiplies corresponding coordinates and then adds the results. But the meaning is not merely arithmetic. The dot product measures how much two vectors point in compatible directions.

Example

Let
$$u = (2, 3), \qquad v = (4, -1).$$
Then
$$u \cdot v = 2 \cdot 4 + 3 \cdot (-1) = 8 - 3 = 5.$$
The result is a number. It is not a vector, and it should not be interpreted as a point or a displacement.

The dot product and length

The dot product of a vector with itself gives the square of its length:

$$v \cdot v = \|v\|^2.$$

Indeed, if $v = (v_1, v_2)$, then

$$v \cdot v = v_1^2 + v_2^2,$$

while

$$\|v\| = \sqrt{v_1^2 + v_2^2}.$$

Therefore

$$\|v\| = \sqrt{v \cdot v}.$$

This connection is important. It shows that the dot product contains metric information: it knows something about lengths.

Example

For

$$v = (-2, 5),$$

we have

$$v \cdot v = (-2)^2 + 5^2 = 4 + 25 = 29.$$

Therefore

$$\|v\| = \sqrt{29}.$$

The dot product and angles

The deeper geometric meaning of the dot product is expressed by the formula

$$u \cdot v = \|u\|\|v\| \cos\theta,$$

where θ is the angle between the non-zero vectors u and v.

This formula should be read as an interpretation. The dot product is large and positive when the vectors point in similar directions. It is negative when they point more against each other. It is zero when the vectors are perpendicular.

Theorem

For non-zero vectors u and v,

$$\cos\theta = \frac{u \cdot v}{\|u\|\|v\|},$$

where θ is the angle between the vectors.

The denominator $\|u\|\|v\|$ removes the effect of the lengths. What remains is information about direction.

Example

Let

$$u = (1, 0), \qquad v = (0, 1).$$

Then

$$u \cdot v = 1 \cdot 0 + 0 \cdot 1 = 0.$$

Both vectors have length 1. Hence

$$\cos\theta = 0,$$

so

$$\theta = 90^\circ.$$

The vectors are perpendicular.

Perpendicular vectors

The most important special case is orthogonality. Two non-zero vectors are perpendicular exactly when their dot product is zero.

Theorem

Non-zero vectors u and v are perpendicular if and only if

$$u \cdot v = 0.$$

This criterion is one of the main reasons the dot product is so useful in analytic geometry. It turns a geometric condition, perpendicularity, into an algebraic equation.

Example

Let

$$u = (3, 2), \qquad v = (4, -6).$$

Then

$$u \cdot v = 3 \cdot 4 + 2 \cdot (-6) = 12 - 12 = 0.$$

Therefore the two vectors are perpendicular.

This is not only a convenient test. It is a model for much of analytic geometry: translate a geometric relation into an algebraic condition, solve or simplify the condition, and then interpret the result geometrically.

The sign of the dot product

The sign of the dot product gives qualitative information about the angle.

- If $u \cdot v > 0$, then the angle between u and v is acute.
- If $u \cdot v = 0$, then the angle is right.
- If $u \cdot v < 0$, then the angle is obtuse.

This follows from the sign of $\cos\theta$. Positive cosine corresponds to an acute angle, zero cosine to a right angle, and negative cosine to an obtuse angle.

Remark

The dot product does not only help us compute angles. It helps us read the relationship between directions. Sometimes the sign of the dot product is already enough to understand the geometry of a problem.

Why this matters for lines and planes

The dot product will appear repeatedly in the next chapters. A line may be described by a direction vector. A plane may be described by a normal vector. Perpendicularity between a vector and a direction, or between a vector and a plane, can be expressed using the dot product.

For example, if n is a normal vector to a plane, then every vector lying inside that plane is perpendicular to n. Algebraically, this becomes an equation of the form

$$n \cdot v = 0.$$

This is the beginning of the standard equation of a plane.

Summary

When using the dot product, ask:

- Am I trying to measure length, angle, or perpendicularity?
- Is the result supposed to be a number rather than a vector?
- Does a zero dot product indicate a right angle?
- Does the sign of the dot product already tell me something about the angle?
- Can a geometric condition be written as a dot product equation?

The dot product is a bridge between algebra and geometry. It allows angles and perpendicularity to be handled with equations.

6.6 Coordinates as a language

The purpose of this chapter is not only to introduce several formulas. Its purpose is to build a language. Coordinates, points, vectors, lengths, distances, and dot products are the words of that language. Once we can read them correctly, we can use them to describe more complicated geometric objects.

The next chapters will study lines and planes. These objects will not appear from nowhere. They will be built from the ideas already introduced here.

Points give position

A point tells us where we are. In the plane, a point has the form

$$P = (x, y).$$

In space, it has the form

$$P = (x, y, z).$$

A point may be fixed, such as

$$A = (2, -1),$$

or variable, such as

$$P = (x, y).$$

When the coordinates are variable, the point is allowed to move through a set of possible positions.

This distinction is essential for equations. An equation in coordinates usually does not describe one number. It describes all points whose coordinates satisfy a condition.

Example

The condition
$$y = 3$$
describes all points (x, y) in the plane whose second coordinate is 3. The first coordinate is free. Thus the equation describes a horizontal line.
The point $(2, 3)$ is one point on that line, but it is not the whole line.

A common mistake is to confuse one solution with the entire geometric object. A single point may satisfy an equation, but the equation may describe a whole set of points.

Vectors give displacement and direction

A vector tells us how to move. If
$$v = (v_1, v_2),$$
then v_1 and v_2 describe coordinate changes. The vector may be used as an actual displacement from one point to another, or it may be used only as a direction.

For example, the vector
$$v = (2, 1)$$
can mean: move two units in the x-direction and one unit in the y-direction. But it can also mean: consider the direction determined by this displacement.

This distinction will matter for lines. A line can be described by one point on the line and one non-zero direction vector. The point anchors the line. The vector gives the direction in which the line extends.

Remark

A point and a vector may have the same coordinates, but they play different roles. A point answers the question "where?" A vector answers the question "how far and in which direction?"

Equations describe conditions

An equation in analytic geometry is usually a condition on points. It says which points are allowed and which points are not.

For example, the equation
$$x^2 + y^2 = 1$$
describes all points whose distance from the origin is 1. This is a circle of radius 1 centered at the origin.

The formula is algebraic, but the object is geometric. The equation does not merely ask us to solve for x or y. It describes a set of points.

Example

Consider the condition
$$(x - 2)^2 + (y + 1)^2 = 9.$$
A point (x, y) satisfies this condition exactly when its distance from $(2, -1)$ is 3. Therefore the equation describes a circle with center $(2, -1)$ and radius 3.

This example shows the main power of analytic geometry. A geometric statement about distance becomes an algebraic equation. Conversely, an algebraic equation can be read as a geometric statement.

Fixed and variable quantities

Many mistakes in analytic geometry come from not distinguishing fixed objects from variable ones. A point may be fixed. A direction vector may be fixed. But the point (x, y) or (x, y, z) often represents a variable point moving through a set.

When we write a line in the form

$$P = A + tv,$$

the point A is fixed, the vector v is fixed and non-zero, and the parameter t is variable. As t changes, the point P moves along the line.

This formula belongs to the next chapter, but the language is already available. A fixed point gives a starting position. A direction vector gives a possible movement. A parameter controls how far we move.

Remark

Whenever a formula contains letters, ask which letters represent fixed data and which letters are allowed to vary. This question often reveals the meaning of the formula.

From this chapter to lines and planes

The chapter has prepared three ideas that will be used repeatedly.

First, coordinates describe position. Second, vectors describe displacements and directions. Third, equations describe conditions on variable points.

Lines will be described by combining points and direction vectors. Planes will be described either by points and directions or by a point and a normal vector. The dot product will allow us to express perpendicularity, and therefore it will help us write equations of planes.

Summary

When working with coordinates and vectors, ask:

- Am I describing a point, a vector, or a set of points?
- Which quantities are fixed, and which are variable?
- Is a vector being used as a displacement, a direction, or a normal vector?
- Can a geometric condition be translated into an algebraic equation?
- After computing, can I interpret the result geometrically?

Analytic geometry is not a collection of unrelated formulas. It is a way of translating between geometric objects and algebraic statements. The translation is useful only when we understand both languages.

Chapter 7

Lines

7.1 What information defines a line

A line is one of the simplest geometric objects, but it is also one of the easiest objects to misunderstand. In elementary algebra a line is often introduced through an equation such as $y = mx + b$. That equation is useful, but it should not be confused with the object itself. A line is not a formula. A line is a set of points arranged in one direction.

Analytic geometry studies this set of points by translating geometric information into algebraic language. The first question is therefore not "Which formula should I use?" The first question is: what information is enough to determine the line?

A point and a direction

The most natural description of a line uses two pieces of information:

- one point on the line,
- one non-zero vector that gives the direction of the line.

If A is a point and $v \neq 0$ is a direction vector, then the line consists of all points obtained by starting at A and moving by arbitrary multiples of v.

Definition

Let A be a point and let $v \neq 0$ be a vector. The line through A in direction v is the set of points

$$P = A + tv,$$

where t runs through all real numbers.

The parameter t controls the movement along the line. When $t = 0$, we are at the point A. When $t = 1$, we move from A by exactly one copy of the vector v. When $t = -1$, we move by the same vector in the opposite direction.

Example

Let

$$A = (1, 2), \qquad v = (3, -1).$$

The line through A in direction v is

$$P = (1, 2) + t(3, -1).$$

For $t = 0$, we get

$$P = (1, 2).$$

For $t = 1$, we get

$$P = (4, 1).$$

For $t = -1$, we get

$$P = (-2, 3).$$

These are different points, but they all lie on the same line because they are obtained from the same starting point by moving in the same direction.

This description matches the language developed in the previous chapter. A point gives position. A vector gives direction. A parameter tells us how much of that direction to use.

Two distinct points

A second common way to determine a line is to give two distinct points. If the points are different, then the vector from one point to the other gives a direction.

Let

$$A = (a_1, a_2), \qquad B = (b_1, b_2), \qquad A \neq B.$$

Then

$$\overrightarrow{AB} = B - A = (b_1 - a_1,\, b_2 - a_2)$$

is a non-zero direction vector for the line through A and B. Therefore the line can be written as

$$P = A + t(B - A).$$

The condition $A \neq B$ is not a technical decoration. If $A = B$, then the vector $B - A$ is the zero vector, and the zero vector has no direction. One point alone does not determine a unique line.

Example

Let

$$A = (2, -1), \qquad B = (5, 3).$$

Then

$$B - A = (3, 4).$$

The line through A and B can be written as

$$P = (2, -1) + t(3, 4).$$

For $t = 1$, this gives $(5, 3)$, so the point B is indeed on the line.

A point and a normal vector in the plane

In the plane there is another important way to determine a line. Instead of giving a direction along the line, we may give a direction perpendicular to the line. Such a vector is called a normal vector.

If $n \neq 0$ is perpendicular to a line and A is a point on the line, then a point P lies on the line exactly when the displacement $P - A$ is perpendicular to n. Using the dot product, this becomes

$$n \cdot (P - A) = 0.$$

This form is the beginning of the normal description of a line. It is extremely useful because perpendicularity, distance from a point to a line, and the general equation of a line all become natural from it.

Remark

A direction vector points along the line. A normal vector points across the line. Both can describe the same line, but they reveal different information.

The same object, different data

A line may therefore be determined from different kinds of information. We may know a point and a direction. We may know two points. In the plane, we may know a point and a normal vector. We may also know an equation and need to read from it what the point, direction, or normal vector is.

This is why this chapter will not present only one formula for a line. A line has many algebraic descriptions. Each description is a way of reading the same geometric object.

Summary

When first meeting a line problem, ask:

- Am I given a point on the line?
- Am I given a direction vector?
- Am I given two points?
- Am I given a normal vector or an implicit equation?
- Is the line in the plane or in space?
- Which description makes the given information visible?

The goal is not to memorize all possible formulas at once. The goal is to understand what information each formula is carrying.

7.2 Many descriptions of the same line

A line is a single geometric object, but it can be described in many algebraic languages. This is not a defect of geometry. It is a strength. Different descriptions make different features visible.

A good student should not ask only: "What is the formula for a line?" A better question is: "Which description of the line is most useful for the question I am trying to answer?"

In this section we collect the most common descriptions of a line in the plane. Later sections will explain them more carefully and show how to move between them.

Vector form

The vector form is

$$P = A + tv,$$

where A is a point on the line, $v \neq 0$ is a direction vector, and $t \in \mathbb{R}$.

This form makes motion along the line visible. It says: start at a point, then move in a fixed direction by an arbitrary amount.

Coordinate parametric form

If

$$A = (x_0, y_0), \qquad v = (a, b),$$

then the vector form becomes

$$(x, y) = (x_0, y_0) + t(a, b).$$

Equivalently,

$$x = x_0 + ta, \qquad y = y_0 + tb.$$

This is the coordinate parametric form. It is useful when we want to generate points on the line, check membership, or describe a line in space later.

Two-point form

If a line passes through two distinct points A and B, then the vector $B - A$ gives a direction. The line can be written as

$$P = A + t(B - A).$$

This form is useful because two points often occur naturally in geometric problems.

Point-slope form

For a non-vertical line in the plane, if the line passes through (x_0, y_0) and has slope m, then

$$y - y_0 = m(x - x_0).$$

This form makes the slope and one point visible. It is useful in elementary coordinate geometry, but it does not describe vertical lines.

Slope-intercept form

The form

$$y = mx + b$$

shows the slope m and the intersection point $(0, b)$ with the y-axis. It is familiar and convenient, but it is not the most general way to describe a line. A vertical line cannot be written in this form.

Implicit or general form

The general form of a line in the plane is

$$Ax + By + C = 0,$$

where A and B are not both zero.

This form is very important. It describes vertical, horizontal, and oblique lines in one language. It is also useful for intersections, side-of-line tests, and distance from a point to a line.

Normal form

The normal form uses a point P_0 on the line and a normal vector $n \neq 0$, perpendicular to the line:

$$n \cdot (P - P_0) = 0.$$

If $n = (A, B)$, $P = (x, y)$, and $P_0 = (x_0, y_0)$, this becomes

$$A(x - x_0) + B(y - y_0) = 0.$$

After expanding, it becomes an equation of the form

$$Ax + By + C = 0.$$

Thus the general form hides a normal vector.

Why so many forms?

The forms are not separate topics to memorize independently. They are different ways of reading the same object.

Description	Visible information	Useful for
$P = A + tv$	point and direction	movement, parametrization
$P = A + t(B - A)$	two points	building a line from points
$y - y_0 = m(x - x_0)$	point and slope	non-vertical plane lines
$y = mx + b$	slope and intercept	quick graphing
$Ax + By + C = 0$	normal vector hidden in coefficients	intersections, distance
$n \cdot (P - P_0) = 0$	point and normal vector	perpendicularity, distance

This table should not be read as a list of unrelated formulas. It is a map. When a problem gives certain information, one description may be natural. When a problem asks for a certain property, another description may be more useful.

Remark

Changing the description of a line is like translating a sentence from one language into another. The meaning should remain the same, but different words may make different parts of the meaning easier to see.

Summary

When choosing a line description, ask:

- Do I need to see a direction vector?
- Do I need to use a normal vector?
- Do I need to check whether a point lies on the line?
- Do I need to compare this line with another line?
- Do I need to compute a distance?
- Is a special case, such as a vertical line, possible?

The rest of the chapter develops this map into a working method.

7.3 Parametric description of lines

The parametric description is one of the most natural ways to describe a line. It uses the idea that a line can be traced by a moving point.

We start with a fixed point and a fixed direction. Then we allow a parameter to vary. As the parameter changes, the point moves along the line.

Vector parametric form

Let A be a point and let $v \neq 0$ be a direction vector. The line through A in direction v is

$$P = A + tv, \qquad t \in \mathbb{R}.$$

This is a compact formula, but it contains three different roles:

- A is fixed and anchors the line,
- v is fixed and gives the direction,
- t varies and moves the point along the line.

If we do not distinguish these roles, the formula becomes only a string of symbols. If we do distinguish them, the formula becomes a description of motion.

Example

Consider the line

$$P = (2,1) + t(3,4).$$

For $t = 0$, we get

$$P = (2,1).$$

For $t = 1$, we get

$$P = (5,5).$$

For $t = -1$, we get

$$P = (-1,-3).$$

The same formula produces different points on the same line. The parameter controls how far we move in the direction $(3,4)$.

The parameter is not something mysterious. It is a coordinate along the line, measured in copies of the direction vector.

Coordinate parametric form in the plane

If

$$A = (x_0, y_0), \qquad v = (a,b),$$

then

$$P = A + tv$$

means

$$(x,y) = (x_0, y_0) + t(a,b).$$

Therefore

$$x = x_0 + ta, \qquad y = y_0 + tb.$$

These are the parametric equations of the line in the plane.

Example

Let

$$A = (-1,3), \qquad v = (2,-5).$$

Then the line is

$$(x,y) = (-1,3) + t(2,-5),$$

or equivalently

$$x = -1 + 2t, \qquad y = 3 - 5t.$$

For $t = 2$, the point is

$$x = -1 + 4 = 3, \qquad y = 3 - 10 = -7.$$

Thus $(3,-7)$ lies on the line.

The coordinate equations should be read together. The same value of t must be used in both equations. We are not choosing one parameter for x and another parameter for y. One moving point has one parameter.

Checking whether a point lies on a parametric line

To check whether a point belongs to a parametric line, we ask whether there exists a value of the parameter that produces the point.

Example

Consider the line

$$x = 1 + 3t, \qquad y = 2 - t.$$

Does the point $(7, 0)$ lie on the line?
From the first equation,

$$7 = 1 + 3t,$$

so

$$t = 2.$$

Using the same value in the second equation gives

$$y = 2 - 2 = 0.$$

Therefore $(7, 0)$ lies on the line.
Now consider the point $(7, 1)$. The first coordinate again forces $t = 2$, but then the second coordinate would be 0, not 1. Therefore $(7, 1)$ does not lie on the line.

This example shows why the parameter must be shared. A point lies on the line only if all coordinate equations agree on the same parameter value.

Different parametrizations of the same line

A line does not have a unique parametric equation. We may choose a different point on the same line, or a different non-zero vector parallel to the same direction.

For example,

$$P = (2, 1) + t(3, 4)$$

and

$$P = (5, 5) + s(6, 8)$$

describe the same line. The point $(5, 5)$ lies on the first line, and the vector $(6, 8)$ is twice the direction vector $(3, 4)$. The second description moves along the same line, but uses a different starting point and a different scale for the parameter.

Remark

A parametric equation describes a line, but the parametrization itself is not unique. The geometric line remains the same even if we choose another point on it or replace the direction vector by a non-zero scalar multiple.

Why parametric form is useful

Parametric form is especially useful when the natural data are a point and a direction, or two points. It is also the most natural form for lines in space, where slope-intercept equations are no longer available.

Summary

When reading a parametric line, ask:

- What point is fixed on the line?
- What vector gives the direction?
- What does the parameter do geometrically?
- Does a given point arise from one common parameter value?
- Could a different point or parallel direction vector describe the same line?

The parametric form is not merely a formula. It is a way to watch a point move.

7.4 Two points and direction vectors

Many line problems begin with two points. This is natural: in Euclidean geometry, two distinct points determine exactly one line. In analytic geometry we translate this statement into coordinates and vectors.

The translation is simple. Two points give a displacement. That displacement gives a direction.

Building the direction vector

Let

$$A = (a_1, a_2), \qquad B = (b_1, b_2),$$

and suppose $A \neq B$. The vector from A to B is

$$\overrightarrow{AB} = B - A = (b_1 - a_1,\, b_2 - a_2).$$

This vector is non-zero because the points are distinct. It is therefore a valid direction vector for the line through A and B.

The line can then be written as

$$P = A + t(B - A), \qquad t \in \mathbb{R}.$$

Example

Find a parametric equation of the line through

$$A = (1, 2), \qquad B = (4, -1).$$

First compute the direction vector:

$$B - A = (4 - 1,\, -1 - 2) = (3, -3).$$

Therefore the line is

$$P = (1, 2) + t(3, -3).$$

In coordinate form,

$$x = 1 + 3t, \qquad y = 2 - 3t.$$

This procedure should be read as a chain of meanings. The two points determine a displacement. The displacement gives a direction. A point and a direction determine the line.

The order of the points

If we reverse the order of the two points, the direction vector changes sign:

$$A - B = -(B - A).$$

This does not change the line. It only reverses the orientation of the parametrization.

Using the previous example, we found

$$B - A = (3, -3).$$

If we start from B instead, then

$$A - B = (-3, 3).$$

The line may also be written as

$$P = (4, -1) + s(-3, 3).$$

This is the same line, but the parameter moves along it in the opposite orientation.

Remark

Changing the orientation of the direction vector does not change the geometric line. It changes how the line is traced by the parameter.

Non-unique direction vectors

A direction vector for a line is never unique. If v is a direction vector, then every non-zero scalar multiple λv is also a direction vector for the same line.

For instance, the vectors

$$(3, -3), \qquad (1, -1), \qquad (-2, 2)$$

all describe the same direction. They have different lengths and possibly different orientations, but they are parallel.

Example

The line

$$P = (1, 2) + t(3, -3)$$

can also be written as

$$P = (1, 2) + s(1, -1).$$

The second version uses a shorter direction vector. Both equations describe the same set of points, because $(3, -3) = 3(1, -1)$.

This is often useful. We may simplify a direction vector by dividing out a common factor. The line remains the same, but the notation becomes easier to read.

Finding a point from a parametrization

A parametric line already contains a point. In

$$P = A + tv,$$

the point A appears when $t = 0$. But every value of t gives another point on the same line. We may choose any of them as a new anchor point.

Example

Consider

$$P = (2, -1) + t(4, 2).$$

For $t = \frac{1}{2}$, we get

$$P = (2, -1) + \frac{1}{2}(4, 2) = (4, 0).$$

Thus $(4, 0)$ is also a point on the line. We could write the same line as

$$P = (4, 0) + s(4, 2).$$

This flexibility is not a problem. It is part of the geometry. A line has infinitely many points and infinitely many possible non-zero direction vectors.

When the two points are the same

If the two given points are equal, then they do not determine a unique line. The vector between them is the zero vector:

$$B - A = 0.$$

The zero vector cannot be a direction vector, because it gives no direction.

This is a common hidden issue in problems with parameters. If two points depend on a parameter, there may be special parameter values for which the points coincide. Those values must be checked separately.

Summary

When a line is given by two points, ask:

- Are the two points distinct?
- What is the vector from one point to the other?
- Can the direction vector be simplified?
- Does reversing the order only reverse orientation?
- Are there parameter values for which the two points might coincide?

The formula $P = A + t(B - A)$ is not just a shortcut. It expresses the idea that a line is obtained by moving from one point in the direction of another.

7.5 Normal and implicit forms in the plane

The parametric form describes a line by moving along it. The normal form describes a line in a different way: by using a vector perpendicular to it. This change of viewpoint is important. A direction vector tells us how to travel along the line. A normal vector tells us how to stand across the line.

Both viewpoints describe the same geometric object, but they make different properties visible.

Normal vector to a line

A non-zero vector n is called normal to a line if it is perpendicular to every direction vector of the line.

If a line has direction vector v, then a normal vector n satisfies

$$n \cdot v = 0.$$

In the plane, if

$$v = (a, b),$$

then one possible normal vector is

$$n = (-b, a),$$

because

$$(a, b) \cdot (-b, a) = -ab + ab = 0.$$

Another possible normal vector is $(b, -a)$. The normal vector is not unique, just as the direction vector is not unique.

Example

Let a line have direction vector

$$v = (3, 2).$$

Then

$$n = (-2, 3)$$

is a normal vector, because

$$(3, 2) \cdot (-2, 3) = -6 + 6 = 0.$$

Thus $(-2, 3)$ points perpendicular to the line.

Normal form from a point and a normal vector

Suppose a line passes through a fixed point $P_0 = (x_0, y_0)$, and suppose $n = (A, B) \neq (0, 0)$ is normal to the line. A variable point $P = (x, y)$ lies on the line exactly when the displacement from P_0 to P is perpendicular to n.

The displacement is

$$P - P_0 = (x - x_0,\ y - y_0).$$

The perpendicularity condition is

$$n \cdot (P - P_0) = 0.$$

Therefore

$$(A, B) \cdot (x - x_0,\ y - y_0) = 0,$$

so

$$A(x - x_0) + B(y - y_0) = 0.$$

Definition

A line in the plane with point $P_0 = (x_0, y_0)$ and normal vector $n = (A, B) \neq (0, 0)$ can be written in normal form as

$$A(x - x_0) + B(y - y_0) = 0.$$

Equivalently,

$$n \cdot (P - P_0) = 0.$$

This equation is not arbitrary. It says: the vector from the fixed point to the variable point must be perpendicular to the normal vector.

Example

Find an equation of the line through

$$P_0 = (1, -2)$$

with normal vector

$$n = (3, 4).$$

The normal form is

$$3(x - 1) + 4(y - (-2)) = 0.$$

Thus

$$3(x - 1) + 4(y + 2) = 0.$$

Expanding,

$$3x - 3 + 4y + 8 = 0,$$

so

$$3x + 4y + 5 = 0.$$

Implicit or general form

After expanding the normal form, every line in the plane can be written as

$$Ax + By + C = 0,$$

where A and B are not both zero.

This is called the implicit form or general form of a line. The word "implicit" means that the line is described as a condition. A point (x, y) lies on the line if its coordinates satisfy the equation.

Definition

The general form of a line in the plane is

$$Ax + By + C = 0,$$

where $(A, B) \neq (0, 0)$. The vector

$$n = (A, B)$$

is normal to the line.

The last sentence is essential. The coefficients of x and y are not just coefficients. Together they form a normal vector.

Example

Consider the line

$$2x - 5y + 7 = 0.$$

A normal vector is

$$n = (2, -5).$$

A direction vector must be perpendicular to n. One possible choice is

$$v = (5, 2),$$

because

$$(2, -5) \cdot (5, 2) = 10 - 10 = 0.$$

This example shows how information can be read from an equation. The equation does not display a direction vector directly, but it displays a normal vector.

Checking membership

The implicit form is often the easiest way to check whether a point lies on a line. We substitute the coordinates into the left-hand side.

Example

Let

$$\ell : 2x - 5y + 7 = 0.$$

Check whether $P = (4, 3)$ lies on ℓ. We compute

$$2 \cdot 4 - 5 \cdot 3 + 7 = 8 - 15 + 7 = 0.$$

Therefore $(4, 3)$ lies on the line.
For $Q = (1, 1)$, we get

$$2 \cdot 1 - 5 \cdot 1 + 7 = 4.$$

Since the result is not zero, $(1, 1)$ does not lie on the line.

The number obtained after substitution also has meaning. It measures, up to a scale factor, how far the point is from satisfying the line condition. This idea will later become the distance formula.

Why implicit form is useful

The general form is especially useful because it treats all lines in the plane uniformly. It includes vertical lines, horizontal lines, and oblique lines. It also makes normal vectors visible, which helps with perpendicularity and distance.

Summary

When reading an implicit equation $Ax + By + C = 0$, ask:

- Are A and B not both zero?
- What normal vector is visible in the equation?
- Can I find a direction vector perpendicular to that normal vector?
- Does a given point satisfy the equation?
- Would this form help with distance or intersection questions?

The implicit form is not merely another formula. It is the language of conditions, normal vectors, and perpendicularity.

7.6 Slope forms and special cases

Many students first meet lines through equations of the form $y = mx + b$. This form is useful, but it is only one description among several. It works well for non-vertical lines in the plane. It does not describe every line.

The purpose of this section is therefore not to reject slope forms, but to place them correctly.

They are convenient when slope is the natural information, but they should not replace the more general geometric understanding of a line.

Slope as a ratio of changes

For a non-vertical line, the slope measures the ratio of vertical change to horizontal change. If a line has direction vector

$$v = (a, b)$$

with $a \neq 0$, then its slope is

$$m = \frac{b}{a}.$$

This says that when the horizontal coordinate changes by a, the vertical coordinate changes by b.

Example

If a line has direction vector

$$v = (4, 6),$$

then its slope is

$$m = \frac{6}{4} = \frac{3}{2}.$$

The vector $(4, 6)$ and the vector $(2, 3)$ describe the same direction, and both give the same slope.

Slope is therefore not a new kind of direction. It is a way of encoding direction for non-vertical lines.

Point-slope form

Suppose a non-vertical line passes through (x_0, y_0) and has slope m. The point-slope form is

$$y - y_0 = m(x - x_0).$$

This equation says that every point (x, y) on the line has the same ratio of vertical change to horizontal change relative to (x_0, y_0).

Example

Find an equation of the line through $(2, -1)$ with slope 3.
Using point-slope form,

$$y - (-1) = 3(x - 2),$$

so

$$y + 1 = 3x - 6.$$

Thus

$$y = 3x - 7.$$

The computation is short, but the meaning is important. The equation does not simply contain the number 3. It says that vertical change is always three times horizontal change.

Slope-intercept form

If a non-vertical line crosses the y-axis at $(0, b)$ and has slope m, then its equation is

$$y = mx + b.$$

The number b is the value of y when $x = 0$. The number m describes the slope.

Example

The line
$$y = -2x + 5$$
has slope -2 and intersects the y-axis at
$$(0, 5).$$
A possible direction vector is
$$v = (1, -2),$$
because when x increases by 1, y changes by -2.

This form is good for quick graphing and for reading slope. It is not always the best form for distance, intersections, or vertical lines.

Horizontal lines

A horizontal line has slope 0. Its equation has the form
$$y = c.$$
Every point on the line has the same second coordinate.

For example,
$$y = 4$$
describes all points (x, y) with $y = 4$. The first coordinate is free.

In general form, this line is
$$y - 4 = 0.$$
A normal vector is $(0, 1)$, and a direction vector is $(1, 0)$.

Vertical lines

A vertical line has the form
$$x = c.$$
Every point on the line has the same first coordinate. Such a line does not have a finite slope and cannot be written as $y = mx + b$.

For example,
$$x = -2$$
describes all points (x, y) with first coordinate -2. The second coordinate is free.

In general form, this line is
$$x + 2 = 0.$$
A normal vector is $(1, 0)$, and a direction vector is $(0, 1)$.

Remark

The equation $y = mx + b$ is familiar, but it is not universal. Whenever vertical lines may appear, use a description that can include them, such as parametric form or general form.

From slope form to general form

Every non-vertical line written as

$$y = mx + b$$

can be rewritten in general form:

$$mx - y + b = 0.$$

If we prefer integer coefficients, we may multiply by a common denominator.

Example

Rewrite

$$y = \frac{2}{3}x - 5$$

in general form. Move all terms to one side:

$$\frac{2}{3}x - y - 5 = 0.$$

Multiplying by 3, we get

$$2x - 3y - 15 = 0.$$

A normal vector is therefore

$$n = (2, -3).$$

This example shows again that changing form changes what is visible. The slope form made the slope visible. The general form made the normal vector visible.

Summary

When using slope forms, ask:

- Is the line definitely non-vertical?
- Is slope the most useful information in this problem?
- Would a general form make a normal vector visible?
- Is the line horizontal or vertical?
- Which form avoids unnecessary special cases?

Slope forms are useful tools, but they should be used with awareness of their limits.

7.7 Converting between line descriptions

The same line can be written in many forms. Therefore one of the basic skills in analytic geometry is translation. We translate from the form that is given to the form that makes the next question easier.

This translation should not be mechanical. Each conversion has a meaning. When we convert a parametric equation into an implicit equation, we are changing from a direction-based description to a condition-based description. When we convert an implicit equation into a parametric equation, we are finding a point and a direction hidden inside a condition.

From parametric form to implicit form

Suppose a line is given in parametric form:

$$P = P_0 + tv.$$

In the plane, if

$$P_0 = (x_0, y_0), \qquad v = (a, b),$$

then a normal vector may be chosen as

$$n = (-b, a).$$

The implicit equation is then obtained from

$$n \cdot (P - P_0) = 0.$$

Example

Convert the line

$$(x, y) = (1, 2) + t(3, -1)$$

to implicit form.
The direction vector is

$$v = (3, -1).$$

A normal vector is

$$n = (1, 3),$$

because

$$(3, -1) \cdot (1, 3) = 3 - 3 = 0.$$

Using the point $(1, 2)$, the normal form is

$$1(x - 1) + 3(y - 2) = 0.$$

Thus

$$x - 1 + 3y - 6 = 0,$$

so

$$x + 3y - 7 = 0.$$

The direction vector was visible in the parametric form. The normal vector became visible after conversion to implicit form.

From implicit form to parametric form

Now suppose the line is given by

$$Ax + By + C = 0.$$

A normal vector is immediately visible:

$$n = (A, B).$$

A direction vector must be perpendicular to n. One convenient choice is

$$v = (-B, A),$$

because

$$(A, B) \cdot (-B, A) = -AB + AB = 0.$$

We also need one point on the line. This can be found by choosing one coordinate and solving for the other, whenever possible.

Example

Convert

$$2x - 5y + 10 = 0$$

to parametric form.
A normal vector is

$$n = (2, -5).$$

A direction vector is

$$v = (5, 2),$$

because

$$(2, -5) \cdot (5, 2) = 10 - 10 = 0.$$

To find a point, choose $x = 0$. Then

$$-5y + 10 = 0,$$

so

$$y = 2.$$

Thus $(0, 2)$ lies on the line. A parametric equation is

$$(x, y) = (0, 2) + t(5, 2).$$

There are many possible choices of the point. If we choose another point on the same line, we get another parametrization of the same geometric object.

From two points to implicit form

If a line is given by two distinct points, first build a direction vector. Then choose a normal vector perpendicular to that direction vector.

Example

Find an implicit equation of the line through

$$A = (1, -1), \qquad B = (4, 5).$$

The direction vector is

$$B - A = (3, 6).$$

We may simplify the direction to

$$v = (1, 2).$$

A normal vector is

$$n = (-2, 1),$$

because

$$(1, 2) \cdot (-2, 1) = -2 + 2 = 0.$$

Using the point $A = (1, -1)$, the normal form is

$$-2(x - 1) + 1(y - (-1)) = 0.$$

Thus

$$-2(x - 1) + y + 1 = 0.$$

Expanding,

$$-2x + 2 + y + 1 = 0,$$

so

$$-2x + y + 3 = 0.$$

Equivalently,

$$2x - y - 3 = 0.$$

Both final equations describe the same line. Multiplying every coefficient by a non-zero number does not change the set of solutions.

From slope form to direction and normal vectors

If a non-vertical line has equation

$$y = mx + b,$$

then a direction vector is

$$v = (1, m).$$

Indeed, when x increases by 1, the value of y changes by m.

After rewriting as

$$mx - y + b = 0,$$

we also see a normal vector:

$$n = (m, -1).$$

These two vectors are perpendicular, because

$$(1, m) \cdot (m, -1) = m - m = 0.$$

Example

For the line

$$y = -3x + 4,$$

a direction vector is

$$v = (1, -3).$$

Rewriting as

$$3x + y - 4 = 0,$$

we see a normal vector

$$n = (3, 1).$$

Indeed,

$$(1, -3) \cdot (3, 1) = 3 - 3 = 0.$$

Checking the translation

After converting one form into another, it is good practice to check that the descriptions agree. We can check that the chosen point satisfies the new equation. We can check that the direction vector is perpendicular to the normal vector. We can also test a second point generated by the parametrization.

Summary

When converting between descriptions, ask:

- What information is visible in the given form?
- Do I need a direction vector or a normal vector?
- Have I found at least one point on the line?
- Is the direction vector perpendicular to the normal vector?
- Does the converted equation describe the same set of points?

Conversion is not a ritual of rearranging symbols. It is a way of making hidden geometric information visible.

7.8 Parallel, perpendicular, and intersecting lines

Once we can describe one line, we can compare two lines. The comparison depends on the information we choose to read. Direction vectors make parallelism natural. Dot products make perpendicularity natural. Implicit equations make intersections natural because they become systems of linear equations.

The point is not to memorize a separate test for every situation. The point is to understand what each description makes visible.

Parallel lines from direction vectors

Two lines are parallel when their directions are the same. In vector language, this means that their direction vectors are scalar multiples of one another.

Definition

Let ℓ_1 and ℓ_2 have non-zero direction vectors v_1 and v_2. The lines are parallel if there exists a non-zero real number λ such that

$$v_2 = \lambda v_1.$$

Example

Consider the lines

$$\ell_1 : P = (1, 2) + t(2, -3),$$

and

$$\ell_2 : P = (-4, 0) + s(-4, 6).$$

Their direction vectors are

$$v_1 = (2, -3), \qquad v_2 = (-4, 6).$$

Since

$$v_2 = -2v_1,$$

the lines are parallel.

Parallel lines may be distinct, or they may be the same line. To decide which case occurs, we must check whether a point from one line lies on the other.

The same line or distinct parallel lines

If two lines are parallel and share at least one point, then they are the same line. If they are parallel and share no point, then they are distinct parallel lines.

Example

Let
$$\ell_1 : P = (1, 2) + t(2, -3)$$
and
$$\ell_2 : P = (3, -1) + s(4, -6).$$
The direction vectors are parallel because
$$(4, -6) = 2(2, -3).$$
Now check the point $(3, -1)$ from ℓ_2. On ℓ_1, when $t = 1$,
$$(1, 2) + (2, -3) = (3, -1).$$
Thus $(3, -1)$ lies on ℓ_1. The two equations describe the same line.

This example illustrates a common principle: parallel directions alone do not tell us whether the lines are equal. We also need position.

Perpendicular lines

Two lines are perpendicular when their direction vectors are perpendicular. The dot product gives the test.

Definition

Let ℓ_1 and ℓ_2 have direction vectors v_1 and v_2. If
$$v_1 \cdot v_2 = 0,$$
then the lines are perpendicular.

Example

Let the direction vectors of two lines be
$$v_1 = (3, 1), \qquad v_2 = (2, -6).$$
Then
$$v_1 \cdot v_2 = 3 \cdot 2 + 1 \cdot (-6) = 6 - 6 = 0.$$
Therefore the lines are perpendicular.

This is not only a computational shortcut. It is exactly the meaning of perpendicularity translated into algebra.

Using normal vectors

If lines are written in general form,

$$\ell_1 : A_1x + B_1y + C_1 = 0,$$

$$\ell_2 : A_2x + B_2y + C_2 = 0,$$

then their normal vectors are

$$n_1 = (A_1, B_1), \qquad n_2 = (A_2, B_2).$$

The lines are parallel when their normal vectors are parallel. They are perpendicular when their normal vectors are perpendicular.

Example

Consider

$$\ell_1 : 2x - 3y + 1 = 0,$$

and

$$\ell_2 : 4x - 6y - 5 = 0.$$

The normal vectors are

$$n_1 = (2, -3), \qquad n_2 = (4, -6).$$

Since

$$n_2 = 2n_1,$$

the lines are parallel.

Example

Consider

$$\ell_1 : x + 2y - 4 = 0,$$

and

$$\ell_2 : 2x - y + 3 = 0.$$

The normal vectors are

$$n_1 = (1, 2), \qquad n_2 = (2, -1).$$

Their dot product is

$$n_1 \cdot n_2 = 1 \cdot 2 + 2 \cdot (-1) = 0.$$

Therefore the normal vectors are perpendicular. The lines themselves are also perpendicular.

For lines in the plane, using direction vectors or normal vectors gives equivalent information, but the most convenient choice depends on the form in which the line is given.

Intersecting lines as a system of equations

Two non-parallel lines in the plane intersect in exactly one point. If the lines are written in general form, the intersection point is found by solving a system of two linear equations.

Example

Find the intersection of

$$\ell_1 : x + y - 3 = 0,$$

and

$$\ell_2 : 2x - y = 0.$$

The system is

$$\begin{cases} x + y = 3, \\ 2x - y = 0. \end{cases}$$

Adding the equations gives

$$3x = 3,$$

so

$$x = 1.$$

Then from $2x - y = 0$,

$$y = 2.$$

Thus the lines intersect at

$$(1, 2).$$

This example connects analytic geometry with systems of linear equations. The intersection point is the point whose coordinates satisfy both line equations at once.

Reading the result

When comparing two lines, the answer should include interpretation. It is not enough to say that a determinant is zero or that two vectors are proportional. We should say what that means geometrically.

Summary

When comparing two lines, ask:

- Which form are the lines given in?
- Are direction vectors visible?
- Are normal vectors visible?
- Are the directions parallel?
- Are the relevant vectors perpendicular?
- If the lines are parallel, are they the same line or distinct lines?
- If they are not parallel, what is their intersection point?

The relative position of lines is not a separate topic from line descriptions. It is one of the main reasons those descriptions are useful.

7.9 Distance from a point to a line

The distance from a point to a line is not the distance to an arbitrary point on the line. It is the shortest possible distance. Geometrically, the shortest path from a point to a line is perpendicular to the line.

This is why the normal form of a line is so useful. A normal vector already points in the direction in which this shortest distance is measured.

The geometric idea

Let a line be given in general form:

$$\ell : Ax + By + C = 0.$$

A normal vector to the line is

$$n = (A, B).$$

If a point $P_0 = (x_0, y_0)$ does not lie on the line, then the shortest segment from P_0 to the line is parallel to n. The distance should therefore be measured in the normal direction.

The expression

$$Ax_0 + By_0 + C$$

tells us how far the point is from satisfying the equation of the line, but it still depends on the length of the normal vector. To obtain the actual distance, we divide by the length of the normal vector:

$$\sqrt{A^2 + B^2}.$$

Theorem

Let

$$\ell : Ax + By + C = 0$$

be a line in the plane, with $(A, B) \neq (0, 0)$. The distance from the point $P_0 = (x_0, y_0)$ to ℓ is

$$d(P_0, \ell) = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}.$$

The absolute value appears because distance is non-negative. Without the absolute value, the sign tells us on which side of the oriented line the point lies.

Example of the distance formula

Example

Find the distance from the point

$$P_0 = (3, 1)$$

to the line

$$\ell : 2x - y - 4 = 0.$$

Here

$$A = 2, \qquad B = -1, \qquad C = -4.$$

Substitute $(x_0, y_0) = (3, 1)$:

$$Ax_0 + By_0 + C = 2 \cdot 3 + (-1) \cdot 1 - 4 = 6 - 1 - 4 = 1.$$

The length of the normal vector is

$$\sqrt{A^2 + B^2} = \sqrt{2^2 + (-1)^2} = \sqrt{5}.$$

Therefore

$$d(P_0, \ell) = \frac{|1|}{\sqrt{5}} = \frac{1}{\sqrt{5}}.$$

The computation is short, but it should be interpreted. The point does not satisfy the line equation because substitution gives 1, not 0. Dividing by $\sqrt{5}$ corrects for the length of the normal vector.

Why the denominator is needed

The equation of a line is not unique. For example,

$$2x - y - 4 = 0$$

and

$$4x - 2y - 8 = 0$$

describe the same line. The second equation is obtained by multiplying the first by 2.

If we used only $|Ax_0 + By_0 + C|$, the value would also be multiplied by 2. But the geometric distance should not change. The denominator $\sqrt{A^2 + B^2}$ is multiplied by the same factor, so the ratio remains the same.

Remark

The distance formula uses the equation of the line, but the result is geometric. It cannot depend on which equivalent equation we choose.

Signed distance and sides of a line

The expression

$$Ax_0 + By_0 + C$$

without absolute value has a sign. Points on one side of the line give one sign, and points on the other side give the opposite sign. Points on the line give zero.

A normalized signed quantity is

$$\frac{Ax_0 + By_0 + C}{\sqrt{A^2 + B^2}}.$$

Its absolute value is the distance to the line.

Example

For the line

$$\ell : x + y - 1 = 0,$$

consider the points

$$P = (2, 0), \qquad Q = (0, -2).$$

For P,

$$2 + 0 - 1 = 1.$$

For Q,

$$0 + (-2) - 1 = -3.$$

The signs are different, so the points lie on opposite sides of the oriented line.

This interpretation is useful in geometry and in applications. The equation of a line does more than describe the points on the line. It also separates the plane into two sides.

Distance and projection idea

There is another way to understand the formula. Choose any point A on the line. The vector $P_0 - A$ goes from the line to the point. The distance from P_0 to the line is the length of the projection of this vector onto the normal direction.

If $n = (A, B)$, then a unit normal vector is

$$\frac{n}{\|n\|}.$$

The signed length of the projection is

$$(P_0 - A) \cdot \frac{n}{\|n\|}.$$

This is the same idea hidden inside the distance formula.

Summary

When computing distance from a point to a line, ask:

- Is the line written in general form $Ax + By + C = 0$?
- What is the normal vector (A, B)?
- What value do I get after substituting the point?
- Did I divide by the length of the normal vector?
- Does the sign, before taking absolute value, carry side-of-line information?

The distance formula is not an isolated trick. It is the normal-vector interpretation of a line turned into a measurement.

7.10 Lines in space

Lines in space are best understood through points and direction vectors. The parametric form works in the plane and in space, while slope-intercept form belongs only to the plane. This is one reason why the vector description is so important.

In three-dimensional space, a line still needs the same kind of information: one point and one non-zero direction vector.

Parametric form in space

Let

$$A = (x_0, y_0, z_0)$$

be a point in space, and let

$$v = (a, b, c) \neq (0, 0, 0)$$

be a direction vector. The line through A in direction v is

$$(x, y, z) = (x_0, y_0, z_0) + t(a, b, c), \qquad t \in \mathbb{R}.$$

Equivalently,

$$x = x_0 + ta, \qquad y = y_0 + tb, \qquad z = z_0 + tc.$$

Example

The line

$$(x, y, z) = (1, -2, 3) + t(4, 0, -1)$$

has point $(1, -2, 3)$ and direction vector $(4, 0, -1)$. Its coordinate equations are

$$x = 1 + 4t, \qquad y = -2, \qquad z = 3 - t.$$

The second coordinate is constant because the second component of the direction vector is zero.

This example is important. A zero component of a direction vector does not mean that the vector is invalid. It means that the line does not move in that coordinate direction.

A line through two points in space

Two distinct points in space determine a line in the same way as in the plane. If

$$A = (a_1, a_2, a_3), \qquad B = (b_1, b_2, b_3), \qquad A \neq B,$$

then

$$B - A = (b_1 - a_1,\, b_2 - a_2,\, b_3 - a_3)$$

is a direction vector. The line is

$$P = A + t(B - A).$$

Example

Find a parametric equation of the line through

$$A = (2, 1, -1), \qquad B = (5, 3, 4).$$

The direction vector is

$$B - A = (3, 2, 5).$$

Therefore the line is

$$(x, y, z) = (2, 1, -1) + t(3, 2, 5).$$

In coordinates,

$$x = 2 + 3t, \qquad y = 1 + 2t, \qquad z = -1 + 5t.$$

The idea is unchanged: two points give a displacement, and the displacement gives a direction.

Symmetric form

If all components of the direction vector are non-zero, then the parametric equations

$$x = x_0 + ta, \qquad y = y_0 + tb, \qquad z = z_0 + tc$$

can be rewritten as

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

This is called the symmetric form of a line in space.

Example

For the line

$$(x, y, z) = (1, 2, -3) + t(2, -1, 4),$$

all components of the direction vector are non-zero. Therefore

$$\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z+3}{4}.$$

The symmetric form is compact, but it must be used carefully. It comes from solving each parametric equation for the same parameter t. If a component of the direction vector is zero, we cannot divide by it.

Zero components in symmetric form

Suppose

$$(x, y, z) = (1, -2, 3) + t(4, 0, -1).$$

Then

$$x = 1 + 4t, \qquad y = -2, \qquad z = 3 - t.$$

The second component of the direction vector is zero, so the line has constant y-coordinate. We may write

$$\frac{x-1}{4} = \frac{z-3}{-1}, \qquad y = -2.$$

We should not write a fraction with denominator 0. The equation $y = -2$ must be kept separately.

Remark

Symmetric form is convenient only when used honestly. A zero component of the direction vector becomes a constant-coordinate condition, not a division by zero.

Why one equation is not enough in space

In the plane, a single linear equation

$$Ax + By + C = 0$$

describes a line. In space, a single linear equation

$$Ax + By + Cz + D = 0$$

usually describes a plane, not a line.

This is a fundamental difference. In the plane, one linear condition leaves one degree of freedom, which gives a line. In space, one linear condition leaves two degrees of freedom, which gives a plane.

A line in space can often be described as the intersection of two planes, that is, by a system

$$\begin{cases} A_1x + B_1y + C_1z + D_1 = 0, \\ A_2x + B_2y + C_2z + D_2 = 0. \end{cases}$$

However, the parametric form is usually the clearest first description of a line in space.

Example

The equation

$$z = 2$$

in space does not describe a line. It describes all points whose third coordinate is 2. This is a plane parallel to the xy-plane.
To describe a line in that plane, we need another independent condition or a parametric description.

Comparing lines in space

Lines in space may be parallel, intersecting, or skew. Skew lines are lines that do not intersect and are not parallel. This possibility does not occur for lines in the plane.

For now, the most important point is that direction vectors still detect parallelism. If two direction vectors are scalar multiples, the lines are parallel or the same line. If they are not scalar multiples, the lines may intersect or may be skew. To decide, one must solve the parametric equations together.

Summary

When working with lines in space, ask:

- What point lies on the line?
- What non-zero vector gives the direction?
- Are any direction-vector components zero?
- Is symmetric form safe to use?
- Am I mistakenly treating one plane equation as a line?
- Are two lines parallel, intersecting, or skew?

The vector-parametric description is the most reliable language for lines in space. It keeps the point, the direction, and the movement visible.

7.11 Choosing the right description

This chapter introduced many descriptions of a line. At first this may seem like too much notation. But the purpose is not to collect formulas. The purpose is to learn how to choose a description that fits the problem.

The same line can be written in several ways. The best form depends on what is known and what is being asked.

When a point and direction are given

If a problem gives a point and a direction vector, the parametric form is usually the most natural:

$$P = A + tv.$$

It keeps the given information visible. It also makes it easy to generate points and to understand movement along the line.

This form is also the safest first choice in space.

When two points are given

If a problem gives two distinct points A and B, first form the direction vector

$$B - A.$$

Then write

$$P = A + t(B - A).$$

From there, if needed, convert to another form.

The important point is that the two given points are not just coordinates to insert into a memorized formula. They define a displacement, and that displacement defines a direction.

When a normal vector is visible

If a problem gives an equation

$$Ax + By + C = 0,$$

then the normal vector

$$n = (A, B)$$

is immediately visible. This form is useful for distance questions, side-of-line questions, and comparisons using normal vectors.

If a point P_0 and a normal vector n are given, use

$$n \cdot (P - P_0) = 0.$$

This is often more meaningful than jumping directly to expanded coordinates.

When slope is useful

If the line is non-vertical and slope is central to the question, then slope forms may be convenient:

$$y - y_0 = m(x - x_0)$$

or

$$y = mx + b.$$

But these forms should be used with awareness. Vertical lines require a different description, such as

$$x = c,$$

or a parametric or general equation.

Which form helps which question?

The following guide is often useful.

Question	Useful description
Generate points on the line	parametric form
Line through two points	two-point form, then parametric form
Check whether a point lies on the line	implicit or parametric form
Find an intersection in the plane	general form as a system
Check parallelism	direction vectors or normal vectors
Check perpendicularity	dot product of directions or normals
Compute distance from a point	general form $Ax + By + C = 0$
Work in space	parametric form

This table should not be used mechanically. It is a guide for choosing what to read from the line.

A complete reading example

Consider the line

$$\ell : 2x - 3y + 6 = 0.$$

From this form we immediately read a normal vector:

$$n = (2, -3).$$

A direction vector can be chosen perpendicular to n:

$$v = (3, 2),$$

because

$$(2, -3) \cdot (3, 2) = 6 - 6 = 0.$$

To find a point on the line, set $x = 0$. Then

$$-3y + 6 = 0,$$

so

$$y = 2.$$

Thus $(0, 2)$ lies on the line. A parametric form is

$$P = (0, 2) + t(3, 2).$$

The same line can therefore be read in two ways:

$$2x - 3y + 6 = 0$$

shows the normal vector and is good for distance or side tests, while

$$P = (0, 2) + t(3, 2)$$

shows a point and direction and is good for motion along the line.

Remark

A good solution often changes the description of a line. This is not extra work. It is how we choose the language in which the problem becomes clear.

What to remember

A line is a set of points. Equations and parametrizations are descriptions of that set. The main descriptions are not competitors; they are tools.

Summary

When working with lines, ask:

- Is the line in the plane or in space?
- Am I given a point, two points, a direction vector, or a normal vector?
- Which description makes the given information visible?
- Which description makes the required property easiest to study?
- Am I studying membership, intersection, parallelism, perpendicularity, or distance?

- After computing, can I explain what the result means geometrically?

The most important lesson is not that lines have many formulas. The most important lesson is that each formula reveals a different aspect of the same geometric object.

Chapter 8

Planes

This chapter is a placeholder for the study of planes. It will use the language of coordinates, vectors, and lines developed earlier, with the detailed section structure to be decided later.

Chapter 9

Sequences and Functions

9.1 What a function is

The word "function" appears very often in mathematics, but it is sometimes introduced too quickly. A student may first meet functions as formulas such as $f(x) = 2x + 1$, and then begin to think that a function is the same thing as a formula. This is not quite right.

A formula can describe a function, but a function is more basic than a formula. A function is a rule of assignment. It tells us how to assign one output to each allowed input.

Definition

Let X and Y be sets. A function from X to Y is a rule that assigns to every element $x \in X$ exactly one element of Y.
We write

$$f : X \to Y.$$

The set X is called the domain, and the set Y is called the codomain.

The phrase "exactly one" is essential. For each allowed input, the function must give one output, not two different outputs and not no output at all. This is the basic discipline of functions.

Inputs and outputs

If $f : X \to Y$, then an element $x \in X$ is an input. The value assigned to this input is written as $f(x)$. This value belongs to Y.

We may write this assignment as

$$x \mapsto f(x).$$

This notation should be read as: the input x is sent to the output $f(x)$.

Example

Let

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2 + 1.$$

The input 3 is sent to

$$f(3) = 3^2 + 1 = 10.$$

The input -2 is sent to

$$f(-2) = (-2)^2 + 1 = 5.$$

The rule assigns one real number to each real input.

The notation $f(3) = 10$ describes one value of the function. It does not describe the whole function. The whole function is the entire rule that works for every allowed input.

A function is not only a formula

Functions can be described in several ways. A formula is common, but it is not the only possibility. A function may be described by a table, by a graph, by a verbal rule, or by a recursive rule.

Example

Suppose a small machine accepts the inputs $1, 2, 3$ and returns the following outputs:

x	1	2	3
$f(x)$	4	4	7

This table describes a function with domain $\{1, 2, 3\}$. Each input has exactly one output. There is no need for a formula such as $f(x) = \cdots$ in order for this assignment to be a function.

A function is therefore not a decorative name for an algebraic expression. It is a structured assignment from one set to another.

What can go wrong

Not every relation between two sets is a function. The problem occurs when one input is assigned more than one output, or when an allowed input is assigned no output.

Example

Suppose the input 2 is assigned both 5 and 8. Then the assignment is not a function, because one input has two outputs.
Suppose instead that the domain is $\{1, 2, 3\}$, but no output is assigned to 3. Then the assignment is also not a function, because an allowed input has no output.

The definition of a function is strict because it allows us to ask clear questions. If we ask for $f(2)$, there must be one definite answer.

Summary

When reading a function, ask:

- What is the domain?
- What is the codomain?
- What are the allowed inputs?
- What output is assigned to each input?
- Is each allowed input assigned exactly one output?
- Is the function described by a formula, a table, a graph, or another rule?

The main idea is simple but important: a function is a controlled way of assigning outputs to inputs.

9.2 Domain, codomain, and values

The notation

$$f : X \to Y$$

contains more information than it may first seem. It tells us where inputs are taken from and where outputs are expected to live. Before computing with a function, we should learn to read this notation.

Domain and codomain

In the notation

$$f : X \to Y,$$

the set X is the domain. It contains the inputs for which the function is defined. The set Y is the codomain. It is the set in which the outputs are declared to live.

If $x \in X$, then the value of the function at x is written

$$f(x).$$

This value belongs to Y:

$$f(x) \in Y.$$

Example

Let

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = 2x - 1.$$

The domain is $\mathbb{R}$. This means every real number is an allowed input. The codomain is also $\mathbb{R}$. For example,

$$f(4) = 2 \cdot 4 - 1 = 7.$$

The input is 4, and the output is 7.

The domain is part of the function. Two rules may look the same as formulas but describe different functions if their domains are different.

Example

Consider the rule

$$f(x) = x^2.$$

If the domain is $\mathbb{R}$, then negative numbers, zero, and positive numbers are all allowed inputs. If the domain is $[0, \infty)$, then only non-negative inputs are allowed. The formula is the same, but the function is not the same because the domain has changed.

Range or image

The codomain is the declared target set. The range, also called the image, is the set of outputs that are actually obtained.

Definition

For a function $f : X \to Y$, the range or image of f is the set

$$f(X) = \{f(x) : x \in X\}.$$

It consists of all values that the function actually takes.

The range may be smaller than the codomain.

Example

Let
$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = x^2.$$
The codomain is $\mathbb{R}$, but the range is
$$[0, \infty),$$
because a square is never negative and every non-negative real number is the square of some real number.

This distinction matters. The codomain is part of how the function is declared. The range is discovered by studying what values the function actually produces.

The function and its values

A common source of confusion is the difference between f and $f(x)$. The symbol f denotes the whole function. The expression $f(x)$ denotes the value of the function at the input x.

For the function
$$f(x) = x^2 + 1,$$
the value at 3 is
$$f(3) = 10.$$
The value 10 is a number. The function f is not a number; it is the entire assignment rule.

Remark

The expression $f(x)$ should be read as "the value of f at x," not as another name for the whole function. The function is f. The value at an input is $f(x)$.

Arguments and values

The input of a function is often called an argument. If we write $f(2)$, then 2 is the argument. If we write $f(a)$, then a is the argument. If we write $f(x + h)$, then the whole expression $x + h$ is the argument.

This point will become important later. When we substitute into a function, we substitute the entire argument, not only one letter.

Example

Let
$$f(x) = x^2.$$
Then
$$f(3) = 3^2 = 9,$$
$$f(a) = a^2,$$
and
$$f(x + h) = (x + h)^2.$$
The last expression is not $x^2 + h$. The whole argument $x + h$ is placed into the rule.

Summary

When reading function notation, ask:

- What is the domain?
- What is the codomain?
- What values are actually obtained?
- Am I talking about the whole function f or one value $f(x)$?
- What is the entire argument being substituted into the function?

Careful reading of notation prevents many later mistakes. Functions are not difficult because the symbols are complicated; they become difficult when the roles of the symbols are not read.

9.3 Function notation and evaluation

Function notation is compact. This is useful, but it also means that we must read it carefully. Many mistakes with functions are not caused by difficult calculations. They are caused by not reading what is being substituted into the function.

If a function is given by a rule such as

$$f(x) = 2x - 1,$$

then the letter x is a placeholder for an input. When we evaluate the function, we replace that placeholder by the chosen argument.

Evaluating at numbers

To evaluate a function at a number, substitute that number into the rule.

Example

Let

$$f(x) = 2x - 1.$$

Then

$$f(4) = 2 \cdot 4 - 1 = 7.$$

Also,

$$f(-3) = 2 \cdot (-3) - 1 = -6 - 1 = -7.$$

The notation $f(4)$ means the value of the function at the input 4.

The parentheses are not decoration. They indicate the argument of the function.

Evaluating at expressions

The argument of a function does not have to be a number. It may be a letter or a more complicated expression. The rule is the same: replace the input placeholder by the entire argument.

Example

Let

$$f(x) = 2x - 1.$$

Then

$$f(a) = 2a - 1.$$

Also,

$$f(x+h) = 2(x+h) - 1 = 2x + 2h - 1.$$

The whole expression $x + h$ is substituted in place of x.

This habit is especially important in calculus, where expressions such as $f(x+h)$ appear frequently. If the whole argument is not substituted correctly, later formulas become meaningless.

Example

Let

$$g(x) = x^2.$$

Then

$$g(x+h) = (x+h)^2.$$

Expanding,

$$g(x+h) = x^2 + 2xh + h^2.$$

It would be wrong to write $g(x+h) = x^2 + h$, because the function squares the whole input.

A function acts on its entire argument. This is one of the simplest and most important rules of function notation.

Different letters, same role

The letter used in a formula is not sacred. The function

$$f(x) = x^2 + 1$$

can also be described by writing

$$f(t) = t^2 + 1.$$

The input letter changed, but the function did not. What matters is the rule: take the input, square it, and add 1.

Example

If

$$f(x) = x^2 + 1,$$

then

$$f(t) = t^2 + 1, \qquad f(u) = u^2 + 1, \qquad f(3) = 3^2 + 1 = 10.$$

The symbols x, t, u are placeholders. They indicate where the input goes.

This does not mean that letters never matter. In a larger expression, each letter may already have a role. The point is only that the name of the input variable is not the function itself.

Functions with named rules

Sometimes functions are defined without writing $f(x) = \cdots$ first. For example, we may say: let f be the function that assigns to each real number its square plus one. This means exactly the same rule:

$$f(x) = x^2 + 1.$$

A verbal description may be just as precise as a formula if it clearly assigns one output to each input.

Example

Let f assign to each positive integer the number of its positive divisors. Then

$$f(1) = 1, \qquad f(2) = 2, \qquad f(6) = 4,$$

because 6 has the positive divisors $1, 2, 3, 6$.
This function is not introduced by a simple algebraic formula, but it is still a function.

Reading equations involving functions

An equation such as

$$f(x) = 0$$

asks for inputs whose output is zero. It is not asking for the function itself to disappear. It asks for the arguments x at which the value of the function equals zero.

Example

Let

$$f(x) = x^2 - 4.$$

The equation

$$f(x) = 0$$

means

$$x^2 - 4 = 0.$$

Thus

$$x = -2 \qquad \text{or} \qquad x = 2.$$

These are the inputs where the function has value zero.

This kind of question will reappear often. Zeros of a function, intersections with axes, solutions of equations, and sign changes are all connected to evaluating functions carefully.

Summary

When evaluating a function, ask:

- What is the rule of the function?
- What is the entire argument being substituted?
- Am I substituting a number, a letter, or an expression?
- Does the function act on the whole argument?
- Am I looking for a value $f(a)$, or solving an equation such as $f(x) = 0$?

Function notation becomes manageable when we treat it as careful substitution into a rule.

9.4 Graphs as pictures of functions

A graph is a picture of a function. More precisely, it is a picture of input-output pairs. If the input is x and the output is $f(x)$, then the corresponding point on the graph is

$$(x, f(x)).$$

This simple idea is the key to reading graphs. A graph is not a drawing to guess from. It is a geometric representation of values of a function.

Definition

Let $f : D \to \mathbb{R}$, where $D \subseteq \mathbb{R}$. The graph of f is the set

$$\{(x, f(x)) : x \in D\}.$$

Each point on the graph has first coordinate equal to an input and second coordinate equal to the corresponding output.

If the point $(2, 5)$ lies on the graph of f, then

$$f(2) = 5.$$

If $f(2) = 5$, then $(2, 5)$ lies on the graph. These are two ways of saying the same thing.

Reading values from a graph

To read $f(a)$ from a graph, we look at the input a on the horizontal axis and find the height of the graph above that input. That height is the value $f(a)$.

Example

Suppose a graph contains the points

$$(-1, 3), \qquad (0, 1), \qquad (2, 5).$$

Then the graph tells us that

$$f(-1) = 3, \qquad f(0) = 1, \qquad f(2) = 5.$$

The first coordinate is the input. The second coordinate is the output.

This reading is basic, but it is not trivial. Many later questions about functions depend on reading coordinates of points on a graph correctly.

Intercepts

The point where a graph meets the vertical axis has input 0. Therefore, if the point exists, it has the form

$$(0, f(0)).$$

This is called the y-intercept.

A point where the graph meets the horizontal axis has output 0. Such points have the form

$$(x, 0).$$

They correspond to solutions of the equation

$$f(x) = 0.$$

These are called zeros or roots of the function.

Example

Let

$$f(x) = x^2 - 4.$$

The y-intercept is found by computing

$$f(0) = 0^2 - 4 = -4.$$

So the graph meets the y-axis at

$$(0, -4).$$

The zeros are found from

$$x^2 - 4 = 0,$$

so

$$x = -2 \qquad \text{or} \qquad x = 2.$$

Thus the graph meets the x-axis at

$$(-2, 0) \qquad \text{and} \qquad (2, 0).$$

The graph and the equation give the same information in different forms. The graph shows the intercepts visually. The equation allows us to compute them exactly.

The vertical line test

A graph represents a function of x only if each input x has at most one output. Visually, this means that no vertical line should intersect the graph more than once.

Theorem

A curve in the plane represents the graph of a function of x if every vertical line intersects the curve at most once.

The reason is the definition of a function. One input cannot have two different outputs.

Example

A circle is not the graph of a function of x on its full domain, because many vertical lines meet the circle in two points. For the same value of x, there may be two possible values of y.
This does not mean that circles are not important. It means that a full circle is not the graph of one function $y = f(x)$.

Graphs and domains

The domain of a function is visible in the graph as the set of x-values for which the graph exists. If there is no point above a certain value of x, then that value is not in the domain.

The range is visible as the set of y-values reached by the graph.

Example

For the function

$$f(x) = \sqrt{x},$$

real outputs exist only for $x \geq 0$. Therefore the graph begins at $x = 0$ and continues to the right. The domain is

$$[0, \infty),$$

and the range is also

$$[0, \infty).$$

Summary

When reading a graph, ask:

- What input does the horizontal coordinate represent?
- What output does the vertical coordinate represent?
- What does the point (a, b) say about the value of the function?
- Where does the graph meet the axes?
- What x-values belong to the domain?
- What y-values belong to the range?
- Does the graph pass the vertical line test?

A graph should be read as information, not only as a shape.

9.5 Real functions and basic examples

In this course we will often study functions whose inputs and outputs are real numbers. Such functions are called real functions of a real variable. They are central in calculus because they allow us to describe quantities that depend on one real parameter.

A typical notation is

$$f : D \to \mathbb{R}, \qquad D \subseteq \mathbb{R}.$$

This means that the inputs are taken from a set D of real numbers, and the outputs are real numbers.

Natural domains

Sometimes the domain is given explicitly. Sometimes it is understood from the formula. When a formula is given and no domain is stated, we often look for the largest set of real inputs for which the formula makes sense. This is called the natural domain.

Example

Consider

$$f(x) = 2x + 1.$$

This formula makes sense for every real number x. Its natural domain is

$$\mathbb{R}.$$

Example

Consider
$$g(x) = \frac{1}{x}.$$
This formula does not make sense for $x = 0$, because division by zero is not defined. Therefore the natural domain is
$$\mathbb{R} \setminus \{0\}.$$

Example

Consider
$$h(x) = \sqrt{x}.$$
If we work with real numbers, the expression $\sqrt{x}$ is real only for $x \geq 0$. Therefore the natural domain is
$$[0, \infty).$$

These examples show that the formula is not the whole story. We must also ask where the formula is meaningful.

Linear functions

A linear function has the form
$$f(x) = mx + b.$$
Its graph is a line, provided we view the function as a function of one real variable. The number m is the slope, and b is the value at $x = 0$:
$$f(0) = b.$$

Example

Let
$$f(x) = 3x - 2.$$
Then
$$f(0) = -2, \qquad f(1) = 1, \qquad f(2) = 4.$$
When x increases by 1, the value of the function increases by 3. This constant rate of change is the slope.

A linear function is one of the simplest examples of dependence. The output changes at a constant rate as the input changes.

Quadratic functions

A quadratic function has the form
$$f(x) = ax^2 + bx + c, \qquad a \neq 0.$$
Its graph is a parabola. The simplest example is
$$f(x) = x^2.$$

Example

For

$$f(x) = x^2,$$

we have

$$f(-2) = 4, \qquad f(-1) = 1, \qquad f(0) = 0, \qquad f(1) = 1, \qquad f(2) = 4.$$

The values at -2 and 2 are the same because squaring removes the sign.

Quadratic functions are already more flexible than linear functions. Their graphs can turn around, and their values need not change at a constant rate.

Reciprocal and square root functions

Two basic functions show why domains matter:

$$f(x) = \frac{1}{x}$$

and

$$g(x) = \sqrt{x}.$$

The first excludes $x = 0$. The second excludes negative inputs when we work over the real numbers.

Example

For

$$f(x) = \frac{1}{x},$$

we can compute

$$f(2) = \frac{1}{2}, \qquad f(-4) = -\frac{1}{4}.$$

But $f(0)$ is not defined.

Example

For

$$g(x) = \sqrt{x},$$

we have

$$g(0) = 0, \qquad g(4) = 2, \qquad g(9) = 3.$$

But $g(-1)$ is not a real number.

These examples are elementary, but they teach an important habit: before evaluating a function, check whether the input belongs to the domain.

Piecewise functions

A function may use different formulas on different parts of its domain. Such functions are called piecewise-defined functions.

Example

Define

$$f(x) = \begin{cases} x+1, & x<0, \\ x^2, & x \geq 0. \end{cases}$$

To compute $f(-2)$, we use the first rule because $-2 < 0$:

$$f(-2) = -2 + 1 = -1.$$

To compute $f(3)$, we use the second rule because $3 \geq 0$:

$$f(3) = 3^2 = 9.$$

The input decides which rule applies. A piecewise definition is still a function as long as each allowed input receives exactly one output.

Summary

When studying a real function, ask:

- What is the domain?
- Is the domain stated explicitly, or should I find the natural domain?
- Are there inputs excluded by division by zero, square roots, or other restrictions?
- What type of function am I looking at?
- If the function is piecewise, which rule applies to the given input?

Real functions are the main objects of calculus, but their study begins with reading their domains, formulas, and values carefully.

9.6 Reading information from graphs

A graph can contain a large amount of information, even when we do not know the formula of the function. Reading a graph means extracting that information carefully.

The aim is not to guess. The aim is to connect visible geometric features with statements about the function.

Reading values

If a point (a, b) lies on the graph of f, then

$$f(a) = b.$$

The first coordinate is the input. The second coordinate is the output.

Example

Suppose the graph of f contains the point $(3, -2)$. Then

$$f(3) = -2.$$

If the graph also contains $(-1, 4)$, then

$$f(-1) = 4.$$

This is the most basic skill in reading graphs. It should become automatic: point on graph means input-output pair.

Zeros and signs

A zero of a function is an input where the output is zero. On the graph, zeros are the points where the graph meets the horizontal axis.

If

$$f(a) = 0,$$

then $(a, 0)$ is on the graph.

The sign of $f(x)$ can also be read from the graph. Where the graph lies above the horizontal axis, $f(x) > 0$. Where it lies below the horizontal axis, $f(x) < 0$.

Example

Suppose a graph crosses the horizontal axis at $x = -2$ and $x = 5$. Then

$$f(-2) = 0, \qquad f(5) = 0.$$

If the graph is above the axis between these points, then

$$f(x) > 0 \qquad \text{for } -2 < x < 5.$$

If it is below the axis to the right of 5, then

$$f(x) < 0 \qquad \text{for } x > 5$$

on the visible part of the graph.

A graph therefore answers not only numerical questions, but also qualitative questions about positivity and negativity.

Increasing and decreasing behavior

A function is increasing on an interval if its values go upward as the input moves to the right. It is decreasing if its values go downward as the input moves to the right.

At this stage, we do not need a formal definition with quantifiers. The basic reading is visual: moving from left to right, does the graph rise or fall?

Example

If a graph rises from left to right on the interval $[1, 4]$, then the function is increasing on that interval. This means that larger inputs in that interval give larger function values. If the graph falls from left to right on $[4, 7]$, then the function is decreasing on that interval.

Later, derivatives will give a powerful method for studying increasing and decreasing behavior. But the geometric idea should be understood first.

Maximum and minimum values

A graph can also show where a function has high or low values. A local maximum is a point where the graph is higher than nearby points. A local minimum is a point where the graph is lower than nearby points.

Example

Suppose a graph reaches a visible peak at the point $(2, 5)$. Then near $x = 2$, the function has a local maximum value 5.
If the graph reaches a visible valley at $(-1, -3)$, then near $x = -1$, the function has a local minimum value -3.

The input tells us where the maximum or minimum occurs. The output tells us what the maximum or minimum value is.

Graph reading is not formula guessing

Sometimes a graph is given without a formula. This is not a weakness. A graph can still answer many questions.

From a graph we may read:

- values such as $f(2)$,
- zeros of the function,
- intervals where the function is positive or negative,
- intervals where the function is increasing or decreasing,
- approximate maximum and minimum values,
- the domain and range visible in the picture.

But we should be honest about precision. A graph may show approximate values, while a formula may give exact values. Both forms of information are useful, but they are not the same.

Remark

Reading a graph means translating geometric features into mathematical statements. It does not always mean finding a formula.

Summary

When reading a graph, ask:

- What is the input coordinate?
- What is the output coordinate?
- Where is the graph equal to zero?
- Where is the graph above or below the horizontal axis?
- Where does the graph rise or fall as we move from left to right?
- Where are visible peaks or valleys?
- Am I reading exact information or approximate information?

A graph is a language. Learning to read it is part of learning functions.

9.7 Sequences as functions

Sequences are often introduced as lists of numbers:

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

or

$$2, 4, 6, 8, \dots$$

This way of writing sequences is useful, but it can hide the most important idea. A sequence is not a completely different kind of object from a function. A sequence is a special kind of function.

The difference is in the domain.

Definition

A real sequence is a function

$$a : \mathbb{N} \to \mathbb{R}.$$

For each natural number n, the function assigns a real number $a(n)$. Instead of writing $a(n)$, we usually write

$$a_n.$$

Thus

$$a_n = a(n).$$

The notation changes, but the idea is the same: one input gives one output. For sequences, the inputs are natural numbers. The values are called terms of the sequence.

Terms of a sequence

If $a : \mathbb{N} \to \mathbb{R}$, then

$$a_1, \quad a_2, \quad a_3, \quad \dots$$

are the values of the function at the inputs $1, 2, 3, \dots$. Depending on convention, some sequences start at $n = 0$. What matters is that the starting index must be clear.

Example

Let

$$a_n = 2n + 1, \qquad n \in \mathbb{N}.$$

Then

$$a_1 = 2 \cdot 1 + 1 = 3,$$

$$a_2 = 2 \cdot 2 + 1 = 5,$$

and

$$a_3 = 2 \cdot 3 + 1 = 7.$$

The sequence begins

$$3, 5, 7, 9, \dots$$

The formula $a_n = 2n + 1$ tells us the value of the function a at the natural number n. It is function notation written in sequence language.

Index and value

In a sequence, the input is called the index. The output is called the term. These two roles should not be confused.

In the sequence

$$a_n = 2n + 1,$$

the number n is the index. The number a_n is the term at index n. For example, when $n = 4$,

$$a_4 = 9.$$

The index is 4, and the term is 9.

Remark

The subscript in a_n is not decoration. It records the input of the sequence-function. The expression a_5 means the value of the sequence at index 5.

Graphs of sequences

Since a sequence is a function, it can have a graph. But its graph looks different from the graph of a function defined on an interval. For a sequence $a : \mathbb{N} \to \mathbb{R}$, the graph consists of points

$$(n, a_n), \qquad n \in \mathbb{N}.$$

These points are separated because the inputs are separated.

Example

For the sequence

$$a_n = \frac{1}{n}, \qquad n \geq 1,$$

the graph consists of the points

$$(1, 1), \quad \left(2, \frac{1}{2}\right), \quad \left(3, \frac{1}{3}\right), \quad \left(4, \frac{1}{4}\right), \quad \ldots$$

We do not automatically draw a continuous curve through these points, because the sequence is not defined for every real number between 1 and 2. It is defined only at natural-number inputs.

This is one of the main differences between ordinary real functions on intervals and sequences. A sequence has a discrete domain.

Sequences as ordered data

Because the domain is $\mathbb{N}$, the values of a sequence come with a natural order:

$$a_1, \quad a_2, \quad a_3, \quad \ldots$$

We can ask how the terms change as n increases. Do they grow? Do they decrease? Do they approach a number? Do they repeat a pattern?

These questions are function questions, but adapted to the special domain $\mathbb{N}$.

Summary

When reading a sequence, ask:

- What is the starting index?
- What is the domain of the sequence?
- What is the formula or rule for a_n?
- What are the first few terms?
- What is the difference between the index n and the value a_n?
- Should the graph be read as separated points rather than a continuous curve?

A sequence is a function with discrete input. This simple fact explains both why sequences belong in a chapter on functions and why they are studied with some different questions.

9.8 Explicit and recursive sequences

A sequence is a function on natural-number inputs. To define such a function, we must say how its terms are obtained. There are two common ways to do this: an explicit formula and a recursive definition.

Both are useful, but they answer different practical questions.

Explicit formulas

An explicit formula gives a_n directly in terms of n. If we want the tenth term, we substitute $n = 10$. If we want the hundredth term, we substitute $n = 100$.

Definition

An explicit formula for a sequence gives the term a_n directly as an expression depending on the index n.

Example

Let

$$a_n = 3n - 2, \qquad n \geq 1.$$

Then

$$a_1 = 3 \cdot 1 - 2 = 1,$$
$$a_2 = 3 \cdot 2 - 2 = 4,$$
$$a_3 = 3 \cdot 3 - 2 = 7.$$

The sequence begins

$$1,\ 4,\ 7,\ 10,\ \ldots$$

If we want the tenth term, we compute

$$a_{10} = 3 \cdot 10 - 2 = 28.$$

The advantage of an explicit formula is direct access. We can compute a distant term without computing all previous terms.

Recursive definitions

A recursive definition describes how to obtain the next term from earlier terms. Such a definition needs a starting value. Without a starting value, the sequence is not determined.

Definition

A recursive definition of a sequence gives one or more initial terms and a rule for computing later terms from earlier terms.

Example

Define a sequence by

$$a_1 = 5, \qquad a_{n+1} = a_n + 2.$$

Then

$$a_1 = 5,$$
$$a_2 = a_1 + 2 = 7,$$
$$a_3 = a_2 + 2 = 9,$$
$$a_4 = a_3 + 2 = 11.$$

The sequence begins

$$5,\ 7,\ 9,\ 11,\ \ldots$$

The recursive rule says how to move from one term to the next. It does not immediately give a_{100} unless we either compute many previous terms or find an explicit formula.

Why the initial term matters

The recurrence relation alone may not define one sequence. It may define a whole family of possible sequences, depending on the starting value.

Example

The rule

$$a_{n+1} = a_n + 2$$

does not determine a unique sequence by itself.
If $a_1 = 1$, the sequence begins

$$1,\ 3,\ 5,\ 7,\ \ldots$$

If $a_1 = 10$, the sequence begins

$$10,\ 12,\ 14,\ 16,\ \ldots$$

The recursive rule is the same, but the sequences are different because the starting value is different.

This is similar to describing motion. A rule for how to move is not enough if we do not know where we start.

Geometric-type recursion

Some recursions multiply instead of add.

Example

Define

$$a_1 = 3, \qquad a_{n+1} = 2a_n.$$

Then

$$\begin{aligned} a_1 &= 3, \\ a_2 &= 2a_1 = 6, \\ a_3 &= 2a_2 = 12, \\ a_4 &= 2a_3 = 24. \end{aligned}$$

The sequence begins

$$3,\ 6,\ 12,\ 24,\ \ldots$$

Each term is twice the previous one.

The rule is simple, but the meaning is important. The next value depends on the current value. The sequence is built step by step.

Reading recursive notation

The expression a_{n+1} means the term after a_n. If $n = 1$, then $a_{n+1} = a_2$. If $n = 2$, then $a_{n+1} = a_3$.

A recurrence relation such as

$$a_{n+1} = a_n + 2$$

should be read as a rule that works repeatedly:

$$a_2 = a_1 + 2, \qquad a_3 = a_2 + 2, \qquad a_4 = a_3 + 2,$$

and so on.

Remark

Recursive notation is compact because one line contains infinitely many instructions. To understand it, expand the first few cases and read what the rule is doing.

Summary

When reading a sequence definition, ask:

- Is the sequence given explicitly or recursively?
- If it is explicit, can I compute a_n directly from n?
- If it is recursive, what is the starting value?
- How is the next term obtained from earlier terms?
- What are the first few terms?
- Does the notation a_{n+1} refer to the next term after a_n?

Explicit and recursive descriptions are two languages for sequences. Each one reveals a different aspect of the same kind of object: a function defined on natural-number inputs.

9.9 Discrete and continuous domains

Sequences and real functions belong to the same general idea: they are functions. But they often have very different domains. This difference changes the kinds of questions we naturally ask.

A function such as

$$f : [0, 1] \to \mathbb{R}$$

has inputs from an interval. Between two different inputs in $[0, 1]$, there are many other inputs. A sequence

$$a : \mathbb{N} \to \mathbb{R}$$

has inputs $1, 2, 3, \ldots$. Between 1 and 2, there is no natural-number input.

Continuous-looking domains

Intervals of real numbers are often called continuous domains in elementary calculus. This word means that the inputs are not separated into isolated points. If an interval contains two numbers, it also contains all real numbers between them.

For example, the interval

$$[0, 1]$$

contains 0, 1, and also

$$\frac{1}{2}, \quad \frac{1}{3}, \quad 0.72, \quad \sqrt{2} - 1,$$

and infinitely many other numbers between 0 and 1.

A function defined on an interval can be studied by asking what happens as x moves gradually through the domain. This is why graphs of such functions are often drawn as curves.

Discrete domains

The set of natural numbers is different:

$$\mathbb{N} = \{1, 2, 3, \ldots\}$$

or, in some conventions,

$$\mathbb{N} = \{0, 1, 2, 3, \ldots\}.$$

The exact convention should be stated when it matters. In either case, the natural numbers are separated. There is a next natural number after 1, and there is no natural number halfway between 1 and 2.

This is why a sequence graph consists of separated points:

$$(n, a_n), \qquad n \in \mathbb{N}.$$

The sequence is not automatically defined at $n = 1.5$, because $1.5 \notin \mathbb{N}$.

Example

Let

$$a_n = n^2, \qquad n \in \mathbb{N}.$$

Then

$$a_1 = 1, \qquad a_2 = 4, \qquad a_3 = 9.$$

The expression $a_{1.5}$ is not part of this sequence, because 1.5 is not a natural-number index.

This does not mean sequences are less important. It means that their input structure is different.

Why this changes the questions

For functions defined on intervals, we can ask how the function behaves near a point from both sides. We can ask whether a graph has a break, whether values approach a limit as $x \to a$, or whether the function is continuous at a.

For sequences, the natural local picture is different. The inputs jump from one natural number to the next. Therefore, in an introductory course, we do not usually study continuity of a sequence at a natural-number input in the same way we study continuity of a real function on an interval.

Instead, for sequences we often ask questions such as:

- What happens to a_n as $n \to \infty$?
- Is the sequence increasing or decreasing?
- Is the sequence bounded?
- Does the sequence approach a number?
- Is the sequence defined recursively or explicitly?

The object is still a function. The domain changes the natural questions.

Remark

A sequence is not outside the world of functions. It is a function with a discrete domain. Because the domain is discrete, we study it with questions adapted to discrete input.

Drawing responsibly

When drawing a function defined on an interval, it may be appropriate to draw a continuous curve if the function behaves continuously. When drawing a sequence, we should usually draw separated points.

Connecting sequence points with line segments may be useful as a visual guide, but it can also be misleading. The line segment between $(1, a_1)$ and $(2, a_2)$ suggests values at non-natural inputs, but the sequence itself does not provide those values.

Example

For

$$a_n = \frac{1}{n},$$

the points

$$(1,1), \quad \left(2, \frac{1}{2}\right), \quad \left(3, \frac{1}{3}\right)$$

belong to the graph of the sequence. A smooth curve through these points may help the eye see a trend, but the sequence itself is defined only at natural-number inputs.

Summary

When comparing ordinary real functions and sequences, ask:

- What is the domain?
- Are the inputs separated or do they fill an interval?
- Should the graph be drawn as points or as a curve?

- Is the natural question about behavior near a real point or behavior as the index goes to infinity?
- Does the notation hide the fact that a sequence is still a function?

The difference between continuous and discrete domains is not cosmetic. It shapes the way we read, draw, and investigate mathematical objects.

9.10 What questions can we ask?

Once we know that functions and sequences are related, we should also understand why they are not studied in exactly the same way. A sequence is a function, but its domain is usually $\mathbb{N}$. A real function in calculus often has a domain that is an interval or a union of intervals. This changes the natural questions.

The purpose of this section is to organize those questions.

Questions about real functions

For a real function

$$f : D \to \mathbb{R}, \qquad D \subseteq \mathbb{R},$$

we often ask questions connected with its graph, values, and behavior as the input varies through real numbers.

Typical questions include:

- What is the domain?
- What is the range?
- What is $f(a)$ for a given input a?
- For which x do we have $f(x) = 0$?
- Where is the function positive or negative?
- Is the function increasing or decreasing on an interval?
- What happens to $f(x)$ as x approaches a point?
- Is the function continuous?

Not all of these questions are answered in this chapter. Some of them will appear later, especially limits and continuity. But the language needed to ask them begins here.

Questions about sequences

For a sequence

$$a : \mathbb{N} \to \mathbb{R},$$

we ask questions adapted to natural-number inputs.

Typical questions include:

- What is the starting index?
- What are the first few terms?
- Is the sequence given by an explicit formula or recursively?

- Is the sequence increasing or decreasing as n grows?
- Is the sequence bounded above or below?
- What happens to a_n as $n \to \infty$?
- Does the sequence approach a limiting value?

The question about $n \to \infty$ is especially important. Since the inputs of a sequence continue without end, we often want to know the long-term behavior of the terms.

A comparison

Object	Typical domain	Natural questions
Real function	interval or subset of $\mathbb{R}$	graph, zeros, signs, continuity, limits as $x \to a$
Sequence	$\mathbb{N}$	terms, recursion, monotonicity, boundedness, limits as $n \to \infty$

This table is not meant to separate functions and sequences completely. It is meant to show that different domains lead to different habits of investigation.

The common idea

Despite these differences, the common idea remains the same. A function assigns outputs to inputs. A sequence also assigns outputs to inputs. The input set is special, but the structure is still functional.

For a real function, we may write

$$f(x)$$

for the value at input x. For a sequence, we usually write

$$a_n$$

for the value at input n. These notations look different, but they express the same idea: value at an input.

Remark

The notation changes because the situation changes. But the underlying idea remains: one input gives one output.

Preparing for limits

The next chapter will study limits and continuity. The present chapter prepares for that in two ways.

First, it gives the language of functions, values, domains, and graphs. Without that language, the expression

$$\lim_{x \to a} f(x)$$

has no clear meaning.

Second, it introduces sequences as functions on $\mathbb{N}$. This prepares the expression

$$\lim_{n \to \infty} a_n.$$

Both limits involve the idea of approaching. But the input variable approaches in different ways: x may approach a real point through real values, while n goes through natural numbers toward infinity.

Summary

At the end of this chapter, remember:

- A function is an assignment from inputs to outputs.
- The domain is part of the function.
- The notation $f(x)$ means the value of f at x.
- A graph is a picture of input-output pairs.
- A sequence is a function whose domain is usually $\mathbb{N}$.
- Because sequences have discrete domains, we ask different questions about them.
- The type of domain shapes the type of mathematics we do.

Functions and sequences are not two unrelated topics placed side by side. Sequences are a special case of functions, and this viewpoint will help us understand limits, continuity, and later calculus.

Chapter 10

Limits and Continuity

10.1 What a limit tries to describe

Limits are one of the central ideas of calculus. They are also one of the first ideas that require a new way of reading mathematical notation. A limit does not ask only what happens at a point. It asks what values are approached near a point, or far along a sequence.

This distinction is essential. The value of a function at a point and the limit of a function at that point are related, but they are not the same question.

Approaching is not the same as arriving

Consider a function f and a point a. The expression

$$f(a)$$

asks for the value of the function at the input a. The expression

$$\lim_{x \to a} f(x)$$

asks what value $f(x)$ approaches when x approaches a.

The variable $x \to a$ means that x gets close to a. It does not mean that $x = a$. This is why a limit can exist even when the function is not defined at the point.

Example

Consider

$$f(x) = \frac{x^2 - 1}{x - 1}, \qquad x \neq 1.$$

The function is not defined at $x = 1$, because the denominator becomes zero. But for $x \neq 1$, we can factor:

$$\frac{x^2 - 1}{x - 1} = \frac{(x-1)(x+1)}{x - 1} = x + 1.$$

When x approaches 1, the expression $x + 1$ approaches 2. Therefore

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = 2.$$

The limit exists even though $f(1)$ is not defined.

This example should be read as a lesson about the question being asked. The limit does not ask what happens after substituting $x = 1$. It asks what values are approached when x gets close to 1.

Limits of sequences and limits of functions

There are two main limit situations in this chapter.

For a sequence (a_n), we ask what happens to the terms when the index grows:

$$\lim_{n \to \infty} a_n.$$

Here the input is discrete. The index n runs through natural numbers.

For a function $f(x)$, we may ask what happens when x approaches a real number a:

$$\lim_{x \to a} f(x),$$

or when x grows without bound:

$$\lim_{x \to \infty} f(x).$$

The notation is different because the input situation is different. But the common idea is the same: we study what values are approached.

What a limit can tell us

Limits help us describe behavior that may not be visible from one single value. They allow us to discuss:

- long-term behavior of sequences,
- behavior of functions near a point,
- holes in graphs,
- vertical and horizontal asymptotes,
- continuity,
- the idea of a derivative in the next chapter.

A limit is therefore not just another calculation. It is a way of reading behavior.

Summary

When first reading a limit, ask:

- What object is being studied: a sequence or a function?
- What input is approaching something?
- Is the input approaching a finite point or infinity?
- Am I being asked about the value at the point or behavior near the point?
- Does direct substitution answer the question, or does it reveal a difficulty?

The main habit is this: before calculating a limit, read what kind of approaching is being described.

10.2 Limits of sequences

A sequence is a function whose input is usually a natural number. Therefore the most natural limiting question for a sequence is not what happens near a finite real input. The natural question is what happens when the index becomes larger and larger.

If (a_n) is a sequence, we write

$$\lim_{n \to \infty} a_n = L$$

when the terms of the sequence approach the number L as n grows.

Definition

We say that a sequence (a_n) converges to L if the terms a_n become arbitrarily close to L for sufficiently large n. We write

$$\lim_{n\to\infty} a_n = L.$$

If no finite limit exists, we say that the sequence diverges.

The phrase "for sufficiently large n" is important. A sequence may behave irregularly at the beginning and still have a limit. Limits describe long-term behavior, not the first few terms.

A first convergent sequence

Consider

$$a_n = \frac{1}{n}.$$

The first terms are

$$1, \quad \frac{1}{2}, \quad \frac{1}{3}, \quad \frac{1}{4}, \quad \ldots$$

These terms get closer and closer to 0. Therefore

$$\lim_{n\to\infty} \frac{1}{n} = 0.$$

The sequence never actually reaches 0, because $\frac{1}{n} > 0$ for every natural number n. But reaching the limit is not required. The question is whether the terms approach the limit.

Remark

A limit is about approaching, not necessarily about becoming equal. The terms $1/n$ approach 0, even though no term of the sequence is equal to 0.

Divergence to infinity

Some sequences do not approach a finite number. For example,

$$a_n = n$$

grows without bound. We write informally

$$\lim_{n\to\infty} n = \infty,$$

meaning that the terms become larger than any fixed bound. This is not convergence to a real number. It is divergence to infinity.

Example

The sequence

$$a_n = n^2$$

has terms

$$1, 4, 9, 16, \ldots$$

The terms grow without bound. The sequence does not converge to a finite real number.

Oscillation

A sequence may also fail to converge because it oscillates.

Example

Consider

$$a_n = (-1)^n.$$

The terms are

$$-1, 1, -1, 1, \dots$$

They do not approach one number. The sequence alternates forever between two values. Therefore the sequence diverges.

Oscillation does not automatically prevent convergence. If the oscillation becomes smaller, a limit may still exist.

Example

Consider

$$a_n = \frac{(-1)^n}{n}.$$

The sign alternates, but the size of the terms becomes smaller:

$$\left|\frac{(-1)^n}{n}\right| = \frac{1}{n} \to 0.$$

Therefore

$$\lim_{n\to\infty} \frac{(-1)^n}{n} = 0.$$

The difference between these two examples is the size of the oscillation. In $(-1)^n$, the oscillation never becomes small. In $(-1)^n/n$, the oscillation is squeezed toward 0.

Reading sequence limits

When reading a sequence limit, we should identify the long-term structure of the formula. Does the numerator and denominator grow at comparable rates? Does a power dominate? Does a term go to zero? Does an alternating sign matter?

Example

Consider

$$a_n = \frac{n}{n+1}.$$

For large n, the numerator and denominator have almost the same size. Divide numerator and denominator by n:

$$\frac{n}{n+1} = \frac{1}{1+\frac{1}{n}}.$$

Since $\frac{1}{n} \to 0$, we get

$$\lim_{n\to\infty} \frac{n}{n+1} = 1.$$

This calculation should be understood conceptually: the extra 1 in the denominator becomes insignificant compared with n when n is large.

Summary

When studying a sequence limit, ask:

- What happens when n becomes large?
- Are the terms approaching a finite number?
- Are the terms growing without bound?
- Is there oscillation?
- If there is oscillation, does its size shrink?
- Which part of the formula dominates for large n?

A sequence limit describes long-term behavior. The first few terms may help us guess, but the limit is determined by what happens as n grows without bound.

10.3 Basic techniques for sequence limits

Many sequence limits can be found by reading which terms dominate when n becomes large. The goal is not to memorize many separate tricks. The goal is to learn how to simplify a formula so that its long-term behavior becomes visible.

Dividing by the highest power

For rational expressions in n, a common method is to divide numerator and denominator by the highest power of n that appears in the denominator or numerator.

Example

Compute
$$\lim_{n\to\infty} \frac{4n-3}{6-5n}.$$
Divide numerator and denominator by n:
$$\frac{4n-3}{6-5n} = \frac{4-\frac{3}{n}}{\frac{6}{n}-5}.$$
Since
$$\frac{3}{n} \to 0, \qquad \frac{6}{n} \to 0,$$
we get
$$\lim_{n\to\infty} \frac{4n-3}{6-5n} = \frac{4}{-5} = -\frac{4}{5}.$$

The method works because lower-order terms become small compared with the highest power of n.

Example

Compute
$$\lim_{n\to\infty} \frac{n^2-1}{3-n^3}.$$

The denominator has degree 3, while the numerator has degree 2. Divide by n^3:

$$\frac{n^2-1}{3-n^3} = \frac{\frac{1}{n}-\frac{1}{n^3}}{\frac{3}{n^3}-1}.$$

The numerator tends to 0, and the denominator tends to -1. Hence

$$\lim_{n\to\infty} \frac{n^2-1}{3-n^3} = 0.$$

This example shows that when the denominator grows faster than the numerator, the quotient may tend to zero.

Using dominant terms

Sometimes we can read the answer by comparing leading terms. For example,

$$\frac{2n^3-4n-1}{6n+3n^2-n^3}$$

has leading term $2n^3$ in the numerator and $-n^3$ in the denominator. Therefore the limit should be

$$\frac{2}{-1} = -2.$$

A careful calculation confirms this:

$$\frac{2n^3-4n-1}{6n+3n^2-n^3} = \frac{2-\frac{4}{n^2}-\frac{1}{n^3}}{\frac{6}{n^2}+\frac{3}{n}-1} o - 2.$$

The phrase "dominant term" means the term that grows fastest as $n \to \infty$.

Roots and rationalization

Expressions with square roots often need to be rewritten before the limit is visible. A standard method is to multiply by the conjugate.

Example

Compute

$$\lim_{n\to\infty} \left(\sqrt{n^2+n}-n\right).$$

Direct reading is not enough because both terms grow without bound. Multiply by the conjugate:

$$\sqrt{n^2+n}-n = \frac{(\sqrt{n^2+n}-n)(\sqrt{n^2+n}+n)}{\sqrt{n^2+n}+n}.$$

The numerator becomes

$$(n^2+n)-n^2 = n.$$

Thus

$$\sqrt{n^2+n}-n = \frac{n}{\sqrt{n^2+n}+n}.$$

Factor n from the square root:

$$\frac{n}{n\sqrt{1+\frac{1}{n}}+n} = \frac{1}{\sqrt{1+\frac{1}{n}}+1}.$$

Therefore

$$\lim_{n\to\infty}\left(\sqrt{n^2+n}-n\right)=\frac{1}{2}.$$

The conjugate method changes a difference of large quantities into a quotient whose behavior is easier to read.

Alternating signs

Terms such as $(-1)^n$ should be read carefully. They may cause oscillation, but if they are multiplied by something tending to zero, the whole sequence may still converge.

Example

Compute

$$\lim_{n\to\infty}\frac{2n+(-1)^n}{n}.$$

Rewrite as

$$\frac{2n+(-1)^n}{n}=2+\frac{(-1)^n}{n}.$$

The term $\frac{(-1)^n}{n}$ tends to 0, because its absolute value is $1/n$. Therefore

$$\lim_{n\to\infty}\frac{2n+(-1)^n}{n}=2.$$

The alternating part does not disappear because signs are unimportant. It disappears because its size tends to zero.

Powers and roots

For expressions involving powers, we compare growth rates. If $|q|<1$, then

$$q^n\to 0.$$

If $q>1$, then q^n grows without bound.

Example

Compute

$$\lim_{n\to\infty}\frac{4^n-1}{2^{2n}-7}.$$

Since

$$2^{2n}=4^n,$$

we have

$$\frac{4^n-1}{2^{2n}-7}=\frac{4^n-1}{4^n-7}.$$

Divide by 4^n:

$$\frac{1-\frac{1}{4^n}}{1-\frac{7}{4^n}}\to\frac{1}{1}=1.$$

Summary

Useful habits for sequence limits:

- For rational expressions, divide by the dominant power of n.
- For square-root differences, consider multiplying by the conjugate.
- For alternating signs, check whether the size of the alternating term tends to zero.
- For powers, compare the dominant exponential terms.
- After transforming the expression, interpret which terms vanish and which remain.

Techniques are useful only when connected with meaning. Each transformation should make the long-term behavior easier to see.

10.4 Limits of functions

A function limit describes what happens to the values $f(x)$ when the input x approaches a certain point. The point may or may not belong to the domain of the function. What matters for the limit is the behavior near the point.

We write

$$\lim_{x \to a} f(x) = L$$

when the values $f(x)$ approach L as x approaches a.

Definition

The notation

$$\lim_{x \to a} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a, with $x \neq a$.

The condition $x \neq a$ is not a technicality. A limit studies nearby values. The value at the point itself is a separate question.

Direct substitution

Sometimes the limit can be found by direct substitution. This often happens for polynomials and many simple functions.

Example

Compute

$$\lim_{x \to 3} (x^2 + 2x - 1).$$

The expression is defined and behaves regularly near $x = 3$, so we substitute:

$$3^2 + 2 \cdot 3 - 1 = 9 + 6 - 1 = 14.$$

Therefore

$$\lim_{x \to 3} (x^2 + 2x - 1) = 14.$$

Direct substitution is not a rule to use blindly. It is a first test. If it gives a meaningful value and the function behaves regularly, it may solve the problem. If it produces a form such as 0/0, more work is needed.

A removable difficulty

Consider

$$\lim_{x\to 3}\frac{x^2-4x+3}{x-3}.$$

Direct substitution gives

$$\frac{9-12+3}{0}=\frac{0}{0},$$

which is not a number. This form tells us that direct substitution has failed. It does not tell us that the limit does not exist.

Factor the numerator:

$$x^2-4x+3=(x-1)(x-3).$$

For $x\neq 3$,

$$\frac{x^2-4x+3}{x-3}=x-1.$$

Therefore

$$\lim_{x\to 3}\frac{x^2-4x+3}{x-3}=\lim_{x\to 3}(x-1)=2.$$

The original expression is not defined at $x=3$, but near 3 it behaves like $x-1$. The limit reads the nearby behavior.

The meaning of $0/0$

The expression $0/0$ is called an indeterminate form. It means that the information from direct substitution is insufficient. It is not an answer.

The limit may exist, may fail to exist, or may be infinite. We must transform the expression or analyze it in another way.

Example

The limit

$$\lim_{x\to 1}\frac{x^2-1}{x-1}$$

gives $0/0$ by direct substitution. After factoring,

$$\frac{x^2-1}{x-1}=x+1$$

for $x\neq 1$, so the limit is 2.

But

$$\lim_{x\to 0}\frac{|x|}{x}$$

also cannot be found by direct substitution, and the two-sided limit does not exist because the left and right behaviors are different.

This is why we must read the situation before choosing a technique.

Limits and graphs

On a graph, the limit $\lim_{x\to a} f(x)$ asks what height the graph approaches near $x = a$. A hole in the graph may not prevent a limit from existing. A jump may prevent a two-sided limit from existing.

Remark

The value $f(a)$ is the height of the graph at $x = a$, if that point exists. The limit $\lim_{x\to a} f(x)$ is the height approached by nearby points of the graph.

Summary

When computing a function limit, ask:

- What point is x approaching?
- Is the function defined at nearby points?
- Does direct substitution give a meaningful value?
- If direct substitution gives $0/0$, can the expression be simplified?
- Is the left-hand behavior the same as the right-hand behavior?
- Does the graph suggest a hole, a jump, or an asymptote?

A function limit is a statement about local behavior. The point itself matters for continuity, but the limit first asks what happens nearby.

10.5 One-sided limits

A two-sided limit asks what happens as x approaches a point from both sides. Sometimes the behavior from the left and from the right is different. To describe this, we use one-sided limits.

The left-hand limit is written as

$$\lim_{x\to a^-} f(x).$$

It describes what happens when x approaches a through values smaller than a.

The right-hand limit is written as

$$\lim_{x\to a^+} f(x).$$

It describes what happens when x approaches a through values greater than a.

When the two-sided limit exists

A two-sided limit exists exactly when the left-hand and right-hand limits both exist and are equal.

Theorem

The limit

$$\lim_{x\to a} f(x) = L$$

exists if and only if

$$\lim_{x\to a^-} f(x) = L \qquad \text{and} \qquad \lim_{x\to a^+} f(x) = L.$$

If the two one-sided limits are different, then the two-sided limit does not exist.

Example

Consider

$$f(x) = \frac{|x|}{x}, \qquad x \neq 0.$$

If $x > 0$, then $|x| = x$, so

$$f(x) = 1.$$

If $x < 0$, then $|x| = -x$, so

$$f(x) = -1.$$

Therefore

$$\lim_{x \to 0^-} \frac{|x|}{x} = -1, \qquad \lim_{x \to 0^+} \frac{|x|}{x} = 1.$$

The one-sided limits are different, so

$$\lim_{x \to 0} \frac{|x|}{x}$$

does not exist.

This example shows why looking from both sides matters. A formula may look simple, but its behavior can depend on the direction of approach.

Piecewise functions

One-sided limits are especially important for piecewise-defined functions. Each side of the point may use a different formula.

Example

Let

$$f(x) = \begin{cases} x + 2, & x < 1, \\ 3x, & x \geq 1. \end{cases}$$

The left-hand limit at 1 uses the formula $x + 2$:

$$\lim_{x \to 1^-} f(x) = 1 + 2 = 3.$$

The right-hand limit at 1 uses the formula $3x$:

$$\lim_{x \to 1^+} f(x) = 3 \cdot 1 = 3.$$

The two one-sided limits are equal, so

$$\lim_{x \to 1} f(x) = 3.$$

Also, because $f(1) = 3 \cdot 1 = 3$, the function is continuous at 1. We will define continuity later in this chapter.

Example

Let

$$g(x) = \begin{cases} x+2, & x<1, \\ 3x+1, & x \geq 1. \end{cases}$$

Then

$$\lim_{x \to 1^-} g(x) = 3,$$

but

$$\lim_{x \to 1^+} g(x) = 4.$$

Since the one-sided limits are different,

$$\lim_{x \to 1} g(x)$$

does not exist.

The point of the calculation is not only to obtain numbers. The calculation tells us whether the two sides of the graph meet at the same height.

Reading from a graph

On a graph, the left-hand limit is the height approached as we move toward the point from the left. The right-hand limit is the height approached as we move toward the point from the right.

If the graph approaches the same height from both sides, the two-sided limit exists. If the graph approaches different heights, there is a jump.

Summary

When studying one-sided limits, ask:

- What happens when x approaches from the left?
- What happens when x approaches from the right?
- Are the two one-sided limits equal?
- Does the formula change on different sides of the point?
- Does the graph show a jump or a meeting point?

One-sided limits are the language we need when behavior depends on direction.

10.6 Limits at infinity

Some limits describe behavior near a finite point. Other limits describe behavior far away. When we write

$$\lim_{x \to \infty} f(x),$$

we ask what happens to $f(x)$ as x becomes larger and larger. When we write

$$\lim_{x \to -\infty} f(x),$$

we ask what happens as x becomes very large in the negative direction.

This is closely related to sequence limits, where we ask what happens as $n \to \infty$. The difference is that n runs through natural numbers, while x runs through real values.

A first example

Consider

$$f(x) = \frac{1}{x}.$$

As x becomes very large and positive, $1/x$ becomes close to 0. Therefore

$$\lim_{x\to\infty} \frac{1}{x} = 0.$$

As x becomes very large and negative, $1/x$ also becomes close to 0. Therefore

$$\lim_{x\to-\infty} \frac{1}{x} = 0.$$

The sign is different on the two sides, but the values still approach the same number 0.

Rational functions

For rational functions, limits at infinity are often found by comparing the highest powers of x.

Example

Compute

$$\lim_{x\to\infty} \frac{3x^2 - 5x + 1}{2x^2 + 7}.$$

Divide numerator and denominator by x^2:

$$\frac{3x^2 - 5x + 1}{2x^2 + 7} = \frac{3 - \frac{5}{x} + \frac{1}{x^2}}{2 + \frac{7}{x^2}}.$$

As $x \to \infty$, the terms $1/x$ and $1/x^2$ tend to 0. Hence

$$\lim_{x\to\infty} \frac{3x^2 - 5x + 1}{2x^2 + 7} = \frac{3}{2}.$$

This is the same dominant-term idea used for sequence limits. The largest powers determine the long-term behavior.

Example

Compute

$$\lim_{x\to\infty} \frac{2x + 1}{x^2 + 3}.$$

The denominator grows faster than the numerator. Divide by x^2:

$$\frac{2x + 1}{x^2 + 3} = \frac{\frac{2}{x} + \frac{1}{x^2}}{1 + \frac{3}{x^2}}.$$

The numerator tends to 0, and the denominator tends to 1. Therefore

$$\lim_{x\to\infty} \frac{2x + 1}{x^2 + 3} = 0.$$

Horizontal asymptotes

If

$$\lim_{x\to\infty} f(x) = L$$

or

$$\lim_{x \to -\infty} f(x) = L,$$

then the graph approaches the horizontal line

$$y = L$$

in that direction. This line is called a horizontal asymptote.

Example

For

$$f(x) = \frac{3x^2 - 5x + 1}{2x^2 + 7},$$

we found

$$\lim_{x \to \infty} f(x) = \frac{3}{2}.$$

Also,

$$\lim_{x \to -\infty} f(x) = \frac{3}{2}.$$

Thus the graph has horizontal asymptote

$$y = \frac{3}{2}.$$

The asymptote is not necessarily a line the graph never touches. It is a line the graph approaches far away.

Infinite limits

Sometimes a function grows without bound as $x \to \infty$. For example,

$$\lim_{x \to \infty} x^2 = \infty.$$

This means the values become larger than any fixed number. It is not convergence to a real number.

Similarly,

$$\lim_{x \to -\infty} x^3 = -\infty.$$

The values become negative with arbitrarily large magnitude.

Summary

When studying limits at infinity, ask:

- Is the input going to ∞ or $-\infty$?
- Which terms dominate for large $|x|$?
- Does the function approach a finite number?
- Does it grow without bound?
- Is there a horizontal asymptote?
- Is the behavior different for $x \to \infty$ and $x \to -\infty$?

Limits at infinity describe long-term behavior. They tell us what the function does far from the origin.

10.7 Algebra of limits and indeterminate forms

Limit computations often use algebra. But algebra should be used with understanding. Some transformations preserve the behavior near a point; others may accidentally change the problem. The guiding question should always be: what happens to the expression as the input approaches the limiting value?

Basic algebra of limits

If two limits exist,

$$\lim_{x\to a} f(x) = L, \qquad \lim_{x\to a} g(x) = M,$$

then we can combine them in predictable ways:

$$\lim_{x\to a}(f(x)+g(x)) = L+M,$$

$$\lim_{x\to a}(f(x)g(x)) = LM,$$

and, if $M \neq 0$,

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{L}{M}.$$

These rules are useful when the individual limits are known and the resulting expression is meaningful.

Example

Compute

$$\lim_{x\to 2}(x^2+3x-1).$$

Using direct substitution and algebra of limits,

$$\lim_{x\to 2}(x^2+3x-1) = 2^2+3\cdot 2-1 = 9.$$

For polynomials, this direct substitution works because polynomial functions are continuous everywhere. Rational functions behave similarly at points where the denominator is not zero.

When direct substitution fails

Direct substitution may produce expressions such as

$$\frac{0}{0}, \qquad \frac{\infty}{\infty}, \qquad 0\cdot\infty, \qquad \infty-\infty.$$

These are called indeterminate forms. They are not answers. They indicate that the expression needs to be transformed or analyzed more carefully.

Remark

An indeterminate form means that competing behaviors are present. The expression may have a finite limit, an infinite limit, or no limit. The form itself does not decide the answer.

Factoring

Factoring is often used when direct substitution gives $0/0$.

Example

Compute

$$\lim_{x\to 3} \frac{x^2-4x+3}{2x-6}.$$

Substitution gives $0/0$. Factor:

$$x^2-4x+3=(x-1)(x-3), \qquad 2x-6=2(x-3).$$

For $x\neq 3$,

$$\frac{x^2-4x+3}{2x-6}=\frac{(x-1)(x-3)}{2(x-3)}=\frac{x-1}{2}.$$

Therefore

$$\lim_{x\to 3} \frac{x^2-4x+3}{2x-6}=\frac{3-1}{2}=1.$$

The cancellation is valid for $x\neq 3$. This is enough for the limit, because the limit studies values near 3, not necessarily at 3.

Rationalizing

Expressions with radicals may require multiplication by the conjugate.

Example

Compute

$$\lim_{x\to 25} \frac{\sqrt{x}-5}{x-25}.$$

Substitution gives $0/0$. Multiply by the conjugate:

$$\frac{\sqrt{x}-5}{x-25}\cdot\frac{\sqrt{x}+5}{\sqrt{x}+5}=\frac{x-25}{(x-25)(\sqrt{x}+5)}.$$

For $x\neq 25$, this becomes

$$\frac{1}{\sqrt{x}+5}.$$

Therefore

$$\lim_{x\to 25} \frac{\sqrt{x}-5}{x-25}=\frac{1}{5+5}=\frac{1}{10}.$$

Rationalizing is not a trick for its own sake. It removes a difference of square roots that hides the limiting behavior.

Rewriting before deciding

A common mistake is to see $0/0$ and conclude that the limit is zero or undefined. This is incorrect. The form $0/0$ tells us that direct substitution has not answered the question.

Summary

When direct substitution gives an indeterminate form, ask:

- Can I factor the numerator or denominator?
- Is there a common factor that cancels for nearby inputs?

- Is there a square root expression that can be rationalized?
- Can I divide by a dominant power for limits at infinity?
- After rewriting, what expression describes the same nearby behavior?

The algebra of limits is useful because it helps us reveal behavior. The algebra should serve the limit, not replace the meaning of the limit.

10.8 Standard limits

Some limits occur so often that they become standard tools. They should not be treated as isolated formulas to memorize without meaning. Each standard limit captures a basic local behavior of an important function.

The basic trigonometric limit

One of the most important limits is

$$\lim_{x \to 0} \frac{\sin x}{x} = 1,$$

where x is measured in radians.

This limit says that, near zero, $\sin x$ behaves like x. The functions are not equal, but their ratio approaches 1.

Example

Compute

$$\lim_{x \to 0} \frac{\sin 3x}{4x}.$$

Rewrite the expression to use the standard limit:

$$\frac{\sin 3x}{4x} = \frac{3}{4} \cdot \frac{\sin 3x}{3x}.$$

As $x \to 0$, also $3x \to 0$, so

$$\frac{\sin 3x}{3x} \to 1.$$

Therefore

$$\lim_{x \to 0} \frac{\sin 3x}{4x} = \frac{3}{4}.$$

The rewriting is the main step. We reshape the expression until the standard limit becomes visible.

Tangent and arcsine type limits

Since

$$\tan x = \frac{\sin x}{\cos x},$$

and $\cos x \to 1$ as $x \to 0$, we also have

$$\lim_{x \to 0} \frac{\tan x}{x} = 1.$$

Similarly, in elementary calculus we use

$$\lim_{x \to 0} \frac{\arctan x}{x} = 1.$$

These limits express the fact that near zero, these functions are locally close to the identity function.

Example

Compute
$$\lim_{x\to 0}\frac{\tan 2x}{\tan x}.$$
Rewrite as
$$\frac{\tan 2x}{\tan x}=2\cdot\frac{\tan 2x}{2x}\cdot\frac{x}{\tan x}.$$
As $x\to 0$, both fractions tend to 1. Therefore
$$\lim_{x\to 0}\frac{\tan 2x}{\tan x}=2.$$

The exponential standard limit

Another standard limit is
$$\lim_{x\to 0}(1+x)^{1/x}=e.$$
More generally, if $u\to 0$, then
$$(1+u)^{1/u}\to e.$$
This limit is the basis for many expressions involving variable exponents.

Example

Compute
$$\lim_{x\to 0}(1-3x)^{1/x}.$$
Let
$$u=-3x.$$
Then $u\to 0$ as $x\to 0$, and
$$\frac{1}{x}=\frac{-3}{u}.$$
Therefore
$$(1-3x)^{1/x}=(1+u)^{-3/u}=\left((1+u)^{1/u}\right)^{-3}.$$
Taking the limit gives
$$e^{-3}.$$

This example shows why substitution is useful: it changes the expression into a recognizable standard form.

Standard limits as local models

The standard limits tell us how certain functions behave near a point. Near zero,
$$\sin x\sim x,\qquad \tan x\sim x,\qquad \arctan x\sim x.$$
The symbol $\sim$ means that the ratio of the two expressions tends to 1. This notation should be used carefully, but the idea is helpful: near zero, these functions can often be compared to simpler expressions.

Summary

Useful standard limits include:

- $\lim_{x \to 0} \frac{\sin x}{x} = 1,$
- $\lim_{x \to 0} \frac{\tan x}{x} = 1,$
- $\lim_{x \to 0} \frac{\arctan x}{x} = 1,$
- $\lim_{x \to 0} (1 + x)^{1/x} = e.$

Standard limits are not a substitute for understanding. They are reliable local models that help us simplify common limiting situations.

10.9 Continuity at a point

Continuity connects the value of a function at a point with the behavior of the function near that point. A function is continuous at a point when there is no break between the nearby behavior and the value at the point.

The definition uses limits.

Definition

A function f is continuous at a point a if all three conditions hold:

1. $f(a)$ is defined,
2. $\lim_{x \to a} f(x)$ exists,
3. $\lim_{x \to a} f(x) = f(a)$.

Each condition matters. If the value is missing, the function cannot be continuous at the point. If the limit does not exist, there is no single nearby behavior. If the limit exists but differs from the value, the graph has a mismatch at the point.

A continuous example

Example

Let

$$f(x) = x^2 + 1.$$

Check continuity at $a = 2$.
First,

$$f(2) = 2^2 + 1 = 5.$$

Second,

$$\lim_{x \to 2} (x^2 + 1) = 5.$$

The limit exists and equals the value of the function. Therefore f is continuous at 2.

Polynomials are continuous at every real point. This is why direct substitution works for polynomial limits.

A removable discontinuity

A function may fail to be continuous because the value at one point is missing or incorrect, even though the nearby behavior has a clear limit.

Example

Let

$$f(x) = \frac{x^2 - 25}{x + 5}$$

for $x \neq -5$. The function is not defined at -5. For $x \neq -5$, factor:

$$\frac{x^2 - 25}{x + 5} = \frac{(x - 5)(x + 5)}{x + 5} = x - 5.$$

Therefore

$$\lim_{x \to -5} f(x) = -10.$$

If we define

$$f(-5) = -10,$$

then the function becomes continuous at -5. The discontinuity is removable.

The word “removable” means that a suitable value at the point can repair continuity.

When the value is wrong

A function may be defined at a point but still fail to be continuous there.

Example

Define

$$f(x) = \begin{cases} \dfrac{x^2 - 1}{x - 1}, & x \neq 1, \\ 5, & x = 1. \end{cases}$$

For $x \neq 1$,

$$\frac{x^2 - 1}{x - 1} = x + 1.$$

Thus

$$\lim_{x \to 1} f(x) = 2.$$

But

$$f(1) = 5.$$

Since the limit and the value are different, f is not continuous at 1. If we changed the value to $f(1) = 2$, continuity would be restored.

This example shows why continuity is not only about having a value. The value must agree with the nearby behavior.

Continuity on intervals

A function is continuous on an interval if it is continuous at every point of that interval. Many familiar functions have this property on their natural domains.

For example:

- polynomial functions are continuous on $\mathbb{R}$,
- rational functions are continuous wherever the denominator is not zero,
- $\sqrt{x}$ is continuous on $[0, \infty)$,
- trigonometric functions such as $\sin x$ and $\cos x$ are continuous on $\mathbb{R}$.

These facts allow many limits to be computed by substitution, but only at points where the function is continuous.

Summary

To check continuity at a, ask:

- Is $f(a)$ defined?
- Does $\lim_{x \to a} f(x)$ exist?
- Is the limit equal to $f(a)$?
- If continuity fails, is it because of a missing value, a wrong value, or different one-sided behavior?

Continuity means agreement between value and nearby behavior. It is the point where limits become a statement about the function itself.

10.10 Types of discontinuities

When a function is not continuous at a point, the failure can happen in different ways. Classifying the discontinuity helps us understand what kind of break appears in the graph and whether it can be repaired.

Removable discontinuity

A removable discontinuity occurs when the limit exists, but the value at the point is missing or does not agree with the limit.

Example

Let

$$f(x) = \frac{x^2 - 4}{x - 2}, \qquad x \neq 2.$$

For $x \neq 2$,

$$f(x) = x + 2.$$

Therefore

$$\lim_{x \to 2} f(x) = 4.$$

The function is not defined at 2, but if we define $f(2) = 4$, the function becomes continuous there. The discontinuity is removable.

Graphically, a removable discontinuity is often seen as a hole.

Jump discontinuity

A jump discontinuity occurs when the left-hand and right-hand limits exist but are different.

Example

Let

$$f(x) = \begin{cases} 0, & x < 1, \\ 2, & x \geq 1. \end{cases}$$

Then

$$\lim_{x \to 1^-} f(x) = 0, \qquad \lim_{x \to 1^+} f(x) = 2.$$

The one-sided limits are different, so the two-sided limit does not exist. The function has a jump discontinuity at 1.

A jump cannot be repaired by changing only the value at the point. The two sides approach different heights.

Infinite discontinuity

An infinite discontinuity occurs when the function grows without bound near a point. This often happens near vertical asymptotes.

Example

Consider

$$f(x) = \frac{1}{x-2}.$$

As $x \to 2^+$, the denominator is small and positive, so

$$f(x) o \infty.$$

As $x \to 2^-$, the denominator is small and negative, so

$$f(x) o - \infty.$$

The function has an infinite discontinuity at 2, and the line $x = 2$ is a vertical asymptote.

This is different from a removable discontinuity. There is no finite value that can be assigned at $x = 2$ to make the function continuous.

Oscillatory behavior

Some functions fail to have a limit because they oscillate more and more rapidly near a point.

Example

The function

$$f(x) = \sin\frac{1}{x}, \qquad x \neq 0,$$

oscillates infinitely many times as $x \to 0$. It does not approach one number, so

$$\lim_{x \to 0} \sin\frac{1}{x}$$

does not exist.

This kind of example is more advanced than a simple jump, but it teaches an important lesson: a limit requires approaching one value.

Summary

Common types of discontinuities:

- Removable: the limit exists, but the value is missing or incorrect.
- Jump: the left-hand and right-hand limits are different.
- Infinite: the function grows without bound near the point.
- Oscillatory: the function fails to approach one value because of persistent oscillation.

A discontinuity is not merely a place where a formula looks difficult. It is a specific kind of failure of continuity.

10.11 Reading limits from graphs

Graphs are one of the best ways to understand limits, provided we read them carefully. A graph can show whether values approach one height, whether there is a hole, whether there is a jump, or whether the function grows without bound.

The graph should be read as information about nearby behavior.

Value versus limit on a graph

The value $f(a)$ is represented by the point of the graph at $x = a$, if such a point is included. The limit $\lim_{x\to a} f(x)$ is represented by the height approached by the graph near $x = a$.

These may be the same, but they do not have to be the same.

Example

Suppose a graph has a hole at $(2, 5)$, but near $x = 2$ the curve approaches height 5 from both sides. Suppose also that the function has a filled point at $(2, 1)$. Then

$$f(2) = 1,$$

but

$$\lim_{x\to 2} f(x) = 5.$$

The function is not continuous at 2, because the value and the limit are different.

This example is one of the most important pictures in the study of continuity. A graph can have a limit at a point even if the value at that point is different.

Reading one-sided limits

To read

$$\lim_{x\to a^-} f(x),$$

follow the graph toward $x = a$ from the left. To read

$$\lim_{x\to a^+} f(x),$$

follow the graph toward $x = a$ from the right.

If the two approached heights agree, the two-sided limit exists. If they do not agree, the two-sided limit does not exist.

Example

Suppose that as x approaches 3 from the left, the graph approaches height 2. As x approaches 3 from the right, the graph approaches height 7. Then

$$\lim_{x \to 3^-} f(x) = 2, \qquad \lim_{x \to 3^+} f(x) = 7.$$

Since these are different,

$$\lim_{x \to 3} f(x)$$

does not exist.

The issue is not the value at $x = 3$. The issue is that the two sides do not approach the same height.

Vertical and horizontal asymptotes

If the graph grows without bound as x approaches a, then there may be a vertical asymptote:

$$x = a.$$

For example, if

$$\lim_{x \to a^+} f(x) = \infty,$$

then the graph rises without bound on the right side of a.

If the graph approaches a fixed height as $x \to \infty$ or $x \to -\infty$, there may be a horizontal asymptote:

$$y = L.$$

This means that far away, the graph gets close to the line $y = L$.

What graphs can and cannot prove

A graph can strongly suggest a limit, and it is an excellent tool for understanding. But a drawn graph may be approximate. If exactness is required, algebraic or theoretical reasoning may be needed.

Remark

Use graphs to understand the question and predict the answer. Use algebra or definitions when an exact justification is required.

Summary

When reading limits from a graph, ask:

- What height is approached from the left?
- What height is approached from the right?
- Is the value at the point included, missing, or different from the limit?
- Is there a hole, a jump, or a vertical asymptote?
- What happens far to the left or far to the right?

- Is the graph exact, or is it only a guide for reasoning?

Reading limits from graphs is not a separate skill from computing limits. It is another language for the same idea: describing approached behavior.

10.12 Choosing the right limit method

This chapter introduced several kinds of limits and several methods for working with them. The purpose is not to create a list of tricks. The purpose is to learn how to recognize what kind of limiting question is being asked and which method makes the behavior visible.

A good limit solution begins by reading the expression.

First identify the type of limit

Before calculating, ask what kind of input is approaching what.

- If the expression is $\lim_{n\to\infty} a_n$, it is a sequence limit.
- If the expression is $\lim_{x\to a} f(x)$, it is a function limit near a finite point.
- If the expression is $\lim_{x\to a^-} f(x)$ or $\lim_{x\to a^+} f(x)$, it is a one-sided limit.
- If the expression is $\lim_{x\to\infty} f(x)$, it is a limit at infinity.

Different kinds of limits naturally suggest different tools.

Then try the simplest reading

For a function limit at a finite point, direct substitution is often the first test. If it gives a meaningful value and the function is continuous there, the limit is found.

If direct substitution gives $0/0$, do not stop. Try to factor, cancel, rationalize, or use a standard limit.

For a sequence or a limit at infinity, direct substitution is not the right idea. Instead, look for dominant terms and long-term behavior.

A practical guide

Situation	Useful method
Polynomial at a finite point	direct substitution
Rational function with non-zero denominator	direct substitution
Form $0/0$ with polynomials	factor and cancel
Form $0/0$ with square roots	multiply by the conjugate
Trigonometric expression near 0	use standard trigonometric limits
Rational expression as $x\to\infty$ or $n\to\infty$	divide by dominant power
Different formulas on two sides	compute one-sided limits
Question about continuity	compare $f(a)$ with $\lim_{x\to a} f(x)$

The table is a guide, not a replacement for thinking. A problem may require more than one step.

A complete reading example

Consider

$$\lim_{x\to 4}\frac{x^2-2x-8}{x^2-9x+20}.$$

First read the type of problem: this is a function limit at a finite point. Try substitution:

$$\frac{16-8-8}{16-36+20}=\frac{0}{0}.$$

This is indeterminate, so we factor:

$$x^2-2x-8=(x-4)(x+2),$$

$$x^2-9x+20=(x-4)(x-5).$$

For $x\neq 4$,

$$\frac{x^2-2x-8}{x^2-9x+20}=\frac{x+2}{x-5}.$$

Now substitute:

$$\frac{4+2}{4-5}=-6.$$

Therefore

$$\lim_{x\to 4}\frac{x^2-2x-8}{x^2-9x+20}=-6.$$

The important point is the order of thought: identify the type of limit, test direct substitution, recognize $0/0$, transform the expression, then compute.

Preparing for derivatives

Limits are not the end of the story. They are the language needed to define derivatives. In the next chapter, we will study expressions such as

$$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}.$$

This is a limit of a quotient. It measures the limiting rate of change of a function.

The work in this chapter prepares that idea. We learned how to read $f(x+h)$, how to interpret $h\to 0$, and how to handle expressions that initially look like $0/0$.

Summary

At the end of this chapter, remember:

- A limit describes approached behavior.
- A sequence limit describes what happens as $n\to\infty$.
- A function limit near a point describes what happens as $x\to a$, with $x\neq a$.
- One-sided limits must agree for a two-sided limit to exist.
- Continuity means $\lim_{x\to a} f(x)=f(a)$.
- Indeterminate forms are not answers; they are signals to transform or analyze further.
- The method should be chosen from the structure of the expression.

A limit calculation should always end with interpretation. The final number is important, but the reasoning tells us what behavior the number describes.

Chapter 11

Derivatives and Function Analysis

This chapter is a placeholder for derivatives and their use in analysing functions. It will combine differentiation with monotonicity, extrema, convexity, asymptotes, and global understanding of function behaviour.

Chapter 12

Integrals

This chapter is a placeholder for integral calculus. It will develop antiderivatives, integration techniques, and the interpretation of integration as accumulation.

Chapter 13

Applications of Calculus

This chapter is a placeholder for applications of differential and integral calculus. It will focus on modelling, optimisation, geometry, motion, and other situations where calculus becomes a practical reasoning tool.